

FOUNDATIONS OF ELECTRONICS

Electrical Engineering

Book 1 of the Electrical Engineering Series

The physics, the mathematics, and the circuit theory that everything else is built on.

Foundations of Electronics
Book 1 of the Electrical Engineering Series

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Preface — How to Use This Book

This is the first book in a series that walks through electrical engineering from the ground up, and it is the longest, because it carries the largest load. Everything in every later volume — devices, radio, power, control, signal processing — rests on the three subjects gathered here.

The book is in three parts, meant to be read in order. Part I covers the physics: charge, fields, potential, current, magnetism, and induction. Part II covers the mathematical language that physics is written in, from SI units and dimensional analysis through logarithms, trigonometry, complex numbers, phasors, and first-order calculus. Part III brings the two together into circuit theory, DC and AC, ending with resonance and frequency response.

You need not read all of it before moving on. A reader comfortable with secondary-school physics can skim Part I; a reader already fluent with complex numbers can skim Part II. But the parts are cross-referenced throughout, and a chapter reference anywhere in this book points to a chapter in this book — the numbering runs continuously from 1 to 38 across all three parts.

Each chapter follows the same shape. It opens with the question the chapter answers, builds the picture in words first, then introduces the mathematics needed to make that picture precise, and closes with worked examples and practice problems. Three kinds of boxed panel appear throughout. "Key Idea" panels flag the handful of facts in each chapter worth memorising outright. "Worked Example" panels take a representative problem all the way to a number, showing the reasoning at each step rather than just the arithmetic. "Common Misconception" panels address specific wrong turns that show up reliably among people meeting the material for the first time, before those wrong turns calcify.

Almost every chapter ends with a short set of practice problems — usually four, occasionally three or five — and full solutions to all 149 of them are collected in Appendix B. The three bridging chapters that close each part (12, 26, and 38) carry review questions instead, or none at all. The solutions are worth more than the answers, so resist looking until you have made a real attempt: the value of a worked solution lies almost entirely in comparing it against your own reasoning and noticing where the two diverged.

A note on units and notation. SI units are used exclusively — no cgs, no mixed systems. Vectors are written in bold and their magnitudes in italics. The imaginary unit is written j rather than i , because i denotes current; Part II explains this properly when the time comes.

How This Book Is Organised

Part I — The Physics of Electricity and Magnetism (Chapters 1-12)

Begins with charge and Coulomb's law, builds the electric field and Gauss's law, then potential, capacitance, current, and resistance. The second half turns to magnetism: the magnetic field, its sources, and electromagnetic induction, closing with a first look at Maxwell's equations. No prior physics coursework is assumed, and calculus is used sparingly and explained in place.

Part II — Mathematics and Measurement (Chapters 13-26)

The mathematical toolkit, in the order an engineer needs it: SI units and dimensional analysis, engineering notation, measurement uncertainty, the algebra of circuit formulas, linear systems, graph reading, exponentials and decibels,

trigonometry and the sinusoid, complex numbers, Euler's formula, phasors and impedance, and first-order calculus. This is the part you will return to most often, and Appendices A and C are built for that.

Part III — DC and AC Circuit Theory (Chapters 27-38)

Kirchhoff's laws, equivalent resistance and dividers, nodal and mesh analysis, Thevenin and Norton equivalents, superposition, then capacitors and inductors as circuit elements and the first-order transients they produce. The remainder is sinusoidal steady state: phasor analysis, AC power and power factor, and resonance, frequency response, and transformers.

Key Idea

Chapter numbering is continuous across the three parts. A reference to "Chapter 20" means Chapter 20 of this book, which happens to sit in Part II. The part boundaries fall between chapters 12 and 13, and between 26 and 27.

About This Series

This book is the first of twenty-one, together covering the full breadth of electrical engineering from first principles through to its specialized and emerging edges. Each of the twenty-one books corresponds to one tier of a single master topic list, built so that each tier's material is gathered into one volume rather than scattered across a shelf of disconnected textbooks, and so that each book assumes only the books before it — never the ones after.

The overall arc runs from foundations (this book) through core circuits and semiconductor devices, out into RF, microwave, and mmWave engineering, across the wireless, radar, and signal-processing systems built from them, through power and control engineering, and finally into the practical disciplines — layout, simulation, debugging — and the emerging technologies that will define the field's next decade. The full list:

Book	Covers
1. Foundations	This book — physics, mathematics, and circuit theory
2. Core Circuits and Components	Network and transient analysis, resistors/capacitors/inductors as real components
3. Semiconductor Physics and Devices	Diodes, BJTs, MOSFETs, advanced materials, fabrication and packaging
4. RF and High-Frequency Engineering	Transmission lines, S-parameters, RF components and measurements
5. Noise and System Performance	Thermal, shot, and flicker noise; noise figure; SNR and BER
6. Modulation and Communication Theory	AM/FM/PM, digital modulation, coding, Shannon capacity
7. PLL and Frequency Synthesis	Phase detectors, loop filters, VCOs, frequency synthesizers
8. Microwave Engineering (1–30 GHz)	Waveguides, resonators, microstrip and stripline design
9. mmWave Engineering (30–300 GHz)	Propagation, packaging, phased arrays, 5G/6G systems
10. Antennas and Radiation	Antenna theory, radiation patterns, antenna simulation tools
11. EMC, EMI, and Emissions	Conducted and radiated emissions, shielding, regulatory standards
12. Wireless Systems and Protocols	Wi-Fi, Bluetooth, cellular architecture, MIMO systems
13. Radar Systems	Pulsed, CW, and FMCW radar; SAR; automotive radar
14. DSP and Software-Defined Radio	Sampling, digital filters, the FFT, SDR platforms
15. Power and Energy	Power systems, transformers, three-phase systems, renewables

16. Control and Automation	Feedback systems, stability, PID control, robotics
17. Oscillators, Modulators, and Transceivers	Oscillator topologies, receiver architectures, power-amplifier techniques
18. Technologies, Layout, and Packaging	PCB design, high-speed layout, substrates, advanced packaging
19. Simulation Tools and Methods	SPICE, ADS, HFSS, CST, and circuit/EM co-simulation
20. Debug and Validation	Oscilloscope, spectrum analyzer, and VNA debugging; failure analysis
21. Emerging Technologies	6G, terahertz electronics, quantum devices, photonics, AI-driven RF design

Key Idea

You do not need the other twenty books to use this one. This book, and each book after it, is written to stand on its own given its stated prerequisites — almost always just the books before it in the list above. Treat the table as a map of where you have been and where a given later interest can take you, not as a syllabus you must complete in order.

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Part I — The Physics of Electricity and Magnetism

Chapter 1 — Electric Charge

1.1 What Charge Is

Electric charge is one of a small number of properties, alongside mass, that a particle can carry without that property being made of anything more fundamental. You cannot explain charge in terms of something simpler; you can only describe what it does. What it does is this: particles with charge exert forces on other particles with charge, and those forces are enormously stronger than gravity. Two protons separated by a meter repel each other with a force about 10^{36} times stronger than they attract each other gravitationally. Electromagnetism, not gravity, is the force that dominates chemistry, materials, and — for our purposes — every circuit ever built.

It is worth pausing on just how strange this is. Gravity is the force most people build their physical intuition around, because it is the only fundamental force whose effects are obvious in daily life without any special equipment. But gravity is, by a spectacular margin, the weakest of the four fundamental forces. The only reason gravity dominates the large-scale structure of the universe — planets, stars, galaxies — is that ordinary matter is, to very high precision, electrically neutral: the positive and negative charges in any large object nearly perfectly cancel, leaving gravity, weak as it is, as the only force with nothing to cancel it out. Get that cancellation even slightly wrong — charge a balloon by rubbing it on your hair — and the electric force wins over gravity by such a wide margin that the balloon sticks to the ceiling, plainly overpowering the gravitational pull of an entire planet.

Charge comes in two kinds, arbitrarily labeled positive and negative by Benjamin Franklin in the 1740s, before anyone knew what was actually moving inside a wire. Franklin guessed wrong about which kind of charge does the moving in metals — conventional current, the direction we still use in circuit diagrams today, points the opposite way electrons actually travel — but the labels stuck, and the sign convention is now so deeply embedded in circuit theory that no one has seriously proposed fixing it. You will get used to the fact that current “flows” one way while the electrons carrying it move the other.

1.2 The Elementary Charge and Quantization

Charge is quantized: it comes only in integer multiples of a smallest unit, the elementary charge, denoted e . Every free particle you will ever measure carries a charge that is a whole-number multiple of e (quarks are the exception, but they never appear alone outside of a nucleus, so this does not affect circuit-level physics).

$$e = 1.602 \times 10^{-19} \text{ C}$$

The proton carries charge $+e$; the electron carries charge $-e$. A neutral atom has equal numbers of each and so carries no net charge. When we talk about a material becoming “charged,” we mean it has acquired a surplus or deficit of electrons relative to protons — the protons themselves are locked in the nucleus and essentially never move in ordinary electrical processes.

Key Idea

Charge is conserved. The total charge in an isolated system never changes. Charge can move from one object to another, and positive and negative charge can be created or destroyed together in equal amounts (as happens in particle physics), but the net total is always constant. This single fact underlies Kirchhoff's current law, which you will meet in the circuit-theory volume of this series.

1.3 The Coulomb

The SI unit of charge is the coulomb (C), defined operationally through electric current: one coulomb is the charge transported past a point by a steady current of one ampere flowing for one second. This means the coulomb is a derived unit, defined in terms of the ampere rather than the other way around — a historical quirk that traces back to how electrical measurements were first standardized using the forces between current-carrying wires, not by counting individual charges.

A coulomb is a very large amount of charge at the scale of individual particles: it corresponds to about 6.24×10^{18} elementary charges. In practical circuits you will almost never move a full coulomb through a wire in a hurry — a typical LED might draw twenty milliamps, meaning it moves about 0.02 coulombs of charge every second. Getting a feel for this scale early pays off: when a later chapter states that a certain wire carries a current of a few amperes, you should be able to picture, at least roughly, the staggering number of individual electrons that represents moving past any given cross-section every second, even though — as Chapter 7 will make clear — none of them individually moves especially fast.

Worked Example 1.1

Problem: A balloon rubbed on hair acquires a charge of -8.0 nC. How many excess electrons does it carry, and what fraction is this of the roughly 10^{23} atoms in a typical balloon's rubber?

Solution: $n = Q/e = (8.0 \times 10^{-9}) / (1.602 \times 10^{-19}) \approx 5.0 \times 10^{10}$ excess electrons. Comparing to the $\sim 10^{23}$ atoms making up the balloon's rubber, this is a fraction of roughly $5 \times 10^{10} / 10^{23} \approx 5 \times 10^{-13}$ — about one atom in two trillion ends up carrying an extra electron. A charge easily strong enough to stick a balloon to the ceiling involves rearranging only a vanishingly small fraction of the material's total electrons — the same “spectacular margin” theme from Section 1.1: it takes only the tiniest imaginable imbalance for the electric force to overwhelm gravity entirely.

1.4 Charging by Friction, Contact, and Induction

Objects acquire net charge through several everyday mechanisms, all of which amount to moving electrons from one material to another.

- Friction (triboelectric charging): rubbing two different insulating materials together strips electrons from one surface and deposits them on the other. A balloon rubbed on hair becomes negatively charged because it pulls electrons off the hair more readily than the hair pulls them back.
- Contact charging: touching a charged conductor to a neutral one allows charge to redistribute across both until they reach a common electric potential (a concept formalized in Chapter 5).
- Charging by induction: bringing a charged object near a neutral conductor, without touching it, causes the conductor's own free electrons to redistribute — piling up on the near side if the inducing charge is positive, leaving a net positive region on the far side. Grounding the far side while the inducing charge is still present, then removing the ground and finally the inducing charge, leaves the conductor with a net charge opposite in sign to the one that induced it.

These three mechanisms look different in practice but share a single underlying idea: nothing is ever created or destroyed, only moved. It is worth checking, for each of the three, that you could point to exactly where the electrons involved came from and where they ended up — an exercise that pays dividends once Kirchhoff's current law asks you to do the same bookkeeping, far more precisely, for entire circuit networks.

1.5 Conductors and Insulators, Revisited at the Charge Level

The difference between a conductor and an insulator is entirely a difference in how tightly a material's electrons are bound. In a conductor — copper, aluminum, silver, most metals — the outermost electrons of each atom are shared collectively across the whole material rather than staying attached to individual atoms, forming what is often called

a “sea” of free electrons. These electrons move readily in response to even a small electric field, which is exactly what makes a conductor useful for carrying current. In an insulator — rubber, glass, most plastics — every electron stays bound to its parent atom or molecule, and only a catastrophic field (well beyond anything found in ordinary electronics) will tear enough of them loose to allow appreciable current.

Semiconductors sit between these extremes, and the entire discipline of solid-state electronics exists because that middle ground can be controlled precisely through doping — a subject taken up in full in the semiconductor-devices volume of this series.

Common Misconception

It is tempting to think of a “fully charged” insulator as somehow saturated, unable to hold any more charge. In reality, an insulator's inability to conduct has nothing to do with how much charge it is holding — it is about whether the charge it already has is free to redistribute. A heavily charged insulating rod holds its charge in whatever pattern it was deposited in (often quite unevenly), precisely because the charge cannot move to spread out, unlike on a conductor, where excess charge always redistributes itself until it reaches electrostatic equilibrium (Chapter 6).

Practice Problems

1. A charged rod carries a charge of -8.0 nC . How many excess electrons does it carry?
2. Two identical small conducting spheres carry charges of $+6.0\text{ nC}$ and -2.0 nC . They are touched together and separated. What charge does each sphere carry afterward?
3. Estimate how many elementary charges pass through a wire in one minute if the wire carries a steady current of 2.0 A .
4. A third identical, initially neutral sphere is touched first to a sphere carrying $+9.0\text{ nC}$, then separated, then touched to a different sphere carrying -3.0 nC . Find the final charge on all three spheres.

Chapter 2 — Coulomb's Law and the Electric Force

2.1 Stating the Law

Charles-Augustin de Coulomb established experimentally, in 1785, exactly how the force between two point charges depends on their charge and their separation, using a delicate torsion balance sensitive enough to measure forces far smaller than the weight of the charged spheres themselves. Coulomb's law states that the magnitude of the electric force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them:

$$F = k q_1 q_2 / r^2$$

where q_1 and q_2 are the magnitudes of the two charges, r is the distance between them, and k is the Coulomb constant:

$$k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

The force is directed along the line connecting the two charges. If the charges have the same sign, the force is repulsive; if opposite, attractive. This is a vector equation in disguise — writing it as a magnitude is convenient, but you must always track direction separately, especially once more than two charges are involved.

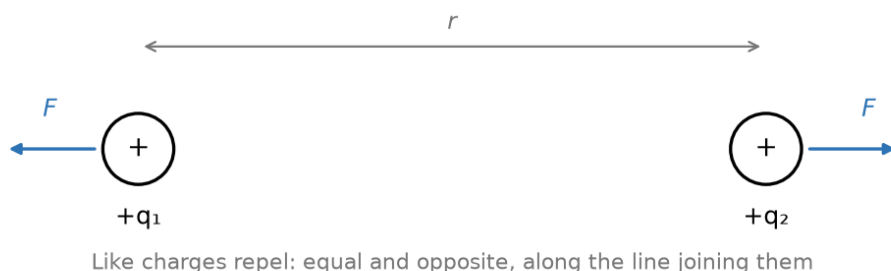


Figure 2.1 — Coulomb's law: like charges repel, with equal and opposite forces along the line joining them.

The Coulomb constant is often written in terms of a more fundamental quantity, the permittivity of free space, ϵ_0 :

$$k = 1 / (4\pi\epsilon_0), \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$$

This looks like an unnecessary complication when you are only dealing with two point charges, but the 4π factor is there so that later results — Gauss's law in particular — come out clean. You will see the payoff in Chapter 4.

2.2 The Inverse-Square Behavior

The $1/r^2$ dependence is the same functional form that governs Newton's law of gravitation, and for the same underlying geometric reason: both forces are carried by something spreading outward from a point source over the surface of an ever-expanding sphere, and the surface area of a sphere grows as r^2 , so the intensity of anything spread evenly over that surface falls as $1/r^2$. This geometric argument is made precise in Chapter 4 and is one of the most useful pieces of physical intuition you will carry into the rest of this series — the same $1/r^2$ falloff will reappear when you study radiation from antennas.

Key Idea

Doubling the distance between two charges cuts the force to one quarter. Tripling the distance cuts it to one ninth. This scaling, not the specific value of k , is the fact worth internalizing.

2.3 Superposition

When more than two charges are present, the net force on any one of them is simply the vector sum of the Coulomb forces exerted by every other charge individually, as if each pair existed alone. This is the principle of superposition, and it holds because Coulomb's law is linear in each charge — nothing about the presence of a third charge alters the force law between the first two.

Superposition sounds almost too obvious to state, until you notice how much it is quietly assumed. It is not, in fact, a logical necessity — one could imagine a universe where three charges interact through some genuinely three-body force that cannot be decomposed into pairwise contributions, and such three-body effects do arise in more exotic physical settings. That electromagnetism happens not to work that way is an experimental fact, not a mathematical inevitability, and it is one of the reasons circuit analysis is tractable at all: without superposition, adding a second component to a circuit could, in principle, change how every existing component behaved with every other, rather than simply adding its own well-defined contribution.

Worked Example 2.1

Problem: Charges of $+3.0\ \mu\text{C}$ and $+5.0\ \mu\text{C}$ sit on the x -axis at $x = 0$ and $x = 4.0\ \text{cm}$. Find the net force on a $+2.0\ \mu\text{C}$ charge placed at $x = 2.0\ \text{cm}$.

Solution: The test charge is $2.0\ \text{cm}$ from each source charge. The force from the $+3.0\ \mu\text{C}$ charge is repulsive, pushing the test charge in the $+x$ direction: $F_1 = k(3.0 \times 10^{-6})(2.0 \times 10^{-6})/(0.020)^2 \approx 135\ \text{N}$. The force from the $+5.0\ \mu\text{C}$ charge is repulsive as well, pushing in the $-x$ direction: $F_2 = k(5.0 \times 10^{-6})(2.0 \times 10^{-6})/(0.020)^2 \approx 225\ \text{N}$. These act along the same line in opposite directions, so the net force is $225 - 135 = 90\ \text{N}$, directed toward $x = 0$ (the smaller source charge is pushed away from harder, in a manner of speaking, by the larger one).

Worked Example 2.2

Problem: Three identical charges of $+4.0\ \mu\text{C}$ sit at the corners of an equilateral triangle with sides of $10\ \text{cm}$. Find the magnitude of the net force on any one of the charges.

Solution: By symmetry, the force from each of the other two charges has the same magnitude: $F = kq^2/r^2 = (8.99 \times 10^9)(4.0 \times 10^{-6})^2/(0.10)^2 \approx 14.4\ \text{N}$. These two forces act along the two sides of the triangle meeting at the corner in question, separated by the triangle's interior angle of 60° . Adding them as vectors (using the law of cosines, or equivalently resolving into components), the net force magnitude works out to $2F \cos(30^\circ) \approx 2(14.4)(0.866) \approx 25.9\ \text{N}$, directed straight out along the triangle's line of symmetry through that corner, away from the triangle's center.

2.4 When Coulomb's Law Stops Being Enough

Coulomb's law as stated applies exactly to point charges (or equivalently, spherically symmetric charge distributions, measured from outside the sphere) that are not accelerating. Real charge distributions — a charged rod, a charged disk, the electron cloud of an atom — must be handled by dividing the distribution into infinitesimal point charges and integrating. That machinery is exactly what Gauss's law, introduced in Chapter 4, is built to avoid having to do by brute force whenever the charge distribution has enough symmetry.

Common Misconception

Students frequently write Coulomb's law with the separation r substituted as if it were always measured from a fixed origin, rather than the actual distance between the two charges in question. Once you move beyond a single pair of charges arranged conveniently on an axis, r for any given pair is always the straight-line distance between that specific pair, recomputed freshly for each pair under consideration — not a coordinate borrowed from somewhere else in the problem.

Practice Problems

1. Two point charges of $+4.0\ \mu\text{C}$ and $-6.0\ \mu\text{C}$ are separated by 30 cm. Find the magnitude and direction of the force each exerts on the other.
2. At what separation would two identical charges of $+1.0\ \mu\text{C}$ repel each other with a force of exactly 1.0 N?
3. Three charges lie on a line: $+2.0\ \mu\text{C}$ at $x = 0$, $-3.0\ \mu\text{C}$ at $x = 10\ \text{cm}$, and $+1.0\ \mu\text{C}$ at $x = 20\ \text{cm}$. Find the net force on the middle charge.
4. Two identical small spheres, each of mass 1.0 g, hang from strings of equal length attached to the same point, and repel each other so that the strings make an angle of 10° with the vertical when each sphere carries the same charge q . Set up (you need not fully solve) the equation you would use to find q , identifying every force acting on one sphere.

Chapter 3 — The Electric Field

3.1 Why Introduce a Field at All

Coulomb's law describes a force between two charges, but it leaves an uncomfortable question hanging: how does one charge “know” the other is there, especially across empty space, with no visible connection between them? The field concept resolves this by proposing that a charge does not reach out and grab a second charge directly. Instead, every charge alters the space around it, creating a condition — the electric field — that exists whether or not a second charge happens to be present to feel it. A second charge, placed anywhere in that altered space, simply responds to the local field at its own location.

This is more than philosophical tidiness. The field concept is what lets electromagnetism handle radiation, time delay, and energy storage in empty space — none of which fit comfortably into an “action at a distance” picture. If a source charge is suddenly moved, a distant test charge does not respond instantaneously; the change in the field propagates outward at the speed of light (Chapter 12), and until it arrives, the test charge continues to respond to where the source charge used to be. Coulomb's law alone, treated as a direct force between two charges, has no room in it for this kind of delay; the field, treated as a physically real thing occupying the space between them, does. By the time you reach the RF and antenna volumes of this series, the field will have become the primary object of study, with charges relegated to being merely its sources.

3.2 Defining the Electric Field

The electric field at a point in space is defined operationally: place a small positive “test charge” q_0 at that point, measure the force F on it, and divide by q_0 .

$$E = F / q_0$$

The field is a vector, pointing in the direction of the force that would be felt by a positive test charge. Its units are newtons per coulomb (N/C), equivalently volts per meter (V/m) once potential is introduced in Chapter 5. The test charge is taken to be small enough that it does not itself disturb the charge distribution being measured — a limiting idealization, since strictly any nonzero q_0 will push the source charges around a little if they are free to move.

From Coulomb's law, the field of a single point charge q at distance r is:

$$E = kq / r^2$$

directed radially outward from a positive source charge, and radially inward toward a negative one.

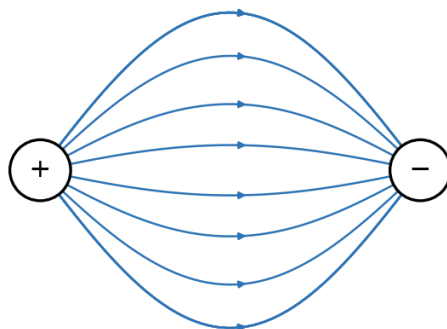
3.3 Superposition of Fields

Just as forces from multiple charges add as vectors, so do the fields they produce. The net field at any point is the vector sum of the fields each source charge would produce there individually. This is the field-picture restatement of the same superposition principle from Chapter 2, and it is what makes the field concept practically useful: once you know how to add vectors, you can find the field of any charge distribution, however complicated, by summing (or integrating) the contributions of its pieces.

3.4 Field Lines

Field lines are a visualization tool, not a physical object: a diagram of curves drawn so that the tangent to the curve at any point gives the field direction there, and the density of lines (how close together they are drawn) represents

field strength. Field lines originate on positive charges and terminate on negative charges (or run off to infinity if there is a net charge imbalance in the region). They never cross, because the field at any point has one and only one direction.



Field lines run from positive to negative charge

Figure 3.1 — Electric field lines run from positive charge to negative charge.

Key Idea

Field line density encodes field strength. Wherever field lines bunch closer together, the field is stronger; wherever they spread apart, it is weaker. This is exactly the intuition Gauss's law makes quantitative in the next chapter.

3.5 The Field of a Dipole

An electric dipole — two equal and opposite charges $+q$ and $-q$ separated by a small distance d — is worth working through explicitly because dipole fields reappear constantly: in molecular physics, in antenna theory, and as the building block for understanding how insulating (dielectric) materials respond to fields, which matters directly for capacitors in Chapter 6. The dipole is characterized by its dipole moment, a vector $\mathbf{p}$ of magnitude $p = qd$, pointing from the negative charge to the positive one. Far from the dipole (at distances much larger than d), the field falls off as $1/r^3$ rather than $1/r^2$ — faster than a single point charge, because the $+q$ and $-q$ fields nearly cancel except for the small asymmetry due to their separation.

A dipole placed in an external, uniform electric field feels no net force (the forces on $+q$ and $-q$ are equal and opposite, since both charges sit in the same external field), but it does feel a net torque, which acts to rotate the dipole until $\mathbf{p}$ aligns with the external field. This torque-without-net-force behavior is exactly how a compass needle responds to Earth's magnetic field, and it is the mechanism by which an external field polarizes and aligns the molecular dipoles inside a dielectric material — the microscopic picture behind the macroscopic dielectric constant introduced in Chapter 6.

Worked Example 3.1

Problem: Find the electric field 5.0 cm from a point charge of $+8.0$ nC.

Solution: $E = kq/r^2 = (8.99 \times 10^9)(8.0 \times 10^{-9})/(0.050)^2 \approx 2.9 \times 10^4$ N/C, directed radially away from the charge.

Worked Example 3.2

Problem: A $+4.0$ nC charge sits at the origin and a -4.0 nC charge sits at $x = 6.0$ cm. Find the net electric field at the point $x = 3.0$ cm (the midpoint between them).

Solution: Field from the $+4.0\text{ nC}$ charge, 3.0 cm away, points in the $+x$ direction (away from a positive charge): $E_1 = kq/r^2 = (8.99 \times 10^9)(4.0 \times 10^{-9})/(0.030)^2 \approx 3.996 \times 10^4\text{ N/C}$. Field from the -4.0 nC charge, also 3.0 cm away, points in the $+x$ direction as well (toward a negative charge, and the negative charge is further along $+x$, so the field at the midpoint points toward it): $E_2 \approx 3.996 \times 10^4\text{ N/C}$, same direction. Because the two fields point the same way at the midpoint of a dipole (this is a general feature of dipole fields directly between the charges), they add rather than cancel: $E_{\text{net}} \approx 7.99 \times 10^4\text{ N/C}$ in the $+x$ direction.

Common Misconception

It is a common first guess that the field exactly midway between a $+q$ and a $-q$ charge of equal magnitude should be zero, by some vague appeal to symmetry. The worked example above shows this is exactly backward for points between the charges: the two fields point the same direction there and add. The field actually vanishes only off to the sides, along the perpendicular bisector at exactly the right distance for the two individual field vectors to cancel — a distance that a quick sketch, not a symmetry argument alone, is needed to locate.

Practice Problems

1. Find the electric field at a point 10 cm from a -12 nC point charge.
2. Two point charges, $+5.0\text{ nC}$ and -5.0 nC , are 4.0 cm apart. Find the field at the midpoint between them.
3. A uniform field of 500 N/C points in the $+x$ direction. Find the force on an electron placed in this field, and state its direction.
4. A small dipole with dipole moment $2.0 \times 10^{-29}\text{ C}\cdot\text{m}$ is placed in a uniform external field of $3.0 \times 10^5\text{ N/C}$, oriented perpendicular to the field. Find the magnitude of the torque on the dipole (torque on a dipole in a uniform field has magnitude $\tau = pE \sin\theta$, where θ is the angle between p and E).

Chapter 4 — Electric Flux and Gauss's Law

4.1 Electric Flux

Electric flux is a measure of how much electric field “flows through” a surface — the field-line-counting intuition from Chapter 3, made precise. For a flat surface of area A in a uniform field E , with the surface oriented so its normal (the perpendicular direction) makes angle θ with the field, the flux is:

$$\Phi_E = E A \cos\theta$$

Flux is maximized when the field is perpendicular to the surface ($\theta = 0$) and zero when the field runs parallel to the surface ($\theta = 90^\circ$), since in that case no field lines pierce through it at all. For a curved surface or a nonuniform field, this generalizes to a surface integral, $E \cdot dA$ integrated over the surface — but for this book's purposes, the cases that matter are the ones with enough symmetry that the integral becomes trivial, which is exactly the point of Gauss's law.

4.2 Gauss's Law

Gauss's law states that the total electric flux through any closed surface is proportional to the net charge enclosed by that surface:

$$\Phi_E = Q_{\text{enc}} / \epsilon_0$$

This is one of the four Maxwell equations (met formally in Chapter 12) and it is exactly equivalent to Coulomb's law — nothing new physically is being added — but it is often dramatically easier to use. The closed surface in question, called a Gaussian surface, is entirely a mathematical construction you get to choose; it does not need to correspond to any physical boundary, and it does not need to contain any actual material at all.

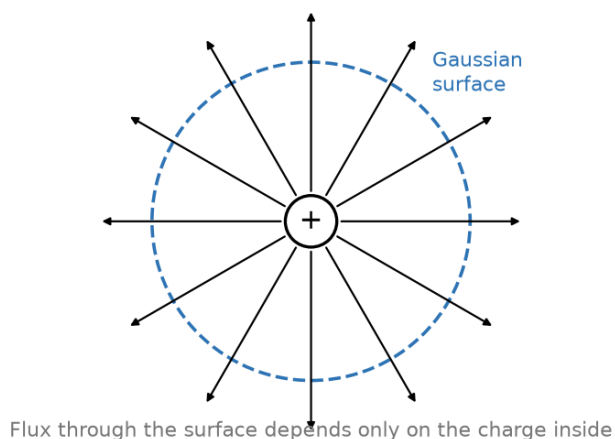


Figure 4.1 — A Gaussian surface enclosing a point charge: the flux through it depends only on the enclosed charge.

Key Idea

Choose a Gaussian surface that matches the symmetry of the charge distribution. For a point charge or any spherically symmetric distribution, use a sphere. For an infinite line or cylinder of charge, use a coaxial cylinder. For an infinite sheet of charge, use a “pillbox” straddling the sheet. Get the shape right, and the field comes out of the surface integral as a constant that can be pulled outside it, turning the integral into simple algebra.

4.3 Deriving Coulomb's Law from Gauss's Law

As a consistency check, and as the cleanest illustration of the method, take a single point charge q and surround it with an imaginary sphere of radius r centered on the charge. By symmetry, the field must point radially and have the same magnitude E everywhere on the sphere's surface — there is no direction along the sphere that is special, so the field cannot depend on where on the sphere you look. The flux integral therefore reduces to E multiplied by the sphere's total surface area, $4\pi r^2$:

$$E (4\pi r^2) = q / \epsilon_0$$

$$E = q / (4\pi\epsilon_0 r^2) = kq / r^2$$

which is exactly the point-charge field from Chapter 3 — and this is precisely why the 4π was folded into the definition of k back in Chapter 2. Choosing ϵ_0 that way makes Gauss's law come out with no stray factors of 4π in it, at the cost of Coulomb's law carrying that factor instead. Physicists made that trade deliberately because Gauss's law, not Coulomb's law, is the one that generalizes to Maxwell's equations.

4.4 The Field of an Infinite Charged Sheet, a Line Charge, and a Charged Conductor

Three results from this method are used constantly in circuit-adjacent physics and are worth having on hand. First, an infinite sheet of charge with uniform surface charge density σ (charge per unit area) produces a field of constant magnitude on either side, independent of distance from the sheet:

$$E = \sigma / (2\epsilon_0)$$

Second, an infinite line of charge with uniform linear charge density λ (charge per unit length), found using a coaxial cylindrical Gaussian surface of radius r , produces a field:

$$E = \lambda / (2\pi\epsilon_0 r)$$

falling off as $1/r$ rather than the $1/r^2$ of a point source — a direct consequence of the source being an infinite line rather than a point, so that the “surface” over which the field spreads out grows only as the cylinder's circumference (proportional to r), not as a sphere's full surface area (proportional to r^2).

Third — and this result underlies why charge sits on the outer surface of a conductor and why the interior of a conductor in electrostatic equilibrium has exactly zero field — a Gaussian surface drawn entirely inside a conducting material, where charge is free to move, must enclose zero net charge, because if there were any field inside the conductor, its free electrons would still be accelerating, contradicting the assumption of equilibrium. All excess charge on a charged conductor therefore resides entirely on its outer surface, a fact this book will use again in Chapter 6 when introducing capacitors.

Worked Example 4.1

Problem: A solid conducting sphere carries a total charge of $+6.0 \mu\text{C}$. Find the electric field at a point 20 cm from the center, given the sphere's radius is 5.0 cm.

Solution: Since 20 cm is outside the sphere, and the sphere is spherically symmetric, Gauss's law gives exactly the point-charge result with all the charge treated as concentrated at the center: $E = kq/r^2 = (8.99 \times 10^9)(6.0 \times 10^{-6})/(0.20)^2 \approx 1.35 \times 10^6 \text{ N/C}$. Note that the field outside a uniformly charged sphere is identical to that of a point charge of the same total charge — the sphere's actual radius plays no role once you are outside it.

Worked Example 4.2

Problem: An infinite sheet carries a uniform surface charge density of $+2.0 \mu\text{C}/\text{m}^2$. Find the field 1.0 cm and 10 cm from the sheet, and comment on what you find.

Solution: $E = \sigma/(2\epsilon_0) = (2.0 \times 10^{-6})/[2(8.85 \times 10^{-12})] \approx 1.13 \times 10^5 \text{ N/C}$ — at both distances, since the field of an ideal infinite sheet does not depend on distance from the sheet at all. This is a genuinely surprising result on first encounter (compare it against the point-charge and line-charge results, which both fall off with distance), and it follows directly from the field-line picture in Chapter 3: for a truly infinite sheet, the field lines run straight out perpendicular to the sheet and never spread apart, so their density — and hence the field strength — never decreases with distance. Real, finite sheets only approximate this behavior close to the sheet, well away from its edges.

Common Misconception

The infinite-sheet result $E = \sigma/(2\epsilon_0)$ is sometimes misapplied to a conductor's surface, where the correct result is instead $E = \sigma/\epsilon_0$ — twice as large, with no factor of 2 in the denominator. The difference is not an error in either formula; it reflects a genuinely different physical situation. An isolated charged sheet (with vacuum or a symmetric environment on both sides) splits its flux evenly, half going each way, which is where the factor of 2 in the sheet result comes from. A conductor's outer surface, by contrast, has zero field on its interior side (Chapter 6) by definition of electrostatic equilibrium, so all of the flux is forced out through the exterior side alone, doubling the field there relative to the two-sided sheet case for the same σ .

Practice Problems

1. A Gaussian sphere of radius 3.0 cm encloses a net charge of +2.0 nC. What is the total electric flux through the sphere?
2. An infinite sheet carries a uniform surface charge density of +4.0 $\mu\text{C}/\text{m}^2$. Find the field 2.0 cm from the sheet.
3. Explain, using Gauss's law, why the electric field just outside the surface of a charged conductor is always perpendicular to that surface.
4. An infinitely long straight wire carries a uniform linear charge density of +5.0 nC/m. Find the electric field at a perpendicular distance of 4.0 cm from the wire.

Chapter 5 — Electric Potential Energy and Voltage

5.1 From Force to Energy

Every conservative force has an associated potential energy, and the electric force is conservative — the work done moving a charge between two points does not depend on the path taken, only on the endpoints. This mirrors gravitational potential energy exactly, and the analogy is worth leaning on: just as an object gains gravitational potential energy when lifted against gravity, a positive charge gains electric potential energy when pushed toward another positive charge, against the repulsive force between them.

For two point charges q and Q separated by distance r , the electric potential energy of the pair is:

$$U = kqQ / r$$

Notice the missing square on r , compared to Coulomb's law's force expression — potential energy for a $1/r^2$ force always goes as $1/r$, the same relationship gravitational potential energy ($U = -GMm/r$) has to gravitational force. U is positive for like charges (energy is stored by pushing them together, against their mutual repulsion) and negative for unlike charges (energy is released by pulling them apart, against their mutual attraction).

5.2 Electric Potential (Voltage)

Just as the electric field is the force per unit charge, electric potential is the potential energy per unit charge:

$$V = U / q_0$$

Potential is a scalar, not a vector — a considerable practical simplification over working with fields directly, since scalars just add algebraically with no direction bookkeeping. Its SI unit is the volt (V), equal to one joule per coulomb, and it is this quantity — not the field, and not the potential energy — that a voltmeter measures and that appears on every component datasheet and circuit diagram you will encounter for the rest of this series.

The potential due to a single point charge q , at distance r , taking the zero of potential to be at infinity (the conventional choice), is:

$$V = kq / r$$

Key Idea

Only differences in potential are ever physically meaningful. The choice of where $V = 0$ is a convention, not a physical fact — exactly like choosing sea level as the zero of height. What every circuit actually cares about is potential *difference*, the voltage between two points, which is why circuit diagrams always show voltage across a component, never a component's voltage in some absolute sense.

5.3 Relating Potential and Field

Potential and field are two descriptions of the same underlying physics, related by the fact that the field points in the direction potential decreases most steeply, and its magnitude is the rate of that decrease. In one dimension:

$$E = -dV/dx$$

For the simplest and most important practical case — a uniform field between two parallel plates separated by distance d , with potential difference ΔV between them — this reduces to:

$$E = \Delta V / d$$

This equation is the workhorse relation for parallel-plate capacitors, taken up in the next chapter, and it is also the reason the field's units, newtons per coulomb, are numerically identical to volts per meter — both are legitimate units for E , and V/m is the one you will see far more often in practice.

5.4 Equipotential Surfaces

An equipotential surface is a surface on which every point has the same potential. Because moving a charge along an equipotential requires no work (there is no change in potential energy), electric field lines must always cross equipotential surfaces at right angles — if a field had any component along an equipotential surface, moving a charge along that direction would change its potential energy at constant potential, a contradiction. For a point charge, the equipotentials are concentric spheres; for a uniform field, they are flat planes perpendicular to the field.

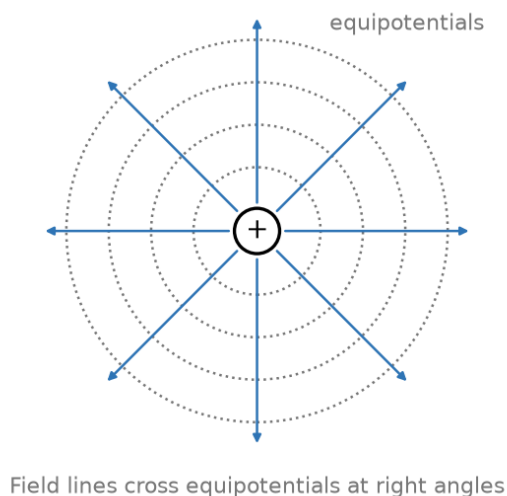


Figure 5.1 — Field lines cross equipotential surfaces at right angles.

This right-angle relationship is a genuinely useful sketching tool: given a diagram of field lines for some charge distribution, you can sketch a reasonable set of equipotential contours simply by drawing curves everywhere perpendicular to the field lines you already have, and vice versa. Topographic maps use exactly this relationship, incidentally, with contour lines of constant elevation (the gravitational analogue of equipotentials) always running perpendicular to the direction water would actually flow downhill.

5.5 Energy Stored in the Electric Field

It is natural, and correct, to think of the potential energy $U = kqQ/r$ as belonging jointly to the pair of charges, rather than to either charge individually — after all, U depends on both charges and vanishes if either one is removed. A more modern and, for this series' purposes, more useful way to think about that same energy is that it is stored in the electric field occupying the space around the charges, at an energy density (energy per unit volume) proportional to E^2 . This field-centered accounting becomes indispensable once fields exist without any nearby charge to “belong” to — as in an electromagnetic wave, where energy propagates through empty space with no charges present at all. Chapter 6 develops the field-energy picture concretely for the case of a capacitor, where it becomes possible to compute a definite, measurable quantity: the total energy stored between two charged plates.

Worked Example 5.1

Problem: Two parallel plates 2.0 mm apart are held at a potential difference of 100 V. Find the electric field between them, and the potential energy change of an electron moved from the negative plate to the positive plate.

Solution: The field magnitude is $E = \Delta V/d = 100 \text{ V} / 0.0020 \text{ m} = 5.0 \times 10^4 \text{ V/m}$. Moving an electron (charge $-e$) from the negative plate to the positive plate means moving it against the field's push on a positive charge — but the electron is negative, so the field actually assists this motion, and the electron's potential energy decreases. Numerically, $\Delta U = q\Delta V = (-1.60 \times 10^{-19} \text{ C})(100 \text{ V}) = -1.60 \times 10^{-17} \text{ J}$: the electron loses $1.60 \times 10^{-17} \text{ J}$ of potential energy, which reappears as kinetic energy — exactly the mechanism that accelerates electrons in a cathode-ray tube or a particle accelerator.

Worked Example 5.2

Problem: Find the electric potential at a point midway between a $+6.0 \text{ nC}$ charge and a -6.0 nC charge, separated by 8.0 cm .

Solution: Potential is a scalar, so it adds algebraically (no vectors needed) — a considerable simplification compared to the equivalent field calculation. Each charge is 4.0 cm from the midpoint. $V_1 = kq/r = (8.99 \times 10^9)(6.0 \times 10^{-9})/(0.040) \approx 1348 \text{ V}$ (from the positive charge). $V_2 = kq/r = (8.99 \times 10^9)(-6.0 \times 10^{-9})/(0.040) \approx -1348 \text{ V}$ (from the negative charge). Net potential: $V_1 + V_2 = 0 \text{ V}$. Contrast this with Worked Example 3.2, which found a nonzero field at the same midpoint of an equal-and-opposite dipole — potential and field do not have to vanish (or fail to vanish) at the same locations, since one is a scalar sum and the other a vector sum, and this particular symmetric case happens to make the potential exactly zero everywhere on the entire perpendicular bisector plane, not just at the midpoint.

Practice Problems

1. Find the electric potential 15 cm from a point charge of $+3.0 \text{ nC}$.
2. How much work is required to bring a $+2.0 \text{ nC}$ charge from infinity to a point 10 cm from a fixed $+5.0 \text{ nC}$ charge?
3. A uniform field of $2.0 \times 10^4 \text{ V/m}$ exists between two plates 5.0 mm apart. Find the potential difference between the plates.
4. An electron is accelerated from rest through a potential difference of 500 V . Find its final kinetic energy in joules, and its final speed (electron mass $9.11 \times 10^{-31} \text{ kg}$).

Chapter 6 — Conductors, Capacitance, and Dielectrics

6.1 Conductors in Electrostatic Equilibrium

Chapter 4 established that a conductor in electrostatic equilibrium has zero field inside it and all excess charge on its outer surface. Two further facts follow directly and are used constantly in circuit reasoning: the entire volume of a conductor in equilibrium is at a single, uniform potential (since $E = 0$ inside means V cannot change from point to point within it), and its surface is therefore an equipotential surface. This is precisely why we can speak of “the potential of a wire” or “the potential of a plate” as a single number, without needing to specify which point on that conductor we mean.

6.2 Capacitance Defined

Any two conductors separated by an insulator (or vacuum) form a capacitor: place equal and opposite charges $+Q$ and $-Q$ on the two conductors, and a potential difference V develops between them. Capacitance is defined as the ratio of charge to voltage:

$$C = Q / V$$

Capacitance is a purely geometric property of a given pair of conductors — it depends on their shape, size, separation, and the material between them, but not on Q or V individually, since doubling Q doubles V proportionally, leaving the ratio fixed. Its SI unit is the farad (F), equal to one coulomb per volt; a farad is an enormous unit for practical components, so real capacitors are almost always specified in microfarads (μF , 10^{-6} F), nanofarads (nF, 10^{-9} F), or picofarads (pF, 10^{-12} F).

6.3 The Parallel-Plate Capacitor

The parallel-plate capacitor is the simplest and most instructive geometry, and its capacitance can be derived directly from the tools already built in this book. Two plates of area A , separated by distance d , carry charges $+Q$ and $-Q$ with surface charge density $\sigma = Q/A$ on each. From Chapter 4, each charged sheet produces a field $\sigma/(2\epsilon_0)$ on either side; between the two plates, these fields add (they point the same direction there), while outside the plates they cancel. The field between the plates is therefore:

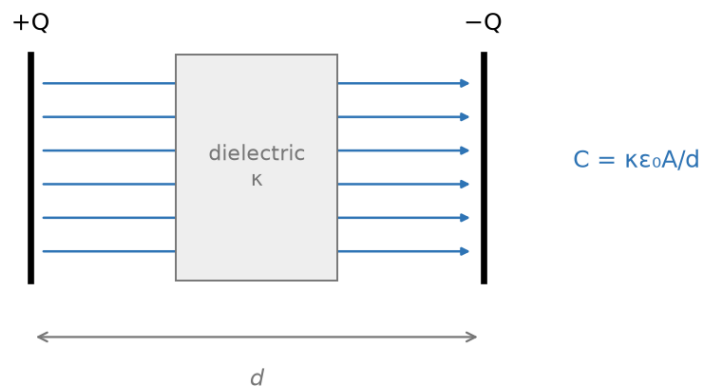


Figure 6.1 — A parallel-plate capacitor with a dielectric filling the gap between the plates.

$$E = \sigma / \epsilon_0 = Q / (\epsilon_0 A)$$

Combining this with $V = Ed$ from Chapter 5 and $C = Q/V$ gives:

$$C = \epsilon_0 A / d$$

Capacitance increases with plate area (more room to store charge at a given field) and decreases with plate separation (a smaller gap produces a larger field, hence a larger voltage, for the same charge — so it takes less charge to reach a given voltage, meaning $C = Q/V$ actually increases as d shrinks; check the algebra above to confirm this rather than trusting intuition alone, since the direction of this dependence trips up many students on a first pass).

6.4 Dielectrics

Filling the gap between the plates with an insulating material — a dielectric — rather than vacuum increases the capacitance by a factor κ , the dielectric constant (also called relative permittivity), a dimensionless number greater than 1 that depends on the material (roughly 1 for air, about 4–7 for common glass, and well over 1000 for some specialized ceramics used in high-value capacitors):

$$C = \kappa \epsilon_0 A / d$$

The physical mechanism is polarization: the dielectric's own molecules, whether permanently polar or only polarized in response to the applied field, align themselves so that their own small dipole fields (Chapter 3) partially cancel the applied field inside the material. A weaker net field for the same free charge Q means a smaller voltage $V = Ed$, and since $C = Q/V$, a smaller V at fixed Q means a larger C . Dielectrics also raise the voltage at which the capacitor breaks down (arcs across), which is often as important in practical component selection as the capacitance value itself, and this maximum safe voltage is what the finished part's datasheet lists as its rated or working voltage. Strictly, that is not quite the same thing as dielectric strength — dielectric strength is a property of the insulating material itself (measured in volts per unit thickness), while the rated voltage is the manufacturer's conservative, thickness-adjusted limit for the finished component, including a safety margin below where breakdown would actually occur.

6.5 Capacitors in Series and Parallel

Just as resistors combine in series and parallel (Chapter 8), so do capacitors, but — reflecting how capacitance is a fundamentally different kind of quantity from resistance — the combination rules come out reversed relative to resistors. Capacitors in parallel (both connected across the same pair of nodes, so both see the same voltage) simply add:

$$C_{\text{parallel}} = C_1 + C_2 + \dots$$

which makes sense physically: connecting capacitors in parallel is equivalent to increasing the effective plate area, and capacitance from Section 6.3 scales directly with area. Capacitors in series (connected end to end, so each carries the same charge Q , with the voltages across each adding to the total) combine by the reciprocal-sum rule instead:

$$1/C_{\text{series}} = 1/C_1 + 1/C_2 + \dots$$

which is the same functional form parallel resistors use, not series resistors — a fact worth committing to memory precisely because it is so easy to reach for the wrong rule by pattern-matching on the word “series” alone.

Key Idea

A capacitor stores energy, not just charge. The energy stored is $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/(2C)$. This stored energy is genuinely present in the electric field occupying the space between the plates — not a bookkeeping convenience — a fact that becomes important once you study energy flow in electromagnetic fields more generally.

Worked Example 6.1

Problem: A parallel-plate capacitor has plates of area 2.0 cm^2 separated by 0.10 mm of air. Find its capacitance, and the capacitance if the gap is instead filled with a dielectric of $\kappa = 5.0$.

Solution: Area in SI units: $A = 2.0 \times 10^{-4} \text{ m}^2$, $d = 1.0 \times 10^{-4} \text{ m}$. Air-filled: $C = \epsilon_0 A/d = (8.85 \times 10^{-12})(2.0 \times 10^{-4})/(1.0 \times 10^{-4}) \approx 17.7 \text{ pF}$. With the dielectric: $C = \kappa \times 17.7 \text{ pF} = 88.5 \text{ pF}$ — a fivefold increase, exactly matching κ .

Worked Example 6.2

Problem: A $6.0 \text{ }\mu\text{F}$ capacitor and a $3.0 \text{ }\mu\text{F}$ capacitor are connected (a) in parallel and (b) in series. Find the equivalent capacitance in each case.

Solution: (a) Parallel: $C = 6.0 + 3.0 = 9.0 \text{ }\mu\text{F}$ — larger than either individual value, as parallel capacitance always is. (b) Series: $1/C = 1/6.0 + 1/3.0 = 1/6.0 + 2/6.0 = 3/6.0$, so $C = 2.0 \text{ }\mu\text{F}$ — smaller than either individual value, as series capacitance always is. Notice this is exactly the opposite pattern from resistors, where series combinations are always larger and parallel combinations are always smaller than any individual value.

Common Misconception

Because capacitors and resistors are both drawn the same way in a circuit diagram — components in a line are “in series,” components bridging the same two nodes are “in parallel” — it is a very natural mistake to apply the resistor combination formulas to capacitors by habit. Resistance and capacitance are related to voltage and charge in inverted ways ($R = V/I$ versus $C = Q/V$), which is exactly why their series and parallel rules come out swapped relative to each other. When in doubt, rederive the rule from first principles — same charge for series capacitors, same voltage for parallel ones — rather than trusting memorized pattern-matching alone.

Practice Problems

1. A capacitor has capacitance $4.0 \text{ }\mu\text{F}$ and is charged to 12 V . Find the charge stored and the energy stored.
2. A parallel-plate capacitor's plate separation is doubled while the charge on it is held fixed. What happens to its capacitance, its voltage, and its stored energy?
3. A 10 pF capacitor is filled with a dielectric of $\kappa = 3.5$. Find the new capacitance.
4. Three capacitors, $2.0 \text{ }\mu\text{F}$, $4.0 \text{ }\mu\text{F}$, and $4.0 \text{ }\mu\text{F}$, are all connected in parallel across a 9.0 V source. Find the total charge supplied by the source.

Chapter 7 — Electric Current and Resistance

7.1 Current Defined

Everything so far in this book has concerned static charge — charge sitting still. Current is what happens when charge moves in an organized way. Electric current is defined as the rate at which charge passes a given cross-section of a conductor:

$$I = dQ/dt$$

with SI unit the ampere (A), equal to one coulomb per second. By long-standing convention (traceable to Franklin's original, backwards guess mentioned in Chapter 1), current is defined as flowing in the direction positive charge would move — which in an ordinary metal wire is opposite to the actual direction the (negatively charged) conduction electrons travel. This convention is used universally in circuit diagrams and will be used throughout the rest of this series without further comment.

7.2 Current Density and Drift Velocity

It is worth being honest about how slowly electrons in a wire actually move, because the result is famously counterintuitive. Current density J is current per unit cross-sectional area, $J = I/A$, and it is related to the drift velocity v_d of the charge carriers (the average velocity they acquire in the direction opposing the field, superimposed on their much larger random thermal motion) by:

$$J = n q v_d$$

where n is the number density of charge carriers and q is the carrier's charge. For a typical copper wire carrying a modest current, the drift velocity works out to a small fraction of a millimeter per second — an electron takes minutes to physically travel the length of a household wire. What makes a lamp turn on essentially instantly when you flip a switch is not any individual electron racing to the bulb; it is that the electric field itself, and the resulting nudge to every electron already present throughout the wire, propagates at a substantial fraction of the speed of light. The signal is fast; the carriers themselves are almost leisurely.

This distinction between the speed of the signal and the speed of the individual carriers is worth sitting with, because it resolves what otherwise looks like a genuine paradox: a wire a few kilometers long, carrying a current whose individual electrons drift at a fraction of a millimeter per second, will still light a bulb at its far end within a small fraction of a second of the switch closing. Nothing in that observation requires any electron to travel the wire's length quickly — only the field's influence, already present all along the wire, needs to propagate, and that is a wave phenomenon governed by the electromagnetic-wave result of Chapter 12, not by how fast any individual charge carrier happens to be moving.

Worked Example 7.0 — How Slow Is Drift Velocity, Really?

Problem: A copper wire of cross-sectional area 2.0 mm^2 carries a steady current of 5.0 A . Copper has roughly 8.5×10^{28} free electrons per cubic meter. Estimate the drift velocity of the conduction electrons.

Solution: From $J = nqv_d$, solve for v_d : $v_d = J/(nq) = I/(nqA)$. Plugging in: $v_d = (5.0) / [(8.5 \times 10^{28})(1.60 \times 10^{-19})(2.0 \times 10^{-6})] \approx 1.8 \times 10^{-4} \text{ m/s}$ — about 0.18 mm per second. At this rate, an individual electron would take roughly 95 minutes to travel just one meter of wire. Every appliance in your home nonetheless responds within a fraction of a second of being switched on, which is only possible because — as this section emphasizes — it is the field's influence propagating through the wire, not the individual electrons' own leisurely progress, that actually carries the “signal.”

Key Idea

The drift velocity of charge carriers and the speed at which an electrical signal propagates are two entirely different things. A typical drift velocity is a fraction of a millimeter per second, yet a lamp switches on essentially instantly — because what reaches the far end of the wire in that instant is not any particular electron, but the electric field's influence on every electron already present throughout the wire, propagating at a substantial fraction of the speed of light. Never confuse how fast the carriers move with how fast the current's effect is felt.

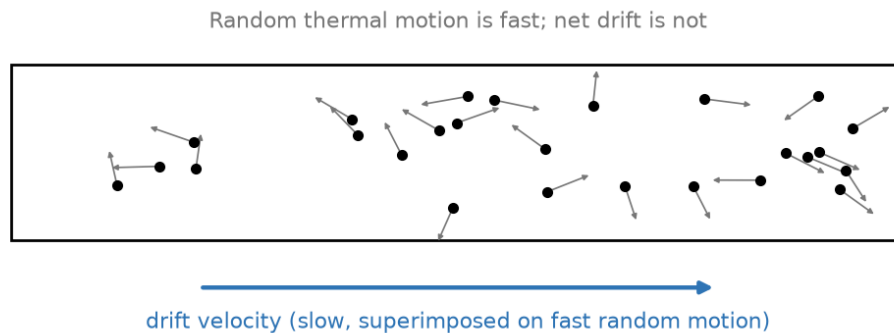


Figure 7.1 — Random thermal motion is fast; the net drift superimposed on it is not.

7.3 Resistance and Resistivity

Most materials resist the flow of current to some degree, converting electrical energy into heat as charge carriers collide with the material's atomic lattice. For a given conductor, resistance R is defined by how much voltage is required to drive a given current through it, and for many materials over a wide range of conditions, R is very nearly constant — this is Ohm's law, the subject of Chapter 8.

Resistance depends on the conductor's geometry and on an intrinsic material property, resistivity ρ :

$$R = \rho L / A$$

where L is the conductor's length and A is its cross-sectional area. Resistance increases with length (more material for carriers to push through) and decreases with cross-sectional area (more parallel paths available). Resistivity is a property of the material alone, independent of shape — copper's resistivity is famously low, which along with its low cost and easy workability is exactly why it dominates wiring; the same reasoning drives the choice of exotic low-resistivity materials like silver in specialized high-frequency applications discussed later in this series.

$$\text{Conductivity: } \sigma = 1/\rho \text{ (units: siemens per meter, S/m)}$$

7.4 Temperature Dependence

Resistivity is temperature dependent, and for most metals it increases with temperature: hotter atoms vibrate more vigorously, scattering conduction electrons more frequently and impeding their drift. Over a modest temperature range, this is well approximated as linear:

$$\rho(T) \approx \rho_0 [1 + \alpha(T - T_0)]$$

where α is the temperature coefficient of resistivity, positive for metals. Semiconductors behave oppositely — their resistivity generally decreases with rising temperature, because additional thermal energy frees more charge carriers to conduct, an effect that dominates over increased scattering. This opposite-sign behavior is one of the first clues,

historically, that semiconductors work by a fundamentally different mechanism than metals, a subject developed fully in the semiconductor-physics volume of this series.

7.5 Superconductivity, Briefly

A small number of materials, cooled below a characteristic critical temperature (for many common superconductors, only a few degrees above absolute zero, though so-called high-temperature superconductors reach tens of degrees above it), undergo a genuinely abrupt transition to a state of exactly zero resistivity — not merely very low resistivity, but immeasurably, and as far as any experiment has ever detected, exactly zero. A current started in a closed superconducting loop has been observed to persist, undiminished, for years without any external source maintaining it. Superconductivity is a quantum-mechanical phenomenon well beyond this book's scope to explain properly, but it is worth knowing it exists: it demonstrates that the ordinary, temperature-dependent resistivity described above is not some unavoidable law of nature, but a specific consequence of how electrons scatter off a lattice's vibrations and impurities in ordinary conducting materials — a mechanism that a superconductor, below its critical temperature, simply does not have.

Worked Example 7.1

Problem: A copper wire ($\rho = 1.68 \times 10^{-8} \Omega \cdot \text{m}$) is 2.0 m long with a cross-sectional area of 1.0 mm^2 . Find its resistance.

Solution: $R = \rho L/A = (1.68 \times 10^{-8} \Omega \cdot \text{m})(2.0 \text{ m}) / (1.0 \times 10^{-6} \text{ m}^2) = 0.0336 \Omega$ — a small resistance, typical of why short lengths of ordinary hookup wire are treated as essentially resistance-free in introductory circuit problems.

Worked Example 7.2

Problem: A nichrome heating element (used, for example, in a toaster) has a resistivity of $1.10 \times 10^{-6} \Omega \cdot \text{m}$. Find the length of nichrome wire, with a cross-sectional area of 0.20 mm^2 , needed to make a resistance of 20Ω , and compare this to the copper wire in the previous example.

Solution: Rearranging $R = \rho L/A$ for length: $L = RA/\rho = (20)(0.20 \times 10^{-6})/(1.10 \times 10^{-6}) \approx 3.6 \text{ m}$. Nichrome's resistivity is roughly 65 times larger than copper's, which is exactly why it is chosen for heating elements rather than copper: a practical, compact length of wire can be made to have substantial resistance (and therefore, by the power relations of Chapter 8, to dissipate substantial heat) without needing an impractically long or thin wire the way copper would require for the same resistance.

Common Misconception

It is easy to conflate resistance and resistivity, since both describe a material “resisting” current in some loose sense, and to therefore treat resistivity as if it changed when a wire is cut shorter or drawn thinner. Resistivity is a fixed property of the material itself, at a given temperature — cutting a copper wire in half does not change copper's resistivity at all, though it does change that particular piece of wire's resistance, because resistance depends on geometry ($R = \rho L/A$) while resistivity does not.

Practice Problems

1. A wire carries a current of 3.0 A. How much charge passes a given point in the wire in 5.0 seconds?
2. A conductor's length is tripled while its cross-sectional area is halved. By what factor does its resistance change?
3. A resistor has resistance 100Ω at 20°C and a temperature coefficient of resistivity of $0.004 / ^\circ\text{C}$. Estimate its resistance at 80°C .
4. A copper wire and an aluminum wire (resistivity $2.65 \times 10^{-8} \Omega \cdot \text{m}$) have the same length and the same resistance. Find the ratio of their cross-sectional areas.

Chapter 8 — Ohm's Law and a First Look at Circuits

8.1 Ohm's Law

Georg Ohm found, in 1827, that for a large class of materials (now called ohmic materials, or resistors when packaged as discrete components), the current through the material is directly proportional to the voltage across it, at a fixed temperature:

$$V = IR$$

This is not a law of nature in the way Coulomb's law or Gauss's law are — it is an empirical approximation that happens to hold extremely well for metals and for manufactured resistors under normal operating conditions, and it fails outright for many devices you will meet later in this series, including diodes and transistors, whose entire usefulness stems from their current-voltage relationship being deliberately non-Ohmic.

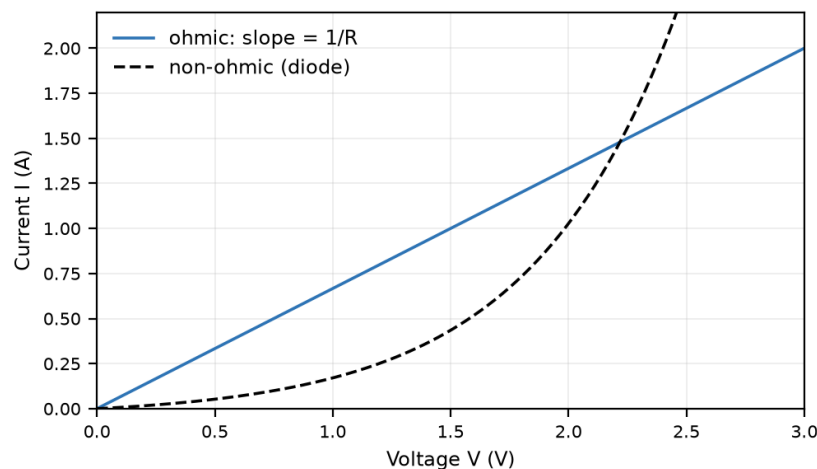


Figure 8.1 — An ohmic device has a straight-line I - V curve; a non-ohmic device (such as a diode) does not.

Key Idea

Ohm's law relates three quantities — memorize the triangle, not just the formula. $V = IR$, so also $I = V/R$ and $R = V/I$. Given any two of voltage, current, and resistance, the third follows immediately. This single relationship, together with the two Kirchhoff laws developed in the dedicated circuit-theory volume of this series, is sufficient to analyze any resistive DC circuit, no matter how complex.

8.2 Power in a Resistor

Electrical power is the rate at which electrical energy is delivered or dissipated:

$$P = IV$$

Combined with Ohm's law, this yields two equally useful alternate forms:

$$P = I^2R = V^2/R$$

For a resistor, this power is dissipated as heat — this is Joule heating, the same physical mechanism (electrons scattering off the lattice, from Chapter 7) that gives rise to resistance in the first place, now viewed from an energy-accounting perspective rather than a current-impeding one. Every resistor in a real circuit has a maximum power

rating, beyond which the resulting heat will damage or destroy it; checking that a resistor's power rating exceeds its expected dissipation is a standard, mandatory step in circuit design that this series will return to repeatedly.

8.3 A First Circuit: Series and Parallel Resistors

Two resistors are in series if the same current must flow through both, one after the other, with no branch point between them. Their combined resistance simply adds:

$$R_{\text{series}} = R_1 + R_2 + \dots$$

Two resistors are in parallel if both are connected across the same pair of nodes, so the same voltage appears across each while the total current splits between them. Their combined resistance follows from the reciprocal sum:

$$1/R_{\text{parallel}} = 1/R_1 + 1/R_2 + \dots$$

A useful special case, worth having memorized, is two resistors in parallel:

$$R_{\text{parallel}} = R_1 R_2 / (R_1 + R_2)$$

Series resistance is always larger than the largest individual resistor (more obstacles, end to end); parallel resistance is always smaller than the smallest individual resistor (more parallel paths for current to take, so less overall opposition). This behavior — and the systematic method for analyzing arbitrary networks that do not reduce to simple series/parallel combinations — is developed in full in this series' dedicated circuit-theory volume, which builds directly on the definitions in this chapter.

8.4 A First Look at Voltage Division

One immediate, practical consequence of the series-resistor rule is worth introducing here, ahead of its full treatment in the circuit-theory volume, because it appears so often in practice: two resistors in series across a voltage source V split that voltage between them in direct proportion to their resistances, since the same current I flows through both:

$$V_1 = V \cdot R_1 / (R_1 + R_2), \quad V_2 = V \cdot R_2 / (R_1 + R_2)$$

This voltage divider is one of the single most common sub-circuits in all of electronics, used to derive a smaller reference voltage from a larger supply, and it is worth internalizing this chapter's result now, since it will reappear as a building block almost immediately in the next book of this series.

Worked Example 8.1

Problem: A 12 V battery is connected across a single 6.0 Ω resistor. Find the current drawn and the power dissipated.

Solution: $I = V/R = 12/6.0 = 2.0$ A. Power: $P = IV = (2.0)(12) = 24$ W (equivalently $P = I^2R = (2.0)^2(6.0) = 24$ W, a useful cross-check).

Worked Example 8.2

Problem: A 10 Ω resistor and a 30 Ω resistor are connected in series across a 20 V source. Find the current through the circuit and the voltage across each resistor.

Solution: Total series resistance: $R = 10 + 30 = 40$ Ω . Current: $I = V/R = 20/40 = 0.50$ A, the same through both resistors since they are in series. Voltage across the 10 Ω resistor: $V_1 = IR_1 = (0.50)(10) = 5.0$ V. Voltage across the 30 Ω resistor: $V_2 = IR_2 = (0.50)(30) = 15$ V. Check: $5.0 + 15 = 20$ V, matching the source voltage exactly, as it must for a single series loop.

Common Misconception

A frequent error when first combining series and parallel resistors is to add parallel resistances directly, the way series resistances add — forgetting the reciprocal step entirely. Two equal $100\ \Omega$ resistors in parallel do not combine to $200\ \Omega$; they combine to $50\ \Omega$, half of either individual value, since two identical parallel paths each carry half the total current at the same voltage, doubling the effective conductance and therefore halving the effective resistance. If a “combined” parallel resistance ever comes out larger than the smallest resistor in the group, the reciprocal step has been dropped somewhere in the calculation.

Practice Problems

1. A $9.0\ \text{V}$ battery drives a current of $0.30\ \text{A}$ through a resistor. Find the resistor's value and the power it dissipates.
2. Two resistors, $4.0\ \Omega$ and $12\ \Omega$, are connected in parallel. Find the equivalent resistance.
3. A resistor rated for a maximum power of $0.25\ \text{W}$ is to be used with a $5.0\ \text{V}$ source. What is the minimum resistance that keeps it within its rating?
4. A voltage divider is built from a $2.0\ \text{k}\Omega$ resistor and an $8.0\ \text{k}\Omega$ resistor in series across a $10\ \text{V}$ supply. Find the voltage across the $8.0\ \text{k}\Omega$ resistor.

Chapter 9 — Magnetism and the Magnetic Field

9.1 A Different Kind of Field

Magnetism was known and used — in lodestones and compass needles — for well over a thousand years before anyone connected it to electricity at all. The connection, once made in the early nineteenth century, turned out to run so deep that magnetism and electricity are now understood as two faces of a single phenomenon, electromagnetism, rather than two separate forces that happen to interact. This chapter introduces the magnetic field on its own terms; the following two chapters show where magnetic fields come from and how electricity and magnetism interconvert.

Unlike the electric field, which exerts a force on any charge simply for being present, the magnetic field exerts a force only on charge that is moving (or, equivalently, on current). A stationary charge sitting in a magnetic field feels nothing at all from that field.

9.2 The Magnetic Force on a Moving Charge

The force on a charge q moving with velocity $\mathbf{v}$ through a magnetic field $\mathbf{B}$ is given by:

$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

The $\times$ denotes a vector cross product, which is the reason the magnetic force behaves so differently from anything encountered so far in this book: its magnitude is $F = qvB \sin\theta$, where θ is the angle between $\mathbf{v}$ and $\mathbf{B}$ (maximal when velocity and field are perpendicular, zero when they are parallel), and — critically — its direction is always perpendicular to both $\mathbf{v}$ and $\mathbf{B}$ simultaneously, found using the right-hand rule: point the fingers of your right hand along $\mathbf{v}$, curl them toward $\mathbf{B}$, and your thumb points along $\mathbf{v} \times \mathbf{B}$ (adjust the sign for a negative charge, which experiences a force in the opposite direction).

Key Idea

A magnetic force can never do work on a charge. Because the magnetic force is always perpendicular to the charge's velocity, it can never have a component along the direction of motion, and force perpendicular to displacement does zero work. A magnetic field can change a charge's direction, but never its speed or its kinetic energy. Any speeding up or slowing down of a charged particle in a real device is always due to an electric field, even in devices — like motors — that are popularly described as “magnetic.”

9.3 Circular Motion in a Uniform Field

Because the magnetic force is always perpendicular to velocity and constant in magnitude for a fixed speed, a charged particle moving perpendicular to a uniform magnetic field travels in a perfect circle, with the magnetic force supplying exactly the centripetal force needed:

$$qvB = mv^2/r \Rightarrow r = mv / (qB)$$

This relationship is the operating principle behind mass spectrometers, cyclotrons, and — in a more elaborate form — the particle accelerators used in high-energy physics. It also explains why charged particles from the solar wind spiral around Earth's magnetic field lines rather than striking the atmosphere directly, redirected toward the poles where the field lines converge — the mechanism behind the aurora.

If the particle's velocity has a component along the field direction as well as perpendicular to it, the perpendicular component still produces circular motion exactly as described, while the parallel component is entirely unaffected by the field (since $\mathbf{v} \times \mathbf{B}$ vanishes when $\mathbf{v}$ is parallel to $\mathbf{B}$). The combined motion is therefore a helix — a circle that

also drifts steadily along the field direction — rather than a closed circle. This helical motion is exactly what channels charged particles along Earth's field lines toward the poles, and it is worth being able to picture clearly.

9.4 The Magnetic Force on a Current-Carrying Wire

A current-carrying wire in a magnetic field experiences a force, because the current is nothing more than a great many moving charges, each individually subject to $F = q\mathbf{v} \times \mathbf{B}$. Summed over a straight wire segment of length L carrying current I in a uniform field $\mathbf{B}$, the total force is:

$$\mathbf{F} = I \mathbf{L} \times \mathbf{B}$$

with magnitude $F = BIL \sin\theta$, where θ is again the angle between the wire (taken in the direction of conventional current flow) and the field. This is the force that turns the rotor of every electric motor: current-carrying wire loops placed in a magnetic field experience a torque that spins the motor's shaft, a device treated in full once this series reaches its power-and-machines material.

9.5 Units

The SI unit of magnetic field is the tesla (T), defined so that a charge of one coulomb moving at one meter per second, perpendicular to a one-tesla field, experiences a force of one newton. A tesla is a substantial field — Earth's own magnetic field at its surface is roughly 0.00005 T ($50 \mu\text{T}$), a small fridge magnet might reach 0.01 T , and MRI machines operate in the range of $1\text{--}3 \text{ T}$. An older, non-SI unit, the gauss ($1 \text{ T} = 10,000 \text{ G}$), still appears frequently in older texts and in some manufacturer datasheets, and is worth recognizing even though this book uses tesla throughout.

Worked Example 9.1

Problem: A proton moves at $2.0 \times 10^6 \text{ m/s}$ perpendicular to a 0.50 T magnetic field. Find the magnitude of the magnetic force on it, and the radius of its circular path.

Solution: $F = qvB = (1.60 \times 10^{-19})(2.0 \times 10^6)(0.50) \approx 1.6 \times 10^{-13} \text{ N}$. Radius: $r = mv/(qB) = (1.67 \times 10^{-27} \text{ kg})(2.0 \times 10^6)/[(1.60 \times 10^{-19})(0.50)] \approx 0.042 \text{ m}$, about 4.2 cm .

Worked Example 9.2

Problem: A straight horizontal wire 0.80 m long carries a current of 6.0 A due east. Find the magnitude and direction of the magnetic force on the wire if it sits in a uniform 0.40 T field pointing straight down (toward the ground).

Solution: Magnitude: $F = BIL \sin\theta$. The wire (east) and the field (down) are perpendicular, so $\theta = 90^\circ$ and $\sin\theta = 1$: $F = (0.40)(6.0)(0.80) = 1.92 \text{ N}$. Direction, by the right-hand rule for $I \mathbf{L} \times \mathbf{B}$: point your fingers east (current direction), curl them downward (field direction) — your thumb points north. The force on the wire is therefore 1.92 N directed due north.

Common Misconception

Because the magnetic force involves a cross product, its direction is one of the more error-prone parts of introductory electromagnetism, and a common shortcut error is to assume the force simply points “along” one of the two vectors involved, or splits the difference between them. It does neither: $F = q\mathbf{v} \times \mathbf{B}$ is always exactly perpendicular to both $\mathbf{v}$ and $\mathbf{B}$, full stop, regardless of the angle between $\mathbf{v}$ and $\mathbf{B}$ (as long as that angle is not 0° or 180° , where the cross product vanishes entirely). There is no partial or intermediate direction to reason about — only perpendicular to both, with the right-hand rule fixing which of the two possible perpendicular directions is correct.

Practice Problems

1. An electron moves at $3.0 \times 10^5 \text{ m/s}$ perpendicular to a 0.20 T field. Find the magnetic force on it.

2. A straight wire 0.50 m long carries a current of 4.0 A, oriented perpendicular to a 0.30 T field. Find the force on the wire.
3. Explain, in one or two sentences, why a magnetic field can change a charged particle's direction of travel but never its speed.
4. A proton travels in a circular path of radius 0.10 m in a magnetic field of 0.80 T. Find the proton's speed.

Chapter 10 — Sources of the Magnetic Field

10.1 Moving Charge Makes a Field

Chapter 9 described how a magnetic field pushes on moving charge. The complementary fact — and the one that reveals magnetism and electricity as genuinely unified — is that moving charge itself produces a magnetic field. A stationary charge produces only an electric field; the same charge, once set in motion, produces a magnetic field as well, without losing its electric field. Current-carrying wires, not permanent magnets, are the fundamental source of magnetic fields in essentially all of modern electrical engineering; permanent magnets turn out to owe their field to countless aligned electron orbits and spins, which are themselves tiny current loops at the atomic scale.

10.2 The Field of a Long Straight Wire

The magnetic field produced by a long, straight, current-carrying wire circles around the wire, with field lines forming closed circles centered on the wire and lying in planes perpendicular to it. The direction is given by a right-hand rule: point your right thumb in the direction of conventional current, and your curled fingers indicate the field's direction. The field's magnitude at a perpendicular distance r from the wire is:

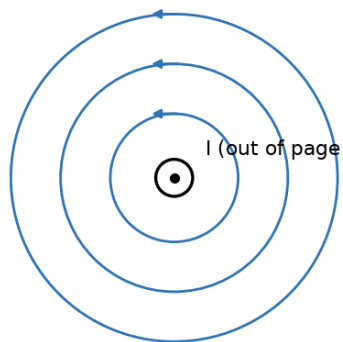
$$B = \mu_0 I / (2\pi r)$$

where μ_0 is the permeability of free space:

$$\mu_0 \approx 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$$

Since the 2019 redefinition of the SI, μ_0 is no longer an exactly defined constant — it is measured, and $4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ is its familiar introductory approximation, accurate to many more decimal places than any circuit calculation in this book will ever need.

Notice the structural resemblance to the electric field of an infinite line charge from Chapter 4 — both fall off as $1/r$ rather than the $1/r^2$ of a point source, because both sources are effectively infinite lines rather than points, and the same expanding-geometry argument that gave $1/r$ for that line charge gives $1/r$ here as well.



B circles the wire; magnitude falls as $1/r$

Figure 10.1 — The magnetic field of a long straight wire circles the wire, falling off as $1/r$.

10.3 Force Between Two Parallel Wires

Two parallel current-carrying wires each produce a field that exerts a force on the other, via the $\mathbf{F} = I\mathbf{L} \times \mathbf{B}$ relation from Chapter 9. Wires carrying current in the same direction attract; wires carrying current in opposite directions repel — a result many students find backward on first encounter, since it is the reverse of the same-charges-repel rule for static charges. This attraction/repulsion between current-carrying wires was, historically, exact enough to measure that it was used for decades to define the ampere itself, before the 2019 SI redefinition tied the ampere to the elementary charge instead.

10.4 Ampere's Law

Just as Gauss's law provides a shortcut to the electric field for sufficiently symmetric charge distributions, Ampere's law provides the analogous shortcut for the magnetic field of sufficiently symmetric current distributions. It states that the line integral of $\mathbf{B}$ around any closed path (an “Amperian loop”) equals μ_0 times the net current enclosed by that path:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Applied to a circular Amperian loop of radius r centered on a long straight wire, where by symmetry $\mathbf{B}$ is constant in magnitude around the loop and everywhere tangent to it, this reduces immediately to $B(2\pi r) = \mu_0 I$, recovering the result of Section 10.2 with almost no work — exactly the payoff Gauss's law provided for the electric field.

Key Idea

Ampere’s law is to the magnetic field exactly what Gauss’s law is to the electric field. Both are always true, but both are only directly solvable for the field when enough symmetry exists to argue the field has constant magnitude around some convenient loop — an infinite straight wire or a solenoid being the standard examples on the magnetic side. Without that symmetry, the law still holds, but finding $\mathbf{B}$ from it requires the far more general, and considerably more work-intensive, Biot–Savart approach instead.

10.5 The Field of a Solenoid

A solenoid — a long coil of wire wound in a tight helix — produces, to a good approximation deep inside the coil (away from its ends), a uniform magnetic field parallel to its axis, closely analogous to the uniform field between parallel-plate capacitor plates. Applying Ampere's law to a rectangular loop that runs partly inside and partly outside the solenoid gives:

$$\mathbf{B} = \mu_0 n \mathbf{I}$$

where n is the number of turns of wire per unit length along the solenoid's axis. Solenoids are the standard way to produce a strong, uniform, and controllable magnetic field in a compact region, and they form the physical basis of the inductor, a component taken up in the next chapter of this book.

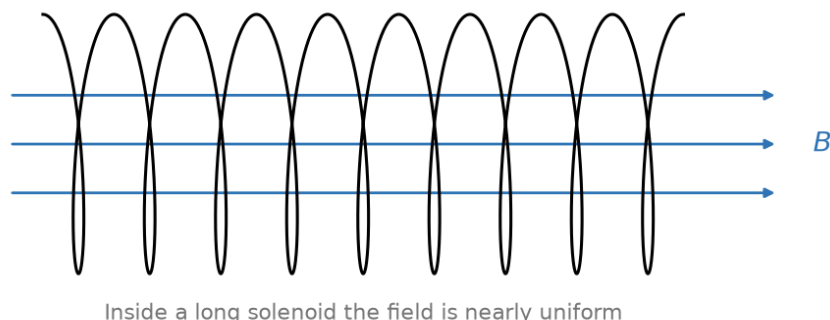


Figure 10.2 — Inside a long solenoid, the magnetic field is nearly uniform.

10.6 Ferromagnetic Materials

Placing a ferromagnetic core — iron being the most familiar example — inside a solenoid dramatically increases the resulting field, often by a factor of several hundred to several thousand, because the core's own atomic-scale current loops (electron spins) align with the applied field and add their own contribution to it, in direct analogy with how a dielectric's molecular dipoles respond to an applied electric field (Chapter 6). This effect is captured by replacing μ_0 with an effective permeability $\mu = \mu_r \mu_0$, where the relative permeability μ_r can be in the thousands for common transformer-grade iron. This is exactly why practical inductors and transformers are wound around iron or ferrite cores rather than left as air-core coils: the same number of turns and the same current produce a vastly stronger field, or equivalently, a much smaller and lighter coil can produce the field a much larger air-core solenoid would need to match.

Worked Example 10.1

Problem: A long straight wire carries a current of 5.0 A. Find the magnetic field at a distance of 2.0 cm from the wire.

Solution: $B = \mu_0 I / (2\pi r) = (4\pi \times 10^{-7})(5.0) / [2\pi(0.020)] = 5.0 \times 10^{-5} \text{ T}$ — exactly the same order of magnitude as Earth's own field, which is why a compass placed near a current-carrying wire is visibly deflected.

Worked Example 10.2

Problem: A solenoid has 800 turns over a length of 20 cm and carries a current of 2.0 A. Find the magnetic field inside it, and find the new field if an iron core with relative permeability 1500 is inserted.

Solution: Turns per unit length: $n = 800/0.20 = 4000 \text{ turns/m}$. Air-core field: $B = \mu_0 n I = (4\pi \times 10^{-7})(4000)(2.0) \approx 1.0 \times 10^{-2} \text{ T} = 10 \text{ mT}$. With the iron core: $B = \mu_r \mu_0 n I = 1500 \times 10 \text{ mT} = 15 \text{ T}$ — an enormous field, and in practice, iron cores saturate (their internal alignment reaches a maximum) well before reaching such an extreme value, so this idealized result overstates what a real core would achieve at this current; the qualitative point, that the core provides a three-orders-of-magnitude boost, still holds at more realistic operating currents.

Common Misconception

The like-charges-repel, unlike-charges-attract rule from electrostatics is so deeply ingrained by this point in the book that it is tempting to assume current-carrying wires must follow an analogous rule based on current direction. They do not follow the electrostatic pattern at all: parallel currents in the *same* direction attract, and parallel currents in *opposite* directions repel — the reverse of the “same repels, opposite attracts” pattern from charge. The two situations are governed by genuinely different force laws (Coulomb's law versus the magnetic force law), and importing intuition from one to the other, without working through the right-hand rule explicitly, reliably produces the wrong sign.

Practice Problems

1. Two long parallel wires, 5.0 cm apart, each carry a current of 3.0 A in the same direction. State whether they attract or repel.
2. A solenoid has 600 turns over a length of 30 cm and carries a current of 2.0 A. Find the magnetic field inside it (air core).
3. At what distance from a wire carrying 10 A does the magnetic field equal 1.0×10^{-4} T?
4. A solenoid produces a field of 8.0 mT with an air core at a given current. Find the field at the same current if a ferrite core of relative permeability 200 is inserted.

Chapter 11 — Electromagnetic Induction

11.1 Faraday's Discovery

Every result so far in this book's magnetism chapters has concerned static, unchanging current distributions producing static magnetic fields. Michael Faraday discovered, in 1831, something qualitatively new: a changing magnetic field, or a changing magnetic flux through a loop of wire more generally, induces an electromotive force — and hence a current, if the loop is closed — even with no battery anywhere in the circuit. This is electromagnetic induction, and it is, without exaggeration, the single physical fact that makes electric generators, transformers, and power grids possible; without it, essentially all large-scale electricity generation as it currently exists would not work.

11.2 Magnetic Flux

Magnetic flux is defined exactly analogously to electric flux from Chapter 4 — for a flat surface of area A in a uniform field B , oriented so its normal makes angle θ with the field:

$$\Phi_B = B A \cos\theta$$

with SI unit the weber (Wb), equal to one tesla-meter-squared. It is a change in this quantity — not the field itself, and not even the presence of current — that induction responds to.

11.3 Faraday's Law

Faraday's law states that the induced electromotive force (EMF) around a closed loop equals the negative rate of change of magnetic flux through the loop:

$$\varepsilon = -d\Phi_B/dt$$

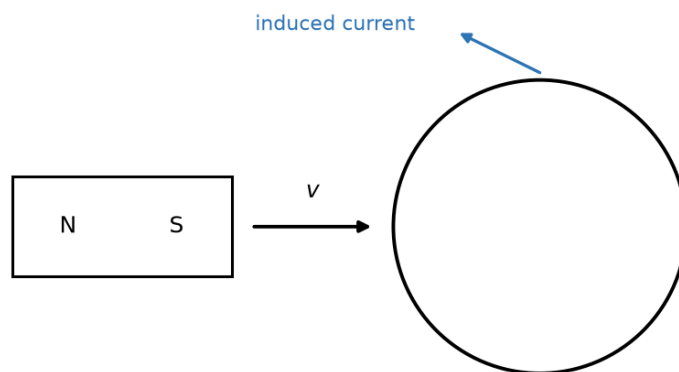
For a coil of N turns, each threaded by the same changing flux, the induced EMFs from every turn add:

$$\varepsilon = -N d\Phi_B/dt$$

Flux can change for several distinct reasons — the field itself changing in time, the loop's area changing, the loop rotating relative to the field, or the loop moving into or out of a field-filled region — and Faraday's law responds identically to all of them, because all of them change Φ^b . This is precisely how an electric generator works: a coil is mechanically rotated within a magnetic field, continuously changing Φ^b as the coil's orientation relative to the field changes, producing an alternating EMF — the direct physical origin of the AC waveform this series' circuit-theory volume takes as its starting point.

11.4 Lenz's Law and the Meaning of the Minus Sign

The minus sign in Faraday's law is not a bookkeeping detail — it encodes Lenz's law, one of the more important qualitative facts in this book. Lenz's law states that the induced current always flows in whatever direction opposes the change in flux that produced it. If flux through a loop is increasing, the induced current flows in the direction that would, by the results of Chapter 10, produce a magnetic field opposing that increase; if flux is decreasing, the induced current flows to reinforce it.



Lenz's law: the induced current opposes the change that made it

Figure 11.1 — Lenz's law: the induced current opposes the change in flux that produced it.

Key Idea

Lenz's law is a direct consequence of energy conservation. If induced current instead reinforced a changing flux, the effect would build on its own cause without limit, generating energy from nothing. Lenz's law guarantees that any electromagnetic induction always opposes the mechanical or electrical change that created it — which is precisely why turning a generator's shaft takes real mechanical effort: you are always pushing against an induced magnetic braking force, and that mechanical work is exactly what becomes the electrical energy delivered by the generator.

11.5 Motional EMF

A useful special case: a conducting rod of length L , moving with velocity v perpendicular to a magnetic field B , develops an EMF along its length:

$$\varepsilon = BLv$$

This can be derived directly from the magnetic force on the rod's own free charges ($F = q\mathbf{v} \times \mathbf{B}$ from Chapter 9) without invoking flux at all, and it is reassuring — and a useful check on your understanding — that both approaches, flux-based and force-based, agree exactly. This is the operating principle behind simple linear generators and behind induced voltages in moving conductors more generally, including eddy currents induced in metal moving through a field, which is exploited deliberately in magnetic braking systems.

11.6 Energy Stored in a Magnetic Field: A First Look at Inductance

Just as a capacitor stores energy in the electric field between its plates (Chapter 6), a current-carrying coil stores energy in its own magnetic field. A coil's ability to do this — to build up a magnetic field, store energy in it, and (by Faraday's law, applied to itself) resist any change in the current that produces it — is quantified by its self-inductance L , defined so that the induced EMF opposing a change in the coil's own current is:

$$\varepsilon = -L \, dI/dt$$

with SI unit the henry (H). Notice the structural parallel to Ohm's law and to the capacitor relation $C = Q/V$: three of this book's core circuit elements — resistor, capacitor, inductor — each relate a pair of circuit quantities (voltage and current, charge and voltage, and now voltage and the rate of change of current) through a single characteristic constant of the component. A full treatment of inductors, RL circuits, and the energy $U = \frac{1}{2}LI^2$ stored in an inductor's magnetic field belongs properly to the circuit-theory volume of this series, which develops all three of

these elements — resistor, capacitor, and inductor — together, once the tools for handling time-varying circuits are in place. What matters here is simply recognizing that Chapter 10's solenoid, once wired into a circuit and thought of as opposing its own changing current via Faraday's law, is exactly what an inductor is.

Worked Example 11.1

Problem: A circular loop of wire, radius 5.0 cm, sits in a magnetic field that increases uniformly from 0.20 T to 0.80 T over 0.30 s, with the field perpendicular to the loop throughout. Find the average induced EMF.

Solution: Loop area: $A = \pi r^2 = \pi(0.050)^2 \approx 7.85 \times 10^{-3} \text{ m}^2$. Change in flux: $\Delta\Phi = A\Delta B = (7.85 \times 10^{-3})(0.80 - 0.20) = 4.71 \times 10^{-3} \text{ Wb}$. Average EMF: $\varepsilon = \Delta\Phi/\Delta t = 4.71 \times 10^{-3}/0.30 \approx 1.57 \times 10^{-2} \text{ V}$.

Worked Example 11.2

Problem: A metal rod 0.40 m long slides along a pair of conducting rails, moving at a constant 3.0 m/s, perpendicular to a uniform 0.60 T magnetic field pointing straight into the plane of the rails. Find the motional EMF developed, and state, using Lenz's law, which way current flows in the circuit formed by the rod and rails as the rod moves.

Solution: Motional EMF: $\varepsilon = BLv = (0.60)(0.40)(3.0) = 0.72 \text{ V}$. To find the current direction with Lenz's law: as the rod slides (say, to the right, into the region enclosed by the rails and rod), the enclosed loop area increases, and since B points into the page, the enclosed flux (into the page) is increasing. The induced current must oppose that increase, meaning it must create its own field pointing out of the page inside the loop — by the right-hand rule for a current loop (Chapter 10), this means the induced current flows counterclockwise as viewed with B pointing into the page toward you.

Common Misconception

Faraday's law is sometimes misremembered as responding to the magnetic field itself, so that a strong but constant field is expected to induce a large EMF. It is the rate of change of flux that matters, not the field's absolute size — a room permeated by an enormous, perfectly steady magnetic field induces no EMF in any wire loop sitting motionless within it, no matter how strong that field is, while a comparatively weak field changing quickly can induce a substantial EMF. The relevant question for any induction problem is always “how fast, and in what way, is the flux through this loop changing?” — never “how strong is the field?” by itself.

Practice Problems

1. A flat coil of 50 turns and area 0.020 m^2 sits perpendicular to a magnetic field that changes at a rate of 0.50 T/s . Find the induced EMF.
2. A conducting rod 0.30 m long moves at 4.0 m/s perpendicular to a 0.50 T field. Find the motional EMF developed along the rod.
3. A bar magnet is pushed, north pole first, toward a loop of wire. Using Lenz's law, state the direction of the induced current in the loop (as viewed from the approaching magnet's side) and explain your reasoning.
4. An inductor with $L = 0.20 \text{ H}$ carries a current that is changing at a rate of 5.0 A/s . Find the magnitude of the induced EMF across the inductor.

Chapter 12 — Toward Maxwell's Equations and the Rest of This Series

12.1 Gathering the Pieces

This book has built up electricity and magnetism largely as two separate stories that occasionally touch — first electrostatics (Chapters 1–6), then steady currents and resistance (Chapters 7–8), then magnetostatics (Chapters 9–10), then the one genuine crossover result, electromagnetic induction (Chapter 11). James Clerk Maxwell's achievement, completed in 1865, was to recognize that four equations — two of which you have already met by name — form a complete, self-consistent description of every electric and magnetic phenomenon covered in this book, and a great many that are not.

12.2 Maxwell's Four Equations, Informally Stated

- Gauss's law for electricity (Chapter 4): electric field lines originate on positive charge and terminate on negative charge; the total flux through any closed surface measures the net enclosed charge.
- Gauss's law for magnetism (not previously stated by this name, but implicit in every field-line diagram drawn in Chapters 9–10): magnetic field lines never begin or end anywhere — there is no such thing as an isolated magnetic “charge,” or magnetic monopole. Every magnetic field line that leaves a region must loop back into it somewhere.
- Faraday's law (Chapter 11): a changing magnetic flux induces an electric field (which in a conducting loop drives a current, but the induced electric field exists even without a wire present to carry current).
- Ampere's law, with Maxwell's own correction (a term beyond Chapter 10's scope, called the displacement current): a changing electric flux, in addition to actual current, also produces a magnetic field.

It is that final correction — the displacement current, Maxwell's own contribution to a law that already bore Ampere's name — that completes the symmetry between the third and fourth equations and, in doing so, predicts something neither Faraday nor Ampere had any reason to expect: a changing electric field creates a magnetic field, which if itself changing creates an electric field, which if changing creates another magnetic field, and so on indefinitely, each one regenerating the next without needing any charge or current physically present to sustain it.

12.3 Electromagnetic Waves and Light

Maxwell worked out that this self-sustaining oscillation propagates through empty space at a speed determined entirely by ϵ_0 and μ_0 — the same two constants that appeared, respectively, in Coulomb's law (Chapter 2) and in the field of a straight wire (Chapter 10), each originally measured in completely unrelated laboratory experiments involving static charges and steady currents:

$$c = 1 / \sqrt{(\epsilon_0 \mu_0)}$$

Plugging in the values from Chapters 2 and 10 gives $c \approx 3.00 \times 10^8$ m/s — the measured speed of light, to within experimental precision, computed from two constants that had nothing to do with optics. Maxwell concluded, correctly, that light itself is an electromagnetic wave: a self-propagating oscillation of electric and magnetic fields, and that visible light is merely the narrow band of electromagnetic radiation that happens to be detectable by the human eye. Radio waves, microwaves, infrared, ultraviolet, X-rays, and gamma rays are all the same physical phenomenon at different frequencies — a fact this series returns to in full when it reaches antennas, RF engineering, and wave propagation.

It is worth appreciating how strange this result must have seemed at the time. Nothing about Coulomb's careful measurements of the force between charged pith balls, and nothing about Ampere's measurements of the force between current-carrying wires, had anything ostensibly to do with light. Maxwell's equations were assembled

entirely from static- and steady-current experiments, plus Faraday's induction results, and yet the mathematics of that assembled system — quite without being asked to — produced a wave equation whose predicted propagation speed matched light exactly. This is among the most celebrated instances in the history of physics of a purely mathematical consequence of an existing theory turning out to describe something no one had built the theory to explain in the first place, and it is difficult to overstate how strong a piece of evidence this was, historically, that Maxwell's four equations were not merely a convenient summary of known results but a genuinely complete and correct description of electromagnetism.

Key Idea

Every subsequent book in this series is, in a real sense, a set of special cases of Maxwell's four equations. DC circuit theory is Maxwell's equations in the limit where nothing changes in time. AC circuit theory is Maxwell's equations for fields that oscillate slowly enough that a circuit's physical dimensions are much smaller than the wavelength involved. RF and antenna theory is what happens once that assumption stops holding. Semiconductor device physics governs how real materials respond to the fields Maxwell's equations describe. There is, in an important sense, only one subject in this entire series — this book has simply built it from the ground up.

12.4 A Historical Aside: Waves Need Something to Wave

Nineteenth-century physicists, having only ever encountered waves that traveled through a physical medium — sound through air, ripples through water — found it almost unthinkable that an electromagnetic wave could travel through genuinely empty space with nothing there to do the waving. This led to decades of searching for the “luminiferous aether,” a hypothesized medium presumed to fill all of space specifically so that light would have something to be a wave in. The most famous search for it, the Michelson-Morley experiment of 1887, attempted to detect Earth's motion through this aether by comparing the speed of light in different directions, and found — to the considerable surprise of everyone involved — no difference whatsoever. It would take Einstein's 1905 special theory of relativity, built in part directly on this null result, to explain why: the speed of light is the same in every direction and in every reference frame, precisely because it is not a wave propagating through any medium at all, but a self-sustaining disturbance in the electric and magnetic fields themselves, exactly as Maxwell's equations describe, needing no material substance to carry it. The aether was never found because there was never anything there to find — but the search for it, and its eventual resolution, is a useful reminder that even a spectacularly successful theory (and Maxwell's equations remain, in their domain, exactly correct today) can still coexist for decades with a widely-held, entirely wrong assumption about what that theory actually implies.

12.5 Where This Series Goes Next

The next book in this series, on DC and AC circuit theory, takes the definitions built here — charge, current, voltage, resistance, capacitance, and (in the closing sections of this book) inductance and induced EMF — and develops the systematic tools (Kirchhoff's laws, Thevenin and Norton equivalents, phasor analysis for AC) needed to analyze real circuits of arbitrary complexity. From there, the series proceeds to semiconductor physics and devices, then to the specialized branches — RF and microwave engineering, digital electronics, power systems, control theory, and communications — that make up the working vocabulary of a practicing electrical engineer. Every one of those later books assumes the physical vocabulary built in this one: charge, field, potential, current, resistance, capacitance, magnetic field, and induction. If any of it still feels unfamiliar, this is the book to return to.

Part I Review Problems

1. State, in your own words, the physical difference between the electric field and the electric potential at a point in space.
2. Explain why Gauss's law and Coulomb's law are considered physically equivalent, even though one is stated as an integral over a closed surface and the other as a force between two points.

3. A capacitor and a resistor are both circuit elements that oppose current in some sense. Explain the physical difference between what each one actually does to the energy delivered to it.
4. Explain, using Lenz's law, why an electric generator becomes harder to turn by hand as more electrical load (more current draw) is placed on it.
5. Using the two constants ϵ_0 and μ_0 , and without looking up the value, explain in words how Maxwell was able to compute the speed of light from experiments that involved neither light nor optics.
6. An inductor resists changes in current, much as a capacitor resists changes in voltage. Explain briefly why these two statements are, in a loose sense, mirror images of each other, referring back to how each component is defined.

Part II — Mathematics and Measurement

Chapter 13 — Quantities, Units, and the SI System

Every number in engineering is a number of something. This sounds too obvious to state, and yet the single most common source of catastrophic error in engineering calculation is a number that lost track of what it was a number of. This chapter is about keeping that attachment intact.

13.1 Quantity, Numerical Value, and Unit

A physical quantity is written as a numerical value multiplied by a unit. When we write $R = 4.7\text{ k}\Omega$, we are making a multiplicative statement: the resistance equals 4.7 times the quantity "one kilohm." The multiplication is real, not notational, and this single realisation is what makes unit algebra work.

Because the multiplication is real, units obey ordinary algebraic rules. They multiply, they divide, they cancel. If a current of 2 A flows through a resistance of 3 Ω , then

$$V = IR = (2\text{ A})(3\text{ }\Omega) = 6\text{ A}\cdot\Omega = 6\text{ V}$$

and the last step is not a convention but an identity: the volt is defined so that one ampere times one ohm is one volt. Every derived SI unit works this way. Learning to carry units through a calculation, rather than dropping them and reattaching them at the end, converts a whole class of silent errors into loud ones.

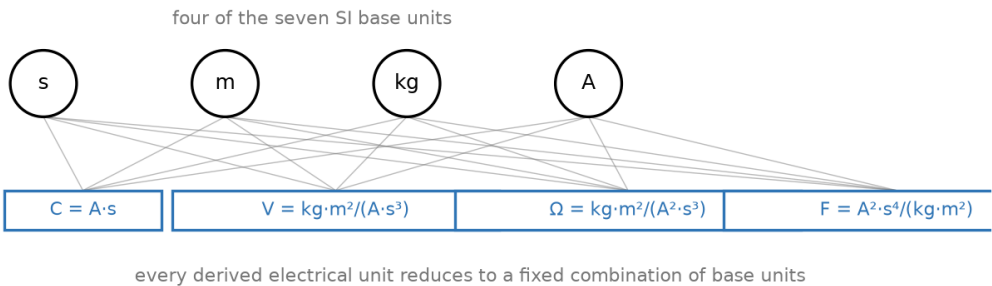


Figure 13.1 — Four SI base units combine to form the derived electrical units.

13.2 The Seven Base Units

The SI system defines seven base quantities and gives each a base unit. Everything else is built from them by multiplication and division.

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Electrical engineering uses four of these constantly — metre, kilogram, second, ampere — and the older name "MKSA system" for that subset still appears in older texts. Note that the base unit of mass is the kilogram, not the gram; the prefix is built into the base unit, which is a historical wrinkle with no deeper meaning and occasionally trips up dimensional bookkeeping.

Since 2019 all seven are defined by fixing the numerical values of seven physical constants — the speed of light, the Planck constant, the elementary charge, and so on — rather than by reference to physical artefacts. The ampere, for instance, is now defined by fixing the elementary charge at exactly $1.602176634 \times 10^{-19}$ C. This matters philosophically and to national measurement institutes; it changes nothing about how you calculate.

13.3 Derived Units in Electrical Engineering

The units you will actually use every day are derived: each is a specific combination of base units that has been given its own name because the combination occurs so often.

Quantity	Unit	Symbol	In base units
Charge	coulomb	C	$A \cdot s$
Voltage	volt	V	$W/A = kg \cdot m^2 / (A \cdot s^3)$
Resistance	ohm	Ω	V/A
Conductance	siemens	S	A/V
Capacitance	farad	F	C/V
Inductance	henry	H	Wb/A
Magnetic flux	weber	Wb	$V \cdot s$
Frequency	hertz	Hz	$1/s$
Power	watt	W	J/s
Energy	joule	J	$W \cdot s$

Two entries deserve comment. The hertz is defined as one per second, and so is the becquerel, and so — dimensionally — is the radian per second; they are dimensionally identical and semantically distinct, which is why you cannot simply cancel Hz against rad/s without thinking. And the siemens is the reciprocal ohm; you will occasionally still meet its charming obsolete name, the mho, written with an upside-down omega.

13.4 Dimensional Analysis as an Error Check

Because units multiply and divide algebraically, any equation you write must balance dimensionally: the units on the left must reduce to the same combination of base units as the units on the right. If they do not, the equation is wrong, and you have found the error without knowing anything about the physics.

This is the cheapest debugging tool in engineering and it is badly underused. Consider the time constant of an RC circuit, $\tau = RC$. Is that dimensionally sensible? Resistance is V/A and capacitance is C/V , so

$$RC = (V/A)(C/V) = C/A = (A \cdot s)/A = s$$

which is a time. Good. Had you misremembered the formula as R/C , the same check would have produced units of $V^2/(A \cdot C) = V^2/(A^2 \cdot s)$, which is not a time, and the error would have surfaced immediately.

Worked Example 13.1

Verify that the resonant frequency formula $f = 1/(2\pi\sqrt{LC})$ has units of hertz.

The constant 2π is dimensionless, so we need only the units of $1/\sqrt{LC}$.

Inductance: $H = Wb/A = (V \cdot s)/A$. Capacitance: $F = C/V = (A \cdot s)/V$.

Their product: $LC = [(V \cdot s)/A] \cdot [(A \cdot s)/V] = s^2$. The volts cancel, the amperes cancel, and we are left with seconds squared.

Therefore $\sqrt{LC}$ has units of seconds, and $1/\sqrt{LC}$ has units of $1/s$, which is the hertz. The formula is dimensionally sound.

Notice what this check does and does not tell you. It confirms the structure of the formula. It cannot confirm the factor of 2π , because 2π is dimensionless — dimensional analysis is blind to every dimensionless constant. A formula can be dimensionally perfect and still numerically wrong by a factor of 2π , which is in fact one of the more common errors in resonance calculations.

Key Idea

Carry units through every calculation. Do not strip them at the start and reattach them at the end. Units that fail to cancel correctly are telling you the equation is wrong, and they are telling you before you have wasted an hour on the arithmetic.

13.5 Units That Are Not Really Units

Some quantities are ratios of two quantities with the same units, and therefore dimensionless. Gain, efficiency, power factor, reflection coefficient, and the Q factor of a resonant circuit are all pure numbers. They are often given a pseudo-unit anyway — "dB", "%", "dimensionless" — as a reminder of what kind of ratio they are.

The radian is the most important of these. An angle in radians is defined as an arc length divided by a radius: metres over metres, dimensionless. This is precisely why angular frequency ω , measured in radians per second, has base units of $1/s$, identical to frequency f in hertz — and precisely why $\omega = 2\pi f$ is not a unit conversion but a genuine numerical relationship between two different quantities that happen to share a dimension.

Common Misconception

It is tempting to treat radians per second and hertz as interchangeable because they share the same base units, and to convert between them by "cancelling" the radian. They are not interchangeable. A signal at 1 kHz has $\omega = 6283$ rad/s, not 1000 rad/s. The factor of 2π between them is real and physical: it is the number of radians in one full cycle. The radian being dimensionless means the units cannot catch this error for you — one of the few places where dimensional analysis offers no protection, and correspondingly one of the places where the mistake is most common.

Practice Problems

1. A capacitor of 100 nF is charged to 12 V. Using $C = Q/V$, find the stored charge, and express your answer in coulombs and in nanocoulombs.
2. Show by dimensional analysis that the quantity L/R has units of time.
3. The energy stored in an inductor is given by $W = \frac{1}{2}LI^2$. Verify that this expression has units of joules.
4. A student writes the impedance of a capacitor as $Z = \omega C$. Use dimensional analysis to show this cannot be correct, and determine what the correct expression must look like dimensionally.

Chapter 14 — Prefixes, Scientific Notation, and Engineering Notation

Electrical engineering spans an absurd range of magnitudes. The gate capacitance of a modern transistor is measured in femtofarads; the capacitance of a power-factor correction bank is measured in millifarads. That is twelve orders of magnitude, in the same quantity, in the same field. This chapter is about writing such numbers without drowning in zeros.

14.1 The SI Prefixes

A prefix multiplies the unit it is attached to by a power of ten. The prefixes used routinely in electrical engineering are these.

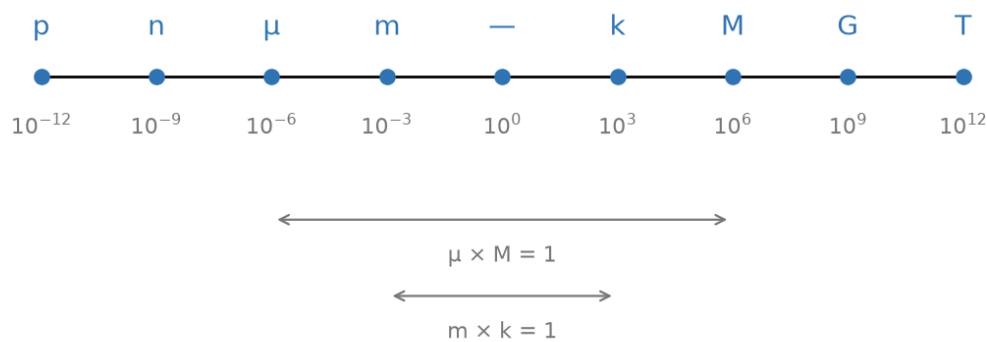


Figure 14.1 — SI prefixes from pico to tera, spaced by their power-of-ten exponent.

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
(none)		10^0
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Case is not decorative. A capital M is mega, a lower-case m is milli, and they differ by a factor of a billion. A capital K is not a valid prefix at all — kilo is always lower-case k — while a capital K is the symbol for the kelvin. Writing 5 KΩ instead of 5 kΩ is a small sin; writing 5 mA when you meant 5 MA is a career-defining one.

Note also that the prefix binds tighter than the exponent. The expression mm² means (mm)², that is, 10⁻⁶ m², not 10⁻³ m². This catches people constantly in current-density and sheet-resistance calculations.

14.2 Scientific Notation

Scientific notation writes every number as a value between 1 and 10, multiplied by a power of ten: 47000 becomes 4.7×10^4 , and 0.00033 becomes 3.3×10^{-4} . The mantissa always has exactly one digit before the decimal point.

Its virtue is unambiguity. The number 4.70×10^4 states plainly that three digits are meaningful, where the plain form 47000 leaves you guessing whether the trailing zeros are measured or merely placeholders — a question Chapter 15 takes up in earnest.

14.3 Engineering Notation

Engineering notation is a variant that restricts the exponent to multiples of three, so that the exponent always corresponds to an SI prefix. Instead of $4.7 \times 10^4 \Omega$ you write $47 \times 10^3 \Omega$, and read it directly as 47 k Ω .

This is what practising engineers actually use, and it is what calculators mean by their "ENG" mode. The reason is entirely practical: component values, datasheet entries, and instrument displays are all expressed with prefixes, so a notation whose exponents map one-to-one onto those prefixes eliminates a translation step, and with it a class of error.

Worked Example 14.1

A current of 0.0000047 A flows through a resistance of 2200000 Ω . Express both in engineering notation, then find the voltage.

The current: $0.0000047 \text{ A} = 4.7 \times 10^{-6} \text{ A} = 4.7 \mu\text{A}$.

The resistance: $2200000 \Omega = 2.2 \times 10^6 \Omega = 2.2 \text{ M}\Omega$.

Now the voltage, keeping the powers of ten explicit:

$$V = IR = (4.7 \times 10^{-6})(2.2 \times 10^6) = (4.7 \times 2.2) \times 10^{-6+6} = 10.34 \times 10^0 = 10.34 \text{ V}.$$

The exponents cancelled exactly, which is the useful thing to notice: micro times mega is unity. Similar pairings are worth committing to memory — milli times kilo is unity, nano times giga is unity, pico times tera is unity. Once you internalise these, a great many circuit calculations can be done in your head, because the prefixes annihilate and you are left with a one-digit multiplication.

Key Idea

Reciprocal prefix pairs cancel. $\mu \times \text{M} = 1$, $\text{m} \times \text{k} = 1$, $\text{n} \times \text{G} = 1$, $\text{p} \times \text{T} = 1$. A resistance in kilohms times a current in milliamps gives a voltage in volts. A capacitance in microfarads times a resistance in megohms gives a time constant in seconds. Recognising these pairings is the single biggest speed-up available in routine circuit arithmetic.

14.4 Arithmetic With Powers of Ten

Multiplication adds exponents; division subtracts them. Raising to a power multiplies the exponent. Taking a square root halves it — which is why square roots in engineering notation are awkward, since halving an exponent that is a multiple of three rarely lands on another multiple of three.

Addition and subtraction are the operations that catch people. You cannot add mantissas unless the exponents match. To compute $4.7 \times 10^3 + 2.2 \times 10^2$, first rewrite the second term as 0.22×10^3 , then add mantissas: 4.92×10^3 . Attempting to add 4.7 and 2.2 directly gives an answer wrong by nearly a factor of two.

Common Misconception

A frequent error is to assume that because prefixes are convenient, they can be mixed freely inside a single expression. They cannot. Adding 4.7 k Ω to 220 Ω requires converting one to match the other — either $4.7 \text{ k}\Omega + 0.22 \text{ k}\Omega = 4.92 \text{ k}\Omega$, or $4700 \Omega + 220 \Omega = 4920 \Omega$. Writing "4.7 k + 220" and reading off 4.92 works only because you performed the conversion silently in your head; write it out, because the day you do it silently and get it wrong is the day it costs you.

Practice Problems

1. Express the following in engineering notation with an appropriate prefix: 0.000000015 F, 68000 Ω , 0.0025 A, 1500000000 Hz.
2. A 4.7 k Ω resistor carries 3.2 mA. Find the voltage across it and the power dissipated in it, using prefix cancellation rather than long decimals.
3. Add the following capacitances, giving the result in nanofarads: 470 pF, 1.2 nF, and 0.01 μ F.
4. A signal has a period of 20 ns. Find its frequency, and express the answer with an appropriate prefix. Then find the angular frequency ω in rad/s.

Chapter 15 — Precision, Significant Figures, and Uncertainty

A measurement is not a number; it is a number together with a statement of how well it is known. An oscilloscope that reads 5.02 V is not telling you the voltage is 5.02 V. It is telling you the voltage is near 5.02 V, with a spread that depends on the instrument, the probe, the temperature, and the day. This chapter is about carrying that spread through a calculation honestly.



Figure 15.1 — Accuracy and precision are independent: a measurement can have either, both, or neither.

15.1 Significant Figures

The significant figures of a number are the digits that carry real information. The rules are mechanical: all non-zero digits are significant; zeros between non-zero digits are significant; leading zeros are never significant; trailing zeros after a decimal point are significant.

So 0.00470 has three significant figures — the leading zeros are placeholders, the trailing zero is a claim of precision. The genuinely ambiguous case is a whole number ending in zeros: 4700 might have two, three, or four significant figures, and the plain notation cannot tell you which. Scientific notation resolves this completely, which is one of its main practical virtues: 4.7×10^3 has two, 4.70×10^3 has three, 4.700×10^3 has four.

15.2 Propagating Precision Through a Calculation

Two rules cover most cases. For multiplication and division, the result carries as many significant figures as the least precise input. For addition and subtraction, the result carries as many decimal places as the input with the fewest decimal places.

The rules differ because the operations degrade precision differently, and it is worth seeing why rather than memorising both. Multiplication scales relative error; addition scales absolute error. If you multiply a quantity known to 1% by one known to 10%, the answer is known to roughly 10%, regardless of the units involved. If you add a quantity known to ± 0.5 V to one known to ± 0.001 V, the sum is known to ± 0.5 V — the precise measurement is swamped, and no amount of decimal places in the second number rescues it.

Common Misconception

The most common error with significant figures is rounding at every intermediate step. Do not. Carry full precision through the calculation and round only the final answer. Rounding intermediate values injects error at each step, and those errors accumulate — in a long chain of calculations, aggressive intermediate rounding can easily shift a final answer by more than the measurement uncertainty you were trying to represent. The significant-figure rules describe how to report a result, not how to compute one.

15.3 Absolute and Relative Uncertainty

Absolute uncertainty is expressed in the units of the quantity: $5.02 \text{ V} \pm 0.05 \text{ V}$. Relative uncertainty is the same thing as a fraction or percentage of the value: $5.02 \text{ V} \pm 1\%$. Converting between them is straightforward, and which one you want depends on the operation you are about to perform.

For products and quotients, relative uncertainties add. If R is known to $\pm 2\%$ and I is known to $\pm 3\%$, then $V = IR$ is known to roughly $\pm 5\%$. For sums and differences, absolute uncertainties add. These are conservative, worst-case rules, useful for a quick hand estimate; when independent random uncertainties are treated statistically, standard uncertainties are more commonly combined in quadrature (the square root of the sum of squares), which gives a smaller, more realistic combined uncertainty than simple addition. This is the formal version of the observation in the previous section, and it has a sharp practical consequence.

Worked Example 15.1

Two voltages are measured as $V_1 = 10.0 \text{ V} \pm 0.1 \text{ V}$ and $V_2 = 9.8 \text{ V} \pm 0.1 \text{ V}$. Find the difference and its uncertainty, as an absolute figure and as a percentage.

The difference: $V_1 - V_2 = 10.0 - 9.8 = 0.2 \text{ V}$.

The absolute uncertainties add: $\pm 0.1 + \pm 0.1 = \pm 0.2 \text{ V}$.

So the difference is $0.2 \text{ V} \pm 0.2 \text{ V}$ — a relative uncertainty of 100%.

Each individual measurement was good to 1%. Their difference is good to 100%. Nothing went wrong in the measurement; this is what subtracting two nearly equal numbers does. The signal is the small residue between two large quantities, and the uncertainty of each large quantity survives into that small residue undiminished.

This phenomenon has a name — catastrophic cancellation — and it is a permanent hazard. It is why four-wire (Kelvin) resistance measurement exists, why instrumentation amplifiers are built the way they are, and why a bridge circuit that measures a small imbalance directly will always beat a scheme that measures two large quantities and subtracts them.

Key Idea

Subtracting nearly equal numbers destroys precision. The absolute uncertainty is preserved while the value shrinks, so the relative uncertainty explodes. Whenever a design calls for the small difference between two large measured quantities, look for a way to measure the difference directly instead.

15.4 Accuracy, Precision, and Resolution

These three words are not synonyms, and instrument datasheets depend on the distinction.

- **Accuracy** is how close a measurement is to the true value. It is limited by calibration, and it drifts with time and temperature.
- **Precision** (or repeatability) is how close repeated measurements are to each other. An instrument can be precise and inaccurate: reading 4.512 V every time when the true value is 5.000 V .
- **Resolution** is the smallest change the instrument can display. A six-digit meter has fine resolution, which says nothing whatever about whether those six digits are accurate. Resolution is free; accuracy is expensive.

A typical bench multimeter specification reads something like " $\pm(0.5\% \text{ of reading} + 3 \text{ counts})$ ". The first term is a relative error that scales with the measurement; the second is an absolute error, in units of the last displayed digit, that dominates at the low end of a range. This is why measuring 0.1 V on a 1000 V range is a poor idea even though the meter will happily display a number.

15.5 Component Tolerance

Real components differ from their marked values. A resistor marked $4.7\text{ k}\Omega$ with a 5% tolerance is guaranteed only to lie between $4.465\text{ k}\Omega$ and $4.935\text{ k}\Omega$. This is not a defect; it is the specification, and designs must work across the whole range.

The standard E-series values — E12, E24, E96 — are chosen as geometric progressions so that consecutive values are spaced by roughly the tolerance. E24 steps by about 10%, matching 5% parts, so that the tolerance bands of adjacent values just meet without large gaps or heavy overlap. This is why the odd-looking sequence 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82 exists: it is a logarithmic ladder, not an arbitrary list, and once you see that, the values stop looking random.

Worst-case analysis assumes every component sits at the tolerance extreme that hurts most. It is pessimistic — the probability of every part being simultaneously worst-case is negligible — but it is the honest way to establish that a design cannot fail, as opposed to establishing that it usually works.

Practice Problems

1. State the number of significant figures in each: 0.00250 , 1200 , 1.200×10^3 , 100.0 , 0.0001 .
2. A resistance of $4.7\text{ k}\Omega \pm 5\%$ carries a current of $2.0\text{ mA} \pm 2\%$. Find the voltage across it and its percentage uncertainty.
3. Two lengths are measured as $100.0\text{ mm} \pm 0.2\text{ mm}$ and $99.5\text{ mm} \pm 0.2\text{ mm}$. Find their difference with its absolute and percentage uncertainty, and comment on the result.
4. A $4\frac{1}{2}$ -digit multimeter with a specification of $\pm(0.05\% \text{ of reading} + 2 \text{ counts})$ is used on its 20 V range, where the last digit represents 1 mV . Find the total uncertainty when measuring 10.000 V , and again when measuring 0.100 V . Express each as a percentage of the reading.

Chapter 16 — Algebra for Circuit Analysis

Circuit analysis produces equations, and equations must be rearranged. The algebra involved is not difficult, but a few specific manipulations recur so often in electrical engineering that they are worth practising until they are automatic — because an engineer who has to stop and think about them loses the thread of the circuit while doing so.

16.1 Rearranging for Any Variable

Ohm's law, $V = IR$, is three formulas in one. Solving for each in turn gives $I = V/R$ and $R = V/I$. Every constitutive relation in this series works the same way, and you should be able to move between forms without hesitation.

The general procedure is to isolate the target variable by applying inverse operations to both sides, working outward from the target. What makes engineering formulas occasionally awkward is that the target often appears inside a reciprocal, inside a square root, or in more than one place at once.

Worked Example 16.1

The parallel resistance of two resistors is $1/R = 1/R_1 + 1/R_2$. Given R and R_1 , solve for R_2 .

Start by isolating the term containing R_2 :

$$1/R_2 = 1/R - 1/R_1$$

Combine the right-hand side over a common denominator:

$$1/R_2 = (R_1 - R)/(R \cdot R_1)$$

Now invert both sides — and this is the step people rush:

$$R_2 = (R \cdot R_1)/(R_1 - R)$$

Inverting both sides of an equation is legitimate provided neither side is zero, but note that you must invert each side as a whole. A frequent error is to invert term by term, turning $1/R - 1/R_1$ into $R - R_1$, which is wrong. The reciprocal of a difference is not the difference of the reciprocals.

Sanity check the result: if $R_1 = R$, the denominator vanishes and R_2 is infinite, which is correct — if the parallel combination already equals one of the resistors, the other must be an open circuit.

Common Misconception

The reciprocal of a sum is not the sum of the reciprocals. $1/(a + b) \neq 1/a + 1/b$. This is the single most common algebraic error in parallel-impedance work, and it is worth testing yourself on numbers once: $1/(2 + 2) = 0.25$, while $1/2 + 1/2 = 1$. The two are not close. The parallel-resistance formula is stated as $1/R = 1/R_1 + 1/R_2$ precisely because it is the reciprocals that add, not the resistances.

16.2 The Product-Over-Sum Shortcut

For exactly two resistors in parallel, the formula above rearranges to a form worth memorising:

$$R = R_1 R_2 / (R_1 + R_2)$$

This is faster than the reciprocal form for hand calculation, and it generalises to any two impedances in parallel, which is why it recurs constantly in later books. Its limitation is that it applies to exactly two elements. For three or more you must either use the reciprocal form or apply product-over-sum repeatedly, combining two at a time.

Two special cases are worth having at your fingertips. Two equal resistors in parallel give half the value. A resistor in parallel with one ten times larger gives approximately 91% of the smaller value — near enough to the smaller value that in a first-pass estimate the larger one can often be ignored entirely.

16.3 Ratios, Dividers, and Fractions

The voltage divider is the most-used formula in electronics, and it is pure algebra:

$$V_{out} = V_{in} \cdot R_2 / (R_1 + R_2)$$

What makes it worth dwelling on is the structure: the output is the input scaled by a dimensionless ratio between zero and one. Recognising this shape lets you read a divider off a schematic without writing anything down. If the two resistors are equal, the output is half the input. If R_2 is nine times R_1 , the output is nine-tenths of the input.

The current divider has the complementary structure, and the thing to notice is that the resistances swap:

$$I_1 = I_{total} \cdot R_2 / (R_1 + R_2)$$

Current through R_1 is proportional to the other resistance. This is not a typographical accident — current preferentially takes the path of lower resistance, so the branch current is larger when its own resistance is smaller, which requires the opposite resistance in the numerator.

Key Idea

Voltage divides in proportion to resistance; current divides in inverse proportion. The voltage-divider numerator is the resistance you are measuring across. The current-divider numerator is the other resistance. Mixing these up is the most common error in first-year circuit analysis, and the check is always the limiting case: as one resistance goes to zero, does the formula give what physics demands?

16.4 Quadratics and Where They Come From

Quadratic equations appear in circuit work whenever power is involved, since power is proportional to the square of voltage or current, and whenever a second-order system's characteristic equation is written down. The solution is the familiar formula:

$$x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$$

The discriminant $b^2 - 4ac$ carries the physical meaning. Positive discriminant gives two distinct real roots — in a second-order circuit, the overdamped case. Zero gives one repeated root, the critically damped case. Negative gives two complex conjugate roots, the underdamped case, which oscillates. Chapters 21 and 22 make the last of these tractable; for now, note only that a negative discriminant is not an error but a signal that the system rings.

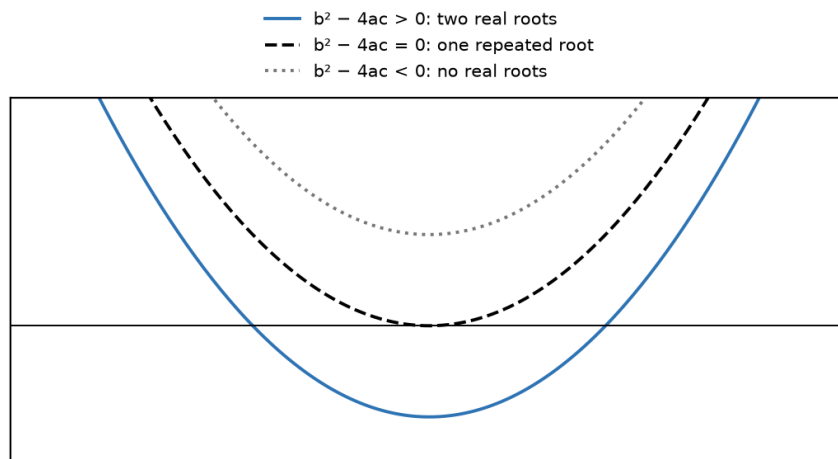


Figure 16.1 — The sign of the discriminant sets the number of real roots of a quadratic.

16.5 Approximations Worth Knowing

Engineers approximate constantly, and two approximations do most of the work. The first is the binomial approximation: for small x ,

$$(1 + x)^n \approx 1 + nx$$

So $\sqrt{1+x} \approx 1 + x/2$ and $1/(1+x) \approx 1 - x$ when x is small. "Small" here means roughly below 0.1, where the error stays under about 1%.

The second is the observation that when two quantities differ greatly in size, the smaller can often be dropped. In $R_1 + R_2$ with R_2 a hundred times R_1 , dropping R_1 costs 1%. Whether 1% matters depends on the tolerance of the components, which is exactly the kind of judgement Chapter 15 was preparing you to make: there is no point computing to 0.01% with 5% resistors.

Practice Problems

1. Solve $V = IR$ for R , and evaluate for $V = 3.3$ V and $I = 150$ μ A.
2. Three resistors of 1 k Ω , 2 k Ω , and 3 k Ω are connected in parallel. Find the equivalent resistance, first with the reciprocal formula and then by applying product-over-sum twice, and confirm the two agree.
3. A voltage divider must produce 2.5 V from a 9 V supply. If $R_2 = 10$ k Ω , find R_1 .
4. Use the binomial approximation to estimate $\sqrt{1.06}$ and $1/0.98$, then compare each with the exact value and state the percentage error.

Chapter 17 — Systems of Linear Equations

Nodal and mesh analysis, introduced in Part III, turn any resistive circuit into a system of simultaneous linear equations. The circuit theory is the interesting part; solving the system is mechanical. This chapter supplies the mechanics so that Part III can concentrate on the circuits.

17.1 What a Linear System Is

A linear equation in several unknowns contains those unknowns only to the first power, multiplied by constants and added. No squares, no products of unknowns, no unknowns inside functions. Two equations in two unknowns look like this:

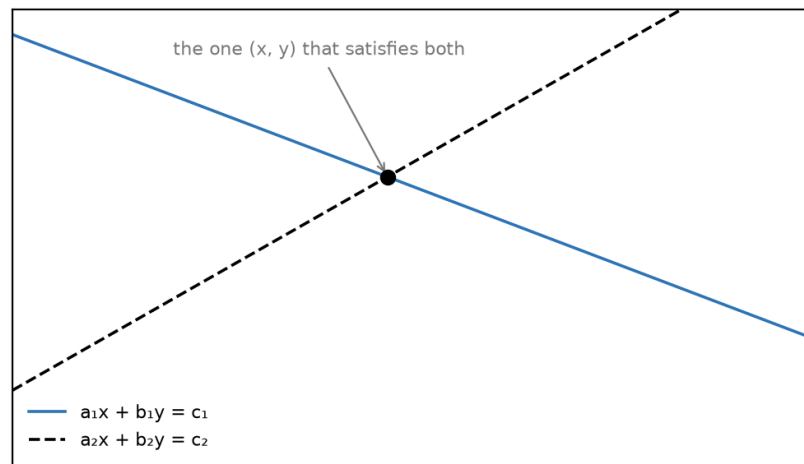


Figure 17.1 — Two linear equations in two unknowns: each is a line, and the solution is where they cross.

$$a_1x + b_1y = c_1 \quad \text{and} \quad a_2x + b_2y = c_2$$

Circuit equations from Kirchhoff's laws are always of this form when the components are resistors and the sources are independent, which is why the systematic methods work at all. Nonlinear elements — diodes, transistors — break linearity and require entirely different techniques, taken up later in the series.

17.2 Substitution and Elimination

For two equations, substitution is usually quickest: solve one equation for one unknown, put that expression into the other, solve the resulting single-variable equation, then back-substitute.

Elimination scales better. Multiply the equations by whatever constants make the coefficients of one unknown equal in magnitude, then add or subtract to eliminate it. For three or more equations, elimination performed systematically becomes Gaussian elimination, which is what every circuit simulator does internally.

Worked Example 17.1

Solve the system: $3x + 2y = 16$ and $5x - y = 9$.

Elimination is convenient here because y has a coefficient of -1 in the second equation. Multiply the second equation by 2:

$$10x - 2y = 18$$

Add this to the first equation, and the y terms cancel:

$(3x + 10x) + (2y - 2y) = 16 + 18$, so $13x = 34$, giving $x = 2.615$.

Back-substitute into the second original equation:

$5(2.615) - y = 9$, so $13.077 - y = 9$, giving $y = 4.077$.

Always verify in the equation you did not use for back-substitution. First equation: $3(2.615) + 2(4.077) = 7.846 + 8.154 = 16.000$. Correct.

That verification step costs ten seconds and catches arithmetic slips reliably. In circuit work, where a sign error in one Kirchhoff equation propagates into every node voltage, it is time exceptionally well spent.

17.3 Determinants and Cramer's Rule

For a two-by-two system, the determinant is defined as

$$D = a_1b_2 - a_2b_1$$

and Cramer's rule gives the solution directly: $x = D_x/D$ and $y = D_y/D$, where D_x is the determinant formed by replacing the x -coefficient column with the constants, and D_y likewise for y .

Cramer's rule is not the fastest method for large systems — its cost grows factorially, which makes it useless beyond about three equations — but for two or three unknowns it is quick, mechanical, and hard to get lost in. Its real value is diagnostic. If $D = 0$, the system has no unique solution, and in a circuit context that almost always means a modelling error: two ideal voltage sources placed in parallel with different values, an ideal current source left in series with an open circuit, or a floating section with no reference to ground.

Key Idea

A zero determinant is a physics error, not an arithmetic one. When the determinant of a circuit's node equations vanishes, treat it as a warning to examine the circuit model and equations carefully, not just the algebra — the equations may simply be redundant (as in a genuinely degenerate mathematical system), but in circuit analysis a vanishing determinant is most often a sign of something physically impossible, such as a node with no DC path to the reference node.

17.4 Matrix Form and What Comes Later

Any linear system can be written compactly as a matrix equation. For circuit work the standard arrangement is

$$G \cdot V = I$$

where G is the conductance matrix, V is the vector of unknown node voltages, and I is the vector of source currents. This is exactly the form SPICE assembles and solves at every operating point, and recognising it now will make the later book on simulation considerably less mysterious.

The conductance matrix has a pleasant structure that makes it easy to write down by inspection: each diagonal entry is the sum of all conductances connected to that node, and each off-diagonal entry is the negative of the conductance directly between the two nodes it indexes. That structure is not a coincidence — it is Kirchhoff's current law, written in matrix form.

Practice Problems

1. Solve by substitution: $2x + 3y = 12$ and $x - y = 1$.
2. Solve by elimination: $4a + 5b = 22$ and $3a - 2b = 5$.

3. Evaluate the determinant of the system $6x + 2y = 10$, $3x + y = 5$, and explain what your result implies about the solution.
4. Two mesh equations for a circuit are $100I_1 - 40I_2 = 5$ and $-40I_1 + 90I_2 = 0$. Solve for both currents using Cramer's rule, and state the physical meaning of the sign of each result.

Chapter 18 — Functions, Graphs, and Limiting Behaviour

Engineers read graphs more often than they read equations. A frequency response, a transfer characteristic, a load line — each is a function drawn rather than written, and the skill being exercised is the ability to extract behaviour from shape. This chapter is about that skill.

18.1 Functions as Input-Output Relations

A function assigns exactly one output to each input. In circuit terms, the input is often a frequency or an input voltage, and the output is a gain or an output voltage. The notation $H(f)$ means "the value of H at frequency f " — it is not H multiplied by f , a confusion that causes real difficulty when the same letters are used for both quantities and functions.

18.2 Linear Functions and Straight Lines

A linear function has the form $y = mx + c$, with m the slope and c the intercept. Ohm's law is linear through the origin, which is why an ohmic resistor plots as a straight line through $(0,0)$ on a current-versus-voltage graph, with slope $1/R$.

A load line is the same idea used in reverse. For a source of voltage V_s with internal resistance R_s driving an unknown element, Kirchhoff gives $V = V_s - I \cdot R_s$: a straight line on the same axes as the device characteristic, with negative slope. Where the two curves cross is the operating point. This graphical technique is how transistor bias points were found before circuit simulators existed, and it remains the clearest way to understand what biasing actually means.

18.3 Asymptotes and Limiting Behaviour

The fastest way to understand an unfamiliar transfer function is to ask what it does at the extremes. Consider the response of a simple low-pass filter:

$$H(f) = 1 / \sqrt{1 + (f/f_c)^2}$$

At very low frequency, f/f_c is nearly zero, so the square root is nearly 1 and $H \approx 1$: the filter passes the signal untouched. At very high frequency, the 1 becomes negligible next to $(f/f_c)^2$, so $H \approx f_c/f$: the response falls in inverse proportion to frequency. At $f = f_c$ exactly, $H = 1/\sqrt{2} \approx 0.707$, which is the definition of the corner frequency.

Three numbers — the low-frequency asymptote, the high-frequency asymptote, and the value at the corner — sketch the entire curve to engineering accuracy, without evaluating the function anywhere else. This is the habit worth building: probe the extremes first, fill in the middle only if you need to.

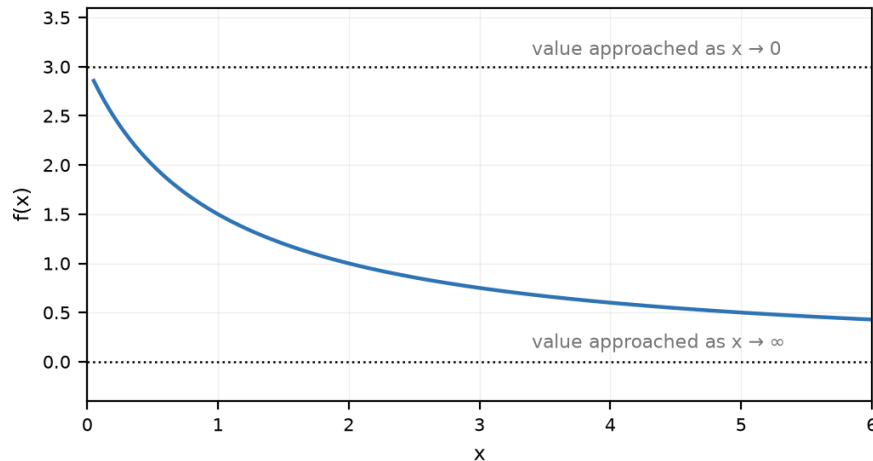


Figure 18.1 — Reading a function's limiting behaviour: the value it approaches at each extreme.

Key Idea

Read a transfer function at its limits before you plot it. What happens as the variable goes to zero, and as it goes to infinity? Those two answers plus the behaviour at the corner give you the shape. Everything else is detail.

Worked Example 18.1

A high-pass filter has transfer function $H(f) = (f/f_c)/\sqrt{1+(f/f_c)^2}$, with corner frequency $f_c = 1.0$ kHz. Find H at $f = 100$ Hz, $f = 1.0$ kHz, and $f = 10$ kHz, and state which asymptote each value sits closest to.

At $f = 100$ Hz: $f/f_c = 0.1$, so $H = 0.1/\sqrt{1.01} \approx 0.0995$ — close to the low-frequency asymptote $H \approx 0$ (a high-pass filter blocks low frequencies). At $f = 1.0$ kHz (the corner): $H = 1/\sqrt{2} \approx 0.707$, matching the general -3 dB corner value exactly. At $f = 10$ kHz: $H = 10/\sqrt{101} \approx 0.995$ — close to the high-frequency asymptote $H \approx 1$ (a high-pass filter passes high frequencies essentially unattenuated). The same three numbers — the low-frequency limit, the high-frequency limit, and the corner value — sketch this curve exactly as they did for the low-pass example above, just with the roles of the two asymptotes reversed.

18.4 Logarithmic Axes

Electrical engineering plots almost everything on logarithmic axes, for a reason that follows directly from Chapter 14: the quantities span many orders of magnitude, and a linear axis cannot show a microvolt and a volt on the same picture.

On log axes, a power-law relationship becomes a straight line, and the exponent becomes the slope. The high-frequency asymptote $H \approx f_c/f$ above is a power law with exponent -1 , so on a log-log plot it is a straight line falling at a constant rate — the familiar 20 dB per decade rolloff. A second-order filter falls at exponent -2 , giving 40 dB per decade. Reading the slope off a log-log plot tells you the order of the system directly, which is a genuinely useful diagnostic when looking at measured data.

Common Misconception

A logarithmic axis has no zero. It cannot: the logarithm of zero is undefined, tending to negative infinity. This surprises people the first time a plotting program refuses to display their data, or silently drops the DC point. It also means you cannot plot a signal that changes sign on a log axis without taking the magnitude first — which is exactly why frequency responses are conventionally plotted as magnitude and phase separately rather than as a single signed quantity.

Practice Problems

1. A resistor obeys Ohm's law. Sketch its current-versus-voltage characteristic for $R = 2 \text{ k}\Omega$, and state the slope in siemens.
2. For the low-pass response $H(f) = 1/\sqrt{1 + (f/f_c)^2}$ with $f_c = 1 \text{ kHz}$, evaluate H at 100 Hz, 1 kHz, and 10 kHz.
3. A 12 V source with $50 \text{ }\Omega$ internal resistance drives a load. Write the load-line equation and find the current when the load voltage is 8 V.
4. A measured response falls by a factor of 100 in magnitude when the frequency increases by a factor of 10. What is the slope on a log-log plot, what is the rolloff in dB per decade, and what order of filter does this suggest?

Chapter 19 — Exponentials, Logarithms, and the Decibel

The exponential function describes every natural charging and discharging process in electronics, and the logarithm is its inverse. Between them they underpin the decibel, which is the unit electrical engineers use for almost every ratio they care about. This chapter builds all three.

19.1 The Exponential Function

The function e^x has the defining property that its rate of change equals its own value. Nothing else does this, and that single property is why it appears in every first-order circuit: a capacitor discharging through a resistor loses charge at a rate proportional to the charge remaining, which is precisely the condition that produces an exponential.

The standard forms are decay and growth toward a limit:

$$v(t) = V_o e^{(-t/\tau)} \quad \text{and} \quad v(t) = V_f (1 - e^{(-t/\tau)})$$

where τ is the time constant, equal to RC for a resistor-capacitor pair and L/R for a resistor-inductor pair — both of which you verified dimensionally in Chapter 13.

The numbers worth memorising are these: after one time constant, a decaying quantity has fallen to 37% of its initial value, and a rising one has reached 63% of its final value. After three time constants, 95%. After five, 99.3%, which is why "five time constants" is the conventional engineering definition of settled.

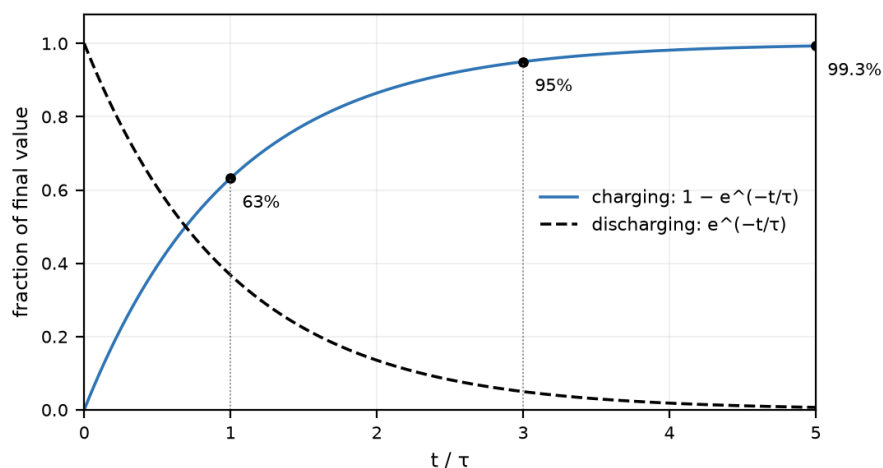


Figure 19.1 — Charging and discharging exponentials, marked at one, three, and five time constants.

Worked Example 19.1

A $10\ \mu\text{F}$ capacitor charges through a $47\ \text{k}\Omega$ resistor toward $5\ \text{V}$. How long until it reaches $4\ \text{V}$?

First the time constant: $\tau = RC = (47 \times 10^3)(10 \times 10^{-6}) = 0.47\ \text{s}$. Note the prefix cancellation from Chapter 14 — kilo times micro is milli, so $47 \times 10 = 470\ \text{ms}$ directly.

The charging equation: $v(t) = 5(1 - e^{(-t/\tau)})$.

Set $v = 4$ and solve. Divide by 5: $0.8 = 1 - e^{(-t/\tau)}$, so $e^{(-t/\tau)} = 0.2$.

Take natural logarithms of both sides: $-t/\tau = \ln(0.2) = -1.609$.

So $t = 1.609\tau = 1.609 \times 0.47 = 0.756\ \text{s}$.

Sanity check: 4 V is 80% of the final value, which should fall between one time constant (63%) and three (95%), and closer to the lower end. It does — 1.6 time constants. That kind of rough check catches sign errors in the logarithm, which are easy to make and otherwise invisible.

19.2 Logarithms

The logarithm answers the question "to what power?" If $10^3 = 1000$, then $\log_{10}(1000) = 3$. Two bases matter in engineering: base 10, written $\log$, and base $e \approx 2.71828$, written $\ln$. Exponential circuit behaviour requires $\ln$; decibels and log-frequency axes use $\log$.

The three properties that do all the work convert multiplication into addition, division into subtraction, and powers into multiplication:

$$\log(ab) = \log a + \log b; \quad \log(a/b) = \log a - \log b; \quad \log(a^n) = n \cdot \log a$$

The first of these is the entire reason decibels exist. Cascaded stages multiply their gains; expressed logarithmically, they add. A chain of amplifiers and losses becomes a sum you can do in your head instead of a product you cannot.

19.3 The Decibel

The decibel expresses a power ratio logarithmically:

$$G(\text{dB}) = 10 \cdot \log_{10}(P_{\text{out}}/P_{\text{in}})$$

For voltage or current ratios, the factor is 20 rather than 10, because power is proportional to the square of voltage and the square becomes a factor of two in front of the logarithm — strictly, this equivalence between the voltage-ratio and power-ratio forms of the decibel holds only when input and output are measured across the same impedance, which is the common case this book assumes unless stated otherwise:

$$G(\text{dB}) = 20 \cdot \log_{10}(V_{\text{out}}/V_{\text{in}})$$

The values worth memorising are few. 3 dB is a factor of 2 in power. 6 dB is a factor of 2 in voltage. 10 dB is a factor of 10 in power. 20 dB is a factor of 10 in voltage. Negative values are the reciprocals: -3 dB is half the power, which is why the corner frequency of a filter — where the response is $1/\sqrt{2}$ in voltage, and therefore half in power — is called the 3 dB point.

Key Idea

Use $10 \cdot \log$ for power ratios and $20 \cdot \log$ for voltage ratios. The factor of two comes from $P \propto V^2$, not from a different definition of the decibel. Applying $20 \cdot \log$ to a power ratio, or $10 \cdot \log$ to a voltage ratio, produces an answer wrong by a factor of two in dB — which is a factor of two in the exponent, and therefore an enormous error in the underlying quantity.

The decibel is a ratio and therefore dimensionless. To express an absolute quantity, a reference must be specified, and it is written into the unit: dBm is decibels relative to one milliwatt, dBV relative to one volt, dB μ V relative to one microvolt. A power of 1 mW is 0 dBm; 1 W is +30 dBm; 1 μ W is -30 dBm. These absolute forms are used constantly in RF work, and a later book in this series depends heavily on fluency with them.

Common Misconception

Decibel values add only when they are ratios. You may add gains in dB along a signal chain because the underlying gains multiply. You may not add two dBm values: adding +10 dBm to +10 dBm does not give +20 dBm, because that would be adding absolute powers, not ratios. Two 10 mW signals combine to 20 mW, which is +13 dBm — a 3 dB increase, because doubling power is always 3 dB regardless of the starting level.

Practice Problems

1. A 100 nF capacitor discharges through 22 k Ω from an initial 9 V. Find the time constant, and the voltage after 5 ms.
2. Express the following power ratios in dB: 2, 10, 100, 0.5, 1000.
3. An amplifier has a voltage gain of 250. Express this in dB. If it is followed by a 6 dB attenuator, what is the overall gain in dB, and as a plain voltage ratio?
4. A receiver input is -95 dBm. Convert this to watts. If an antenna and cable together provide a gain of +12 dB, what is the input power in dBm and in watts at the far end?

Chapter 20 — Trigonometry and the Sinusoid

Alternating current is sinusoidal, and so is every signal that can be built from alternating currents — which, as a later book on Fourier analysis shows, is every signal whatsoever. The sinusoid is therefore not one waveform among many; it is the waveform, and the trigonometry describing it is worth genuine fluency.

20.1 Angles, Degrees, and Radians

An angle in radians is the arc length subtended divided by the radius. A full circle is 2π radians, so $180^\circ = \pi$ rad, and the conversion is a simple proportion. Radians are used throughout analysis because the calculus of trigonometric functions is clean only in radians — the derivative of $\sin x$ is $\cos x$ only when x is in radians, picking up an awkward factor of $\pi/180$ otherwise.

Degrees survive in practice because they are convenient for phase, where quarter and half cycles matter and 90° reads more naturally than $\pi/2$. Both appear constantly, sometimes in the same equation, and the single most common cause of an inexplicably wrong answer in phase calculations is a calculator left in the wrong angular mode.

20.2 The Sinusoid and Its Parameters

A general sinusoid is written

$$v(t) = V_m \sin(\omega t + \phi)$$

with three independent parameters. V_m is the amplitude, the peak departure from zero. ω is the angular frequency in radians per second, related to ordinary frequency by $\omega = 2\pi f$ and to the period by $\omega = 2\pi/T$. ϕ is the phase, which shifts the waveform along the time axis; a positive ϕ advances the waveform, moving it earlier.

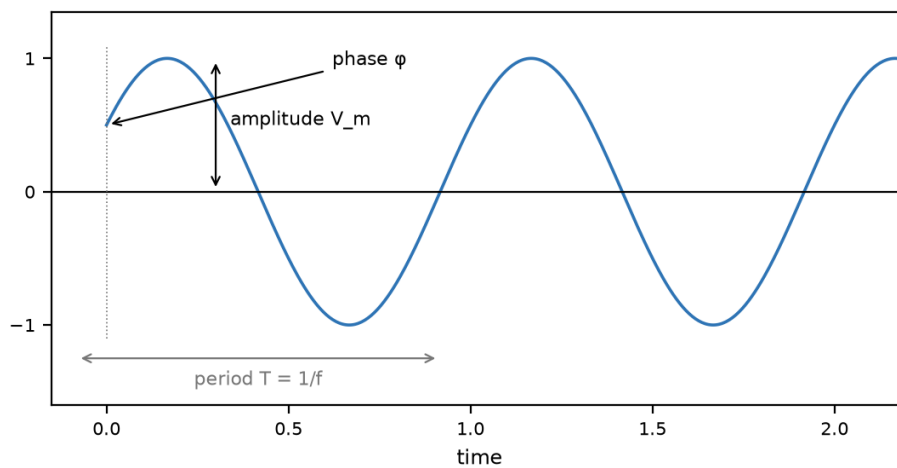


Figure 20.1 — A sinusoid's amplitude, period, and phase, shown on a single cycle.

Only these three numbers are needed. Two sinusoids of the same frequency can differ only in amplitude and phase, and that observation is the seed of the entire phasor method developed in Chapter 23.

20.3 Sine, Cosine, and the Relationship Between Them

Cosine is sine shifted by a quarter cycle: $\cos \theta = \sin(\theta + 90^\circ)$. Which one you use as the reference is arbitrary, and different textbooks choose differently, but you must be consistent within a single problem. This series follows the common convention of taking cosine as the reference for phasor work, for reasons that Chapter 22 makes clear.

The identities that earn their keep in circuit analysis are these:

- **Pythagorean:** $\sin^2\theta + \cos^2\theta = 1$. This is what turns a resistance and a reactance into a magnitude, and it is why impedance magnitudes involve a square root of a sum of squares.
- **Sum of angles:** $\sin(A + B) = \sin A \cos B + \cos A \sin B$. This is the identity behind mixing and modulation — multiply two sinusoids and you get sum and difference frequencies, which is the foundation of every radio receiver.
- **Double angle:** $\sin 2\theta = 2 \sin \theta \cos \theta$, and $\cos 2\theta = 1 - 2\sin^2\theta$. The second form explains why the instantaneous power in an AC circuit oscillates at twice the signal frequency.

Worked Example 20.1

A voltage is described by $v(t) = 15 \cdot \sin(377t + 30^\circ)$. Find the amplitude, angular frequency, ordinary frequency, period, phase, and the instantaneous value at $t = 5$ ms.

Amplitude: $V_m = 15$ V. Angular frequency: $\omega = 377$ rad/s.

Frequency: $f = \omega/2\pi = 377/6.283 = 60.0$ Hz. This is the North American mains frequency, and 377 is a number you will meet often enough to recognise on sight.

Period: $T = 1/f = 16.67$ ms. Phase: $\phi = 30^\circ$, meaning the waveform leads a pure sine by one-twelfth of a cycle.

At $t = 5$ ms, the argument is $\omega t + \phi = (377)(0.005) + 30^\circ = 1.885 \text{ rad} + 30^\circ$.

The mixed units must be reconciled. Converting 1.885 rad to degrees: $1.885 \times 180/\pi = 108.0^\circ$. So the total argument is $108.0^\circ + 30^\circ = 138.0^\circ$.

Therefore $v = 15 \cdot \sin(138.0^\circ) = 15 \times 0.669 = 10.0$ V.

That reconciliation step is where errors happen. The ωt term is inherently in radians because ω is in radians per second; the phase was given in degrees. One must be converted before they can be added, and no calculator will warn you if you skip it.

20.4 Average and RMS Values

The average value of a sinusoid over a whole number of cycles is zero, since it spends as much time positive as negative. That makes the plain average useless for describing AC magnitude, and something better is needed.

The root-mean-square value is defined as the square root of the mean of the square. Squaring makes everything positive, so nothing cancels, and the definition is chosen so that the RMS value of an AC waveform is exactly the DC value that would dissipate the same power in a resistor. For a sinusoid the result is

$$V_{rms} = V_m/\sqrt{2} \approx 0.707 \cdot V_m$$

When a supply is described as 230 V AC, that is an RMS figure; its peak is 325 V, and the peak-to-peak swing is 650 V. Insulation must survive the peak while power calculations use the RMS, and confusing the two has consequences at both ends — under-specified insulation on one side, power estimates wrong by a factor of two on the other.

Key Idea

RMS is defined by equal heating, not by averaging. $V_{rms} = V_m/\sqrt{2}$ holds for sinusoids only. A square wave has $V_{rms} = V_m$; a triangle wave has $V_m/\sqrt{3}$. Applying the 0.707 factor to a non-sinusoidal waveform is simply wrong, and the error grows with how far the waveform departs from a sinusoid.

Practice Problems

1. Convert to radians: 45° , 90° , 120° , 270° . Convert to degrees: $\pi/6$, $2\pi/3$, $3\pi/2$ rad.
2. A current is $i(t) = 2.5 \cdot \sin(2\pi \cdot 1000 \cdot t - 45^\circ)$ A. State its amplitude, frequency, period, and phase, and find its value at $t = 0$.
3. A sinusoidal voltage has a peak-to-peak value of 10 V. Find its amplitude and its RMS value.
4. Two sinusoids of the same frequency have amplitudes 3 V and 4 V and are 90° apart in phase. Use the Pythagorean identity to find the amplitude of their sum, and explain why you may not simply add 3 and 4.

Chapter 21 — Complex Numbers

Complex numbers have an unfortunate name. There is nothing complicated about them, and nothing imaginary about the imaginary part — both terms are historical accidents from a period when mathematicians distrusted the objects they had just invented. What a complex number actually is, for our purposes, is a way of carrying two related pieces of information as a single quantity, and then doing arithmetic on the pair at once.

For electrical engineering the two pieces of information are magnitude and phase. Chapter 20 established that two sinusoids of the same frequency can differ only in those two respects. Complex numbers are the tool that lets us manipulate both at once, which is why they took over AC analysis completely and never gave it back.

21.1 Why j and Not i

The imaginary unit is defined by $j^2 = -1$. Mathematicians write it i ; electrical engineers write j , because i was already taken by current and reusing it would be intolerable. Every textbook in this series uses j . No other difference is implied.

The defining property is worth stating plainly rather than treating as a mystery. There is no real number whose square is negative, so j is not a real number. That is the entire content of the word "imaginary" — it means "not on the real number line," not "fictitious." The number j is exactly as real, in the ordinary sense, as the number 3; it simply lives in a different direction.

21.2 Rectangular Form

A complex number in rectangular form is written $z = a + jb$, where a is the real part and b the imaginary part. Both a and b are ordinary real numbers. The whole object can be pictured as a point in a plane, with a measured along the horizontal axis and b along the vertical — the complex plane, or Argand diagram.

That geometric picture is the useful one. A complex number is a point, or equivalently an arrow from the origin to that point. Its length is the magnitude and the angle it makes with the positive real axis is the phase, which connects directly to the sinusoid parameters of Chapter 20.

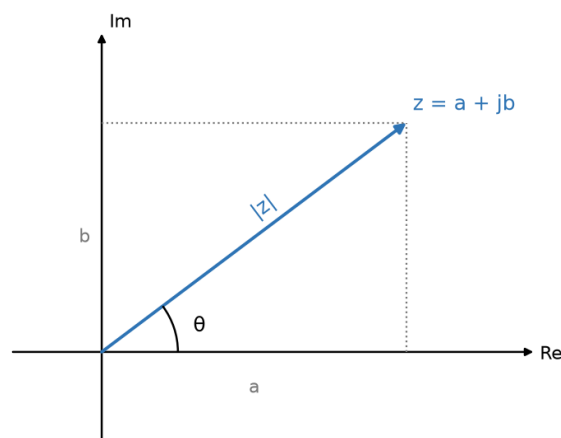


Figure 21.1 — The Argand diagram: a complex number as a point, with magnitude and angle.

Addition and subtraction in rectangular form are componentwise, exactly like adding vectors:

$$(a + jb) + (c + jd) = (a + c) + j(b + d)$$

Multiplication follows the ordinary distributive law, with the single extra rule that $j^2 = -1$:

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc)$$

The cross terms produce the imaginary part; the j^2 in the product of the two imaginary parts flips a sign and lands in the real part. There is nothing to memorise here beyond $j^2 = -1$ — the rest is expansion of brackets.

21.3 The Complex Conjugate and Division

The conjugate of $z = a + jb$, written z^* , is $a - jb$: the same number with the sign of the imaginary part flipped, or geometrically, the point reflected across the real axis.

Its usefulness comes from one fact: a number times its conjugate is always real.

$$z \cdot z^* = (a + jb)(a - jb) = a^2 + b^2$$

This is what makes division possible. To divide by a complex number, multiply top and bottom by the conjugate of the denominator; the denominator becomes real, and the division reduces to dividing by an ordinary number.

Worked Example 21.1

Evaluate $(3 + j4)/(1 - j2)$.

Multiply numerator and denominator by the conjugate of the denominator, which is $(1 + j2)$:

Numerator: $(3 + j4)(1 + j2) = 3 + j6 + j4 + j^2 8 = 3 + j10 - 8 = -5 + j10$.

Denominator: $(1 - j2)(1 + j2) = 1 + 4 = 5$.

So the quotient is $(-5 + j10)/5 = -1 + j2$.

Check by multiplying back: $(-1 + j2)(1 - j2) = -1 + j2 + j2 - j^2 4 = -1 + j4 + 4 = 3 + j4$. Correct.

Note that the denominator became $1^2 + 2^2 = 5$ without any cross terms — that is the conjugate doing its work, and it always happens. If your denominator still has a j in it after multiplying by the conjugate, you have made an arithmetic error.

21.4 Polar Form

The same number can be described by its magnitude and angle instead of its two components. The magnitude is

$$|z| = \sqrt{a^2 + b^2}$$

which is the Pythagorean identity from Chapter 20 doing exactly the job advertised, and the angle is

$$\theta = \arctan(b/a)$$

with a crucial caveat: the arctangent function returns values only between -90° and $+90^\circ$, so it cannot distinguish the second quadrant from the fourth, or the third from the first. You must look at the signs of a and b to place the number in the correct quadrant and adjust by 180° if the real part is negative. Nearly every programming language provides an `atan2` function that handles this; hand calculation does not, and this is a reliable source of answers that are exactly 180° wrong.

Conversion back to rectangular form is straightforward trigonometry: $a = |z| \cdot \cos \theta$ and $b = |z| \cdot \sin \theta$.

Common Misconception

Rectangular form is convenient for addition and subtraction; polar form is convenient for multiplication and division. Trying to multiply in rectangular form is possible but tedious, and trying to add in polar form is a genuine mistake — the magnitudes of a sum are not the sum of the magnitudes unless the two numbers happen to have the same angle. Convert to whichever form suits the operation, then convert back. The conversion is cheap; the error from adding polar magnitudes is not.

21.5 Multiplication and Division in Polar Form

In polar form the two operations become remarkably simple. To multiply, multiply the magnitudes and add the angles. To divide, divide the magnitudes and subtract the angles.

$$|z_1 z_2| = |z_1| \cdot |z_2|, \quad \angle(z_1 z_2) = \angle z_1 + \angle z_2$$

This is the property that makes complex numbers indispensable in AC analysis, and it deserves a moment of attention. Multiplying by a complex number scales and rotates. Multiplying by a number of magnitude 1 and angle 90° — which is to say, multiplying by j — rotates by a quarter turn without changing size at all.

Key Idea

Multiplying by j is a 90° rotation. Multiplying by j twice is a 180° rotation, which is multiplication by -1 — and that is exactly the statement $j^2 = -1$, seen geometrically. The defining property of j is not an arbitrary rule; it is what a quarter-turn does when you apply it twice.

Once you see multiplication as rotation, the behaviour of capacitors and inductors in AC circuits stops being a set of rules to memorise and becomes a single geometric statement, which is where Chapter 23 goes.

Practice Problems

1. Given $z_1 = 4 + j3$ and $z_2 = 2 - j5$, compute $z_1 + z_2$, $z_1 - z_2$, and $z_1 z_2$ in rectangular form.
2. Convert $5 + j12$ to polar form, and convert $10\angle -60^\circ$ to rectangular form.
3. Evaluate $(2 + j3)/(4 - j)$ by the conjugate method, and verify your answer by multiplying back.
4. Compute j^2 , j^3 , j^4 , and $1/j$. Explain each result as a rotation in the complex plane, and state the general rule for j^n .

Chapter 22 — Euler's Formula and the Complex Exponential

This chapter contains one equation. It is arguably the most useful single equation in electrical engineering, and it is the hinge on which the whole of AC analysis turns.

22.1 The Formula

Euler's formula states that

$$e^{j\theta} = \cos \theta + j \sin \theta$$

The left-hand side is an exponential — the function from Chapter 19 whose rate of change equals its own value. The right-hand side is a pair of sinusoids — the functions from Chapter 20. The formula asserts they are the same thing, provided the exponent is imaginary.

Geometrically the statement is transparent. The right-hand side is a complex number with real part $\cos \theta$ and imaginary part $\sin \theta$, so by the Pythagorean identity its magnitude is $\sqrt{(\cos^2 \theta + \sin^2 \theta)} = 1$, and its angle is θ . So $e^{j\theta}$ is simply the point on the unit circle at angle θ . As θ increases steadily, the point travels around the circle at a constant rate.

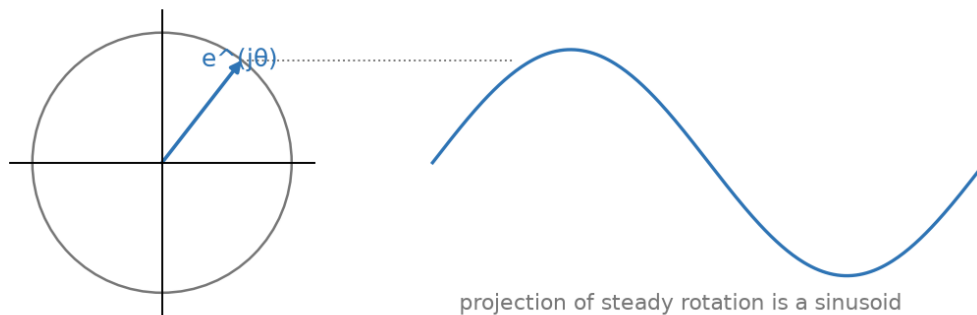


Figure 22.1 — Euler's formula: steady rotation around the unit circle traces out a sinusoid.

That last observation is the whole reason the formula matters to engineers. Steady rotation around a circle, projected onto an axis, is a sinusoid. Euler's formula is the bridge between rotation and oscillation, and AC circuit theory lives on that bridge.

Key Idea

$e^{j\theta}$ is a unit-length arrow at angle θ . Nothing more mysterious than that. Multiplying any complex number by $e^{j\theta}$ rotates it by θ without changing its length, which is the generalisation of "multiplying by j rotates by 90° " from the previous chapter — since j itself is e^{j90° .

22.2 Polar Form, Properly Written

With Euler's formula in hand, the polar form of Chapter 21 can be written as a genuine product rather than as the shorthand angle notation:

$$z = |z| \cdot e^{j\theta}$$

The engineering shorthand $|z|\angle\theta$ means exactly this and nothing more. Writing it as an exponential is not mere notation, though — it means the ordinary rules of exponents now apply, and those rules are why multiplication in polar form works the way it does:

$$|z_1|e^{j\theta_1} \cdot |z_2|e^{j\theta_2} = |z_1||z_2| \cdot e^{j(\theta_1+\theta_2)}$$

Magnitudes multiply and angles add, not as a special rule for complex numbers but as the ordinary law that exponents add when you multiply powers of the same base. The polar multiplication rule of Chapter 21 was a consequence of Euler's formula all along.

22.3 Extracting Sine and Cosine

Applying the formula at $-\theta$ and using $\cos(-\theta) = \cos \theta$, $\sin(-\theta) = -\sin \theta$ gives a companion equation, and adding or subtracting the two isolates the trigonometric functions:

$$\cos \theta = (e^{j\theta} + e^{-j\theta})/2, \quad \sin \theta = (e^{j\theta} - e^{-j\theta})/2j$$

These are worth knowing because they let you replace trigonometric functions with exponentials, and exponentials are far easier to differentiate, integrate, and multiply. A trigonometric identity that takes half a page to prove geometrically often takes two lines in exponential form. The sum-of-angles identity from Chapter 20, for instance, is just the statement that exponents add.

22.4 The Rotating Vector

Now let the angle grow with time at a steady rate, $\theta = \omega t$, and attach an amplitude:

$$v(t) = V_m \cdot e^{j(\omega t + \phi)}$$

This describes an arrow of length V_m rotating counterclockwise at ω radians per second, starting at angle ϕ . Its projection onto the real axis is $V_m \cos(\omega t + \phi)$, which is exactly the sinusoid of Chapter 20.

So a real sinusoidal voltage can be regarded as the real part of a rotating complex quantity. Nothing has been lost — the real signal is recoverable at any moment by taking the real part — but a great deal has been gained, because the rotating exponential is far easier to do algebra with than the cosine is.

Common Misconception

Students often ask what the imaginary part of the rotating vector physically is, expecting it to correspond to some measurable quantity. It does not. The imaginary part is mathematical scaffolding; we deliberately added it because the resulting object is easier to compute with, and we discard it at the end by taking the real part. The physical voltage is, and always was, the real part alone. This is not a defect of the method; it is the method.

22.5 Why This Solves AC Circuits

Here is the payoff, stated now and demonstrated in Chapter 23. Differentiating the rotating exponential with respect to time gives

$$d/dt [V_m \cdot e^{j(\omega t + \phi)}] = j\omega \cdot V_m \cdot e^{j(\omega t + \phi)}$$

The function reproduces itself, multiplied by $j\omega$. Differentiation — a calculus operation — has become multiplication by a constant. Integration likewise becomes division by $j\omega$.

The consequence is enormous. The equations governing capacitors and inductors are differential equations, and differential equations are hard. Written in terms of rotating exponentials, those same equations become algebraic equations in which the derivative has been replaced by the factor $j\omega$. An entire class of calculus problems collapses into complex arithmetic of the kind you practised in Chapter 21.

Worked Example 22.1

Use Euler's formula to prove the identity $\cos(A + B) = \cos A \cos B - \sin A \sin B$.

Start from the exponential of the sum, and use the fact that exponents add:

$$e^{j(A+B)} = e^{jA} \cdot e^{jB}$$

Expand each factor by Euler's formula:

$$e^{jA} \cdot e^{jB} = (\cos A + j \sin A)(\cos B + j \sin B)$$

Multiply out, remembering $j^2 = -1$:

$$= \cos A \cos B + j \cos A \sin B + j \sin A \cos B - \sin A \sin B$$

$$= (\cos A \cos B - \sin A \sin B) + j(\cos A \sin B + \sin A \cos B)$$

Now apply Euler's formula to the left-hand side: $e^{j(A+B)} = \cos(A + B) + j \sin(A + B)$.

Two complex numbers are equal only if their real parts match and their imaginary parts match. Equating real parts gives the identity we wanted; equating imaginary parts gives the sine sum identity for free.

Compare this with the geometric proof, which requires a carefully constructed diagram and several similar triangles. The exponential proof is four lines and produces two identities at once. This is representative of what the complex exponential does throughout the subject.

Practice Problems

1. Evaluate $e^{j\pi}$, $e^{j\pi/2}$, and $e^{j2\pi}$, expressing each as an ordinary complex number.
2. Write $3 + j4$ in the form $|z|e^{j\theta}$, giving θ in both radians and degrees.
3. Use the exponential definitions of sine and cosine to show that $\cos^2\theta + \sin^2\theta = 1$.
4. A signal is $v(t) = \operatorname{Re}\{10 \cdot e^{j(500t + \pi/4)}\}$. Write $v(t)$ explicitly as a cosine, state its frequency in hertz, and find its value at $t = 2$ ms.

Chapter 23 — Phasors and Impedance

This chapter assembles everything the book has built. Units gave us dimensional discipline, algebra gave us rearrangement, trigonometry gave us the sinusoid, complex numbers gave us two-quantity arithmetic, and Euler's formula connected rotation to oscillation. The phasor is what they combine into, and it is the single most powerful labour-saving device in circuit analysis.

23.1 The Central Observation

In a linear circuit driven by a sinusoid at a single frequency, every voltage and every current in the circuit is a sinusoid at that same frequency. The frequency never changes. Only amplitude and phase differ from point to point.

If the frequency is the same everywhere, carrying it through every equation is wasted effort. The phasor method simply stops carrying it. We agree on the frequency once, set it aside, and track only the amplitude and phase — which, as Chapter 21 established, is exactly what a complex number is good for.

23.2 Defining the Phasor

Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$, the corresponding phasor is the complex number

$$V = (V_m/\sqrt{2}) \angle \phi = (V_m/\sqrt{2}) \cdot e^{j\phi}$$

The time dependence has been dropped entirely. To recover the real signal, multiply by $e^{j\omega t}$, take the real part, and scale by $\sqrt{2}$ to undo the RMS normalization: $v(t) = \sqrt{2} \cdot \text{Re}\{V \cdot e^{j\omega t}\}$.

Two conventions must be fixed and then adhered to. First, this series uses cosine as the reference — a phasor angle of zero means a cosine, not a sine. A sine must be converted first, using $\sin \theta = \cos(\theta - 90^\circ)$. Second, phasor magnitudes in this series are RMS values rather than peak values, since that makes power calculations come out without stray factors of two. Other texts use peak amplitudes; neither is wrong, but mixing them within one problem produces answers off by $\sqrt{2}$.

Common Misconception

A phasor is not a function of time, and writing $V(t)$ for a phasor is a category error. The whole point of the phasor is that time has been factored out. Nor is a phasor a vector in physical space — the "arrow" of the complex plane points in a mathematical direction, not a spatial one. A phasor diagram showing a current 90° behind a voltage is saying something about timing, not geometry.

23.3 Impedance

With signals represented as phasors, the relationship between voltage and current across any passive component becomes a simple ratio, and that ratio is called impedance:

$$Z = V/I$$

Impedance is a complex number measured in ohms. Its real part is resistance, which dissipates power; its imaginary part is reactance, which stores and returns energy without dissipating it. Deriving the three basic impedances is now a short exercise, using the result from Chapter 22 that differentiation becomes multiplication by $j\omega$.

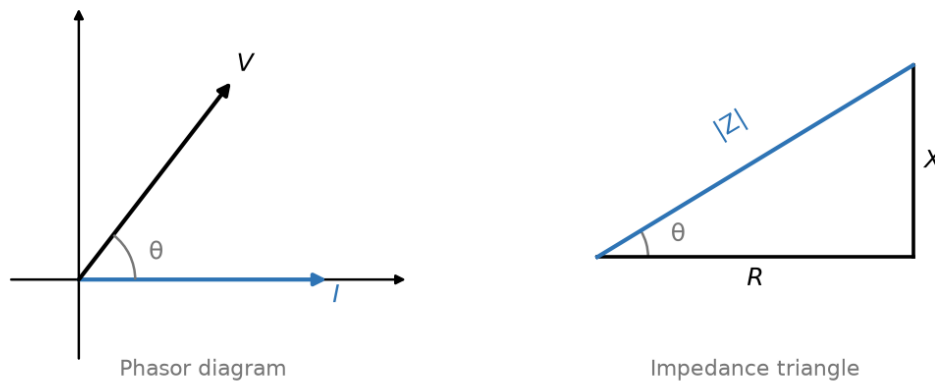


Figure 23.1 — A phasor diagram and its corresponding impedance triangle.

For a resistor, $v = iR$ with no derivative involved, so

$$Z_R = R$$

For an inductor, $v = L \cdot di/dt$. Replacing the derivative with $j\omega$ gives $V = j\omega L \cdot I$, so

$$Z_L = j\omega L$$

For a capacitor, $i = C \cdot dv/dt$, so $I = j\omega C \cdot V$, and inverting,

$$Z_C = 1/(j\omega C) = -j/(\omega C)$$

Three differential relationships have become three algebraic ones. Read the signs: inductive reactance is positive imaginary, capacitive reactance is negative imaginary, and the opposite signs are why they cancel at resonance.

Key Idea

Impedance turns AC circuit analysis into DC circuit analysis with complex numbers. Series impedances add. Parallel impedances combine by product-over-sum. Voltage dividers, current dividers, Thevenin equivalents — every technique from resistive circuit analysis carries over unchanged, with R replaced by Z and real arithmetic replaced by complex arithmetic. Nothing new needs to be learned; the old methods simply work.

23.4 Reading Phase From Impedance

The angle of the impedance is the phase by which voltage leads current. For a pure resistor the impedance is real, its angle is zero, and voltage and current are in phase. For a pure inductor, $Z = j\omega L$ has angle $+90^\circ$, so voltage leads current by a quarter cycle. For a pure capacitor, angle -90° , so voltage lags current.

The traditional mnemonic is "ELI the ICE man": in an inductor (L), voltage E leads current I; in a capacitor (C), current I leads voltage E. It is a silly device and it works.

Real components are never pure, and the impedance angle tells you how far from ideal a component is. A capacitor whose measured impedance angle is -88° rather than -90° has a small resistive component, and that two-degree discrepancy is precisely the loss mechanism quantified later in the series as equivalent series resistance.

Worked Example 23.1

A $1 \text{ k}\Omega$ resistor is in series with a $0.1 \text{ }\mu\text{F}$ capacitor across a 10 V RMS , 1 kHz supply. Find the total impedance, the current, and the voltage across each component.

First the angular frequency: $\omega = 2\pi f = 2\pi(1000) = 6283 \text{ rad/s}$.

Capacitive impedance: $Z_C = -j/(\omega C) = -j/(6283 \times 0.1 \times 10^{-6}) = -j/(6.283 \times 10^{-4}) = -j1592 \Omega$.

Total impedance in series is the sum: $Z = 1000 - j1592 \Omega$.

In polar form: $|Z| = \sqrt{(1000^2 + 1592^2)} = \sqrt{(10^6 + 2.534 \times 10^6)} = \sqrt{(3.534 \times 10^6)} = 1880 \Omega$.

The angle: $\theta = \arctan(-1592/1000) = -57.9^\circ$. The real part is positive, so we are in the fourth quadrant and no 180° adjustment is needed.

Current: $I = V/Z = 10\angle 0^\circ / 1880\angle -57.9^\circ = 5.32\angle +57.9^\circ \text{ mA}$. Dividing means dividing magnitudes and subtracting angles, so the current leads the supply voltage by 57.9° — as expected for a circuit that is predominantly capacitive.

Voltage across the resistor: $V_R = I \cdot R = (5.32 \text{ mA})(1000 \Omega) = 5.32 \text{ V}$, in phase with the current, so at $+57.9^\circ$.

Voltage across the capacitor: $V_C = I \cdot Z_C = (5.32 \text{ mA})(1592 \Omega) = 8.47 \text{ V}$, at an angle of $57.9^\circ - 90^\circ = -33.1^\circ$.

Now the check that catches nearly every error in this kind of problem. The two component voltages are 5.32 V and 8.47 V, which sum arithmetically to 13.79 V — considerably more than the 10 V supply. That is not a violation of Kirchhoff's voltage law. The voltages are 90° apart in phase, so they add as phasors, not as numbers: $\sqrt{(5.32^2 + 8.47^2)} = \sqrt{(29.3 + 71.7)} = \sqrt{100} = 10.0 \text{ V}$. Exactly the supply voltage.

Common Misconception

It looks impossible for the voltages across two series components to exceed the supply voltage, and in a DC circuit it would be. In AC circuits it is routine, because component voltages add as phasors rather than as plain numbers. In a resonant circuit the effect becomes dramatic: the voltage across the inductor and the capacitor individually can exceed the supply by a factor equal to the Q of the circuit, sometimes by hundreds of times. Those voltages are entirely real and will destroy components rated only for the supply voltage. They cancel each other in the sum precisely because they are 180° apart.

23.5 Admittance, and When to Use It

The reciprocal of impedance is admittance, $Y = 1/Z$, measured in siemens. Its real part is conductance and its imaginary part is susceptance.

Admittance exists for the same reason conductance does: for parallel combinations, admittances simply add, whereas impedances require the reciprocal formula. In a circuit with many parallel branches, working in admittance turns a nested reciprocal expression into a plain sum. Nodal analysis, which is formulated around parallel connections at each node, is therefore naturally written in admittances — the conductance matrix of Chapter 17 becomes an admittance matrix, and every technique carries straight over.

Practice Problems

1. Convert to phasor form, using cosine reference and RMS magnitudes: $v(t) = 170 \cdot \cos(377t + 30^\circ) \text{ V}$, and $i(t) = 4.24 \cdot \sin(377t) \text{ A}$.
2. Find the impedance of a 22 mH inductor and a 470 nF capacitor, each at 2 kHz. State the magnitude and angle of each.
3. A 470Ω resistor is in series with a 100 mH inductor across a 24 V RMS, 400 Hz supply. Find the total impedance in polar form, the current, and the phase angle between supply voltage and current.
4. For the series RC circuit of Worked Example 23.1, find the frequency at which the resistive and capacitive impedance magnitudes are equal. What is the phase angle of the total impedance at that frequency, and what is the ratio of output to input voltage if the output is taken across the resistor?

Chapter 24 — The Smith Chart, Impedance Matching, and Bias Networks

This chapter is a preview of where the mathematics of complex impedance leads once RF engineering takes it over — not a transmission-line design course compressed into a foundations book. You do not need to come away able to design a matching network from scratch; the purpose is to see the tools from Chapters 21–23 doing real work, so that when this series reaches RF and microwave engineering in full, none of what follows feels unfamiliar.

24.1 Why Matching Matters: The Conjugate-Match Condition

Chapter 23 introduced impedance and showed that a linear circuit's behavior at one frequency is fully captured by a complex number. A question follows immediately once any real source feeds any real load: how much of the available power actually reaches the load, and what can be done about it if the answer is “not much”? Chapter 31 answered this for purely resistive networks — maximum power transfer requires $R_{\text{load}} = R_{\text{source}}$. The complex generalization, needed the moment either side has any reactance, is the conjugate-match condition:

$$Z_{\text{load}} = Z_{\text{source}}^*$$

That is, the load impedance must equal the complex conjugate of the source impedance — same resistance, equal and opposite reactance. A source with impedance $50 + j30 \, \Omega$ delivers maximum available power to a load of exactly $50 - j30 \, \Omega$, because the load's negative reactance exactly cancels the source's positive reactance (leaving a purely resistive series combination), while the matched resistances split available power exactly as the DC theorem already established. A matching network is nothing more than a lossless circuit — built from reactive elements, which absorb no real power themselves — inserted between an arbitrary source and load specifically to present this conjugate-matched impedance to each side.

24.2 The Reflection Coefficient

Working directly with impedance is awkward once transmission lines enter the picture, because impedance transforms in a complicated way along a line. The reflection coefficient Γ , defined relative to a reference impedance Z_0 (almost always $50 \, \Omega$ in RF work), is better behaved:

$$\Gamma = (Z - Z_0) / (Z + Z_0)$$

Equivalently, working with the normalized impedance $z = Z/Z_0$: $\Gamma = (z - 1)/(z + 1)$. $\Gamma = 0$ means a perfect match (all power delivered, nothing reflected); $|\Gamma| = 1$ means total reflection (an open circuit, a short circuit, or any purely reactive load). For any passive load, $|\Gamma| \leq 1$ always.

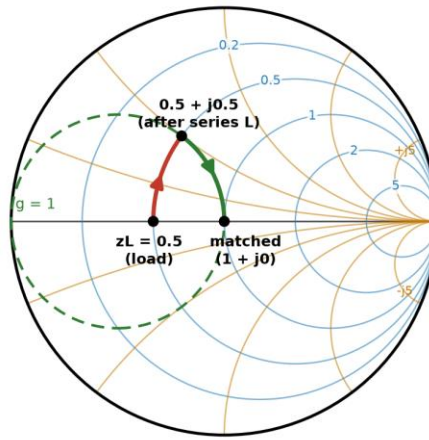
24.3 The Smith Chart

The Smith chart is a graphical picture of the relationship between Γ and z , plotted on the complex Γ -plane, restricted to the unit disk (since $|\Gamma| \leq 1$ for passive loads). Overlaid on that disk are two families of curves: circles of constant normalized resistance r , and arcs of constant normalized reactance x . Every point inside the chart corresponds to exactly one normalized impedance $z = r + jx$, and reading r and x directly off the chart — rather than doing the complex algebra of the $\Gamma \leftrightarrow z$ conversion by hand every time — is the entire practical value of the chart.

The constant- r circles all pass through the point $\Gamma = 1$ (the open-circuit point, $z \rightarrow \infty$), with the $r = 0$ circle being the chart's own outer boundary (the case of a purely reactive load, $|\Gamma| = 1$). The constant- x arcs are centered off-chart,

above the real axis for inductive reactance (positive x) and below it for capacitive reactance (negative x), with the entire real axis itself forming the $x = 0$ line — purely resistive impedances.

Smith chart: normalized impedance $z = r + jx$



Series L moves the point along $r = 0.5$ (red); shunt C moves it along $g = 1$ (green) to the center

Figure 24.1 — The Smith chart, with a two-element matching path plotted (Worked Examples 24.1 and 24.2).

24.4 VSWR and Return Loss

Two practical numbers are computed directly from $|\Gamma|$ and appear on every piece of RF test equipment. The voltage standing wave ratio:

$$\text{VSWR} = (1 + |\Gamma|) / (1 - |\Gamma|)$$

runs from 1 (perfect match) to infinity (total reflection), and is the ratio most commonly quoted for antennas and RF connectors. Return loss, expressed in decibels, is the other common figure of merit:

$$\text{Return Loss (dB)} = -20 \log_{10} |\Gamma|$$

with a larger return loss (in dB) meaning a better match — the opposite sense from VSWR, which is worth keeping straight, since both numbers describe exactly the same underlying $|\Gamma|$.

24.5 Moving on the Chart: Series and Shunt Elements

Adding a series reactance to a load moves its point along a constant- r circle — a series element changes only x , leaving r (and hence the circle it sits on) unchanged, whether the added reactance is inductive (moving clockwise, toward more positive x) or capacitive (moving counterclockwise). Adding a shunt element is most naturally described in admittance terms: $y = 1/z = g + jb$, where a shunt reactance changes only b , moving the point along a constant- g circle. The admittance grid is simply the impedance grid rotated 180° about the chart's center (since $y = 1/z$ corresponds to $\Gamma \rightarrow -\Gamma$), so the same physical chart serves both purposes — a shunt element moves a point along whichever constant- g circle is the point-reflection, through the center, of the corresponding constant- r circle.

Key Idea

Series elements move you along a constant- r circle; shunt elements move you along a constant- g circle. A two-element L-network match is nothing more than finding one circle of each kind that intersect — one step to reach the intersection, one more to reach the center (the matched point, $\Gamma = 0$).

24.6 Lumped-Element (L-Network) Matching

The simplest matching network — an L-network — uses exactly two reactive elements, one in series and one in shunt, arranged so the load's impedance is transformed to Z_0 . Given the discussion above, the design procedure is direct: pick whichever element (series or shunt) moves the load's own r-circle (or g-circle) so that it crosses the $g = 1$ circle (or $r = 1$ circle), then pick the second element to complete the trip to the center.

Worked Example 24.1

Match a purely resistive load $R_L = 25\ \Omega$ to a $Z_0 = 50\ \Omega$ system at $f = 1.0\ \text{GHz}$ using an L-network.

Normalized load: $z_L = 25/50 = 0.5$. Since $R_L < Z_0$, add a series reactance first. Solving for the series reactance X_s that lands exactly on the $g = 1$ circle (algebraically, $X_s^2 = R_L(Z_0 - R_L) = 25(50 - 25) = 625$, so $X_s = 25\ \Omega$): the point moves from $z = 0.5$ to $z = 0.5 + j0.5$ — and a direct check confirms this normalized admittance is $y = 1 - j1$, i.e., $g = 1$ exactly, as required.

A series inductor supplies this positive reactance: $L = X_s/\omega = 25/(2\pi \times 10^9) \approx 3.98\ \text{nH}$. The remaining susceptance, $b = +1$ (normalized) = $0.02\ \text{S}$ (actual), is cancelled by a shunt capacitor: $C = 0.02/\omega \approx 3.18\ \text{pF}$. With both elements in place, the load presents exactly $50 + j0\ \Omega$ to the source.

Worked Example 24.2

A load impedance of $Z_L = 25 + j25\ \Omega$ (not purely resistive) is to be matched to the same $50\ \Omega$ system, also at $1.0\ \text{GHz}$. How many elements are needed?

Normalized: $z_L = 0.5 + j0.5$ — which is exactly the intermediate point reached in Worked Example 24.1 after its series inductor was added. This load's own reactance already does the job a series element would otherwise have to do. Only the shunt capacitor is needed: $C \approx 3.18\ \text{pF}$, exactly as before, bringing the load to a perfect match with a single component.

Common Misconception

It is tempting to assume an L-network always needs exactly one inductor and one capacitor. It needs one series and one shunt reactance, of whichever sign the specific match requires — Worked Example 24.1 happened to need a series L and a shunt C, but the dual solution (series C, shunt L) is equally valid for a resistive load, and which pair of element types is actually needed depends on the load's own reactance, exactly as Worked Example 24.2 shows: a load with the right amount of its own series reactance already present needs only a single shunt element to finish the job.

24.7 Distributed Matching: Microstrip Lines and Stubs

At the frequencies where microstrip circuit boards operate, a short length of transmission line is often a more practical matching element than a discrete inductor, especially once the frequency is high enough that a lumped component's own parasitic behavior becomes significant. The single-stub tuning technique replicates the L-network idea using distributed elements: a length d of line (of the board's own characteristic impedance Z_0) is inserted between the load and a shunt stub, chosen so that at distance d from the load, the line's input admittance has real part exactly $Y_0 = 1/Z_0$ — placing the point on the $g = 1$ circle, exactly as a series lumped element did in Section 24.6. A stub of length l , then, cancels the remaining susceptance, exactly as the shunt capacitor did.

Worked Example 24.3

A purely resistive load $Z_L = 100\ \Omega$ is to be matched to a $Z_0 = 50\ \Omega$ microstrip line using a single short-circuited shunt stub. Find the required distance d from the load to the stub, and the stub length l , both as fractions of a wavelength.

Normalized load: $z_L = 2.0$. Solving the standard single-stub distance equation for a real load gives two valid solutions within one half-wavelength; the shorter one works out to $d \approx 0.152\lambda$. At that point, the line's input impedance is $Z_{in} \approx 33.3 - j23.6\ \Omega$, so its admittance is $Y_{in} \approx 0.020 + j0.0141\ \text{S}$ — the real part matches $Y_0 = 1/50 = 0.020\ \text{S}$ exactly, confirming the point sits on the $g = 1$ circle as intended.

A short-circuited stub of length l presents an input susceptance of $-Y_0 \cot(\beta l)$; setting this equal to -0.0141 S (to cancel the remaining $+j0.0141 \text{ S}$) gives $\beta l \approx 54.7^\circ$, or $l \approx 0.152\lambda$. The second valid solution, $d \approx 0.348\lambda$ with a stub of $l \approx 0.348\lambda$, is equally correct — in practice, whichever pair gives the more convenient physical length on the actual board is used.

An open-circuited stub can be used in place of a short-circuited one — often preferred in microstrip specifically because an open stub needs no via to ground, which a short-circuited stub does — at the cost of a generally different (and here, longer) required stub length for the same match, since an open stub's input susceptance is $+Y_0 \tan(\beta l)$ rather than $-Y_0 \cot(\beta l)$.

24.8 Fine-Tuning a Match with SMD Components

A matching network built from fixed-value components — whether lumped or microstrip — rarely lands exactly on the intended match once real component tolerances, board parasitics, and assembly variation are accounted for. A small additional shunt capacitance, realized as a surface-mount trimmer or a selectable small SMD pad, is the standard way to correct a small residual mismatch after assembly, precisely because shunt capacitive susceptances simply add: adding a second capacitor in parallel with the first increases the total susceptance by exactly $\omega \Delta C$, with no interaction between the two.

Worked Example 24.4

A matching network designed for a perfect match at 1.0 GHz is measured, after assembly, to present an input admittance of $Y_{in} = 0.020 - j0.0010 \text{ S}$ (the real part is correct; a small residual inductive-looking susceptance remains, most likely from stray inductance in the shunt capacitor's own PCB pad). Find the additional SMD shunt capacitance needed to null out this residual susceptance.

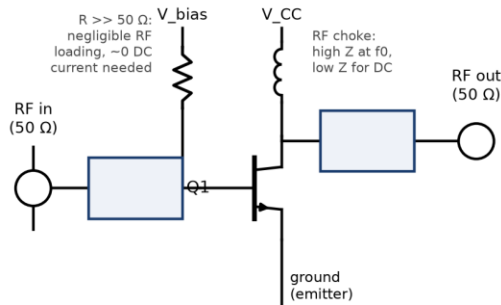
The residual susceptance to cancel is $+0.0010 \text{ S}$ (a shunt capacitor's susceptance is positive, ωC , so it can only add positive susceptance — exactly what is needed here to cancel a negative residual). $\Delta C = 0.0010/\omega = 0.0010/(2\pi \times 10^9) \approx 0.159 \text{ pF}$ — a genuinely tiny trim capacitance, which is exactly why practical RF fine-tuning uses sub-picofarad SMD padders rather than trying to hit the ideal value on the first try with the primary matching components alone.

24.9 Bias Networks: RF Chokes and High-Ohmic Resistors

A practical amplifier stage needs DC bias delivered to its active device without disturbing the RF matching networks built around it — the bias network must be effectively invisible to the signal at the operating frequency, while still providing whatever DC path the device actually needs.

At an output node (a bipolar transistor's collector, or a FET's drain), meaningful DC current must flow, so an RF choke — an inductor sized so its reactance ωL is large compared to the system impedance at the operating frequency — is the standard solution: it looks like a low-resistance DC path (an inductor has zero DC resistance, ideally) while presenting a high impedance, close to an open circuit, to the RF signal, leaving the surrounding matching network's design essentially undisturbed. A common rule of thumb sizes the choke's reactance to at least 5-10 times the system impedance at the design frequency.

At an input node (a FET's gate, in particular), the DC current required is essentially zero — an ideal FET gate draws no DC current at all — so a simple high-value resistor works just as well as a choke, and is simpler and cheaper to implement. A large resistor barely loads the RF path (since it sits in parallel with a much smaller system impedance) and needs to carry essentially no current, so the power dissipated in it is negligible regardless of its value being “too large” by ordinary circuit-design standards.



A high-value resistor biases the (near-zero-current) input; an RF choke biases the output without disturbing either matching network at the design frequency

Figure 24.2 — A single-stage amplifier: a high-value resistor biases the input, an RF choke biases the output, and neither disturbs its adjacent matching network at the design frequency.

Worked Example 24.5

An amplifier operating at $f = 2.4$ GHz uses a $50\ \Omega$ matching network at its output. Using a rule-of-thumb factor of 10, find the minimum RF choke inductance for output biasing. Separately, verify that a $10\ \text{k}\Omega$ gate-bias resistor barely disturbs a $50\ \Omega$ input match.

Choke: required reactance $X_L \geq 10 \times 50 = 500\ \Omega$. $L_{\min} = X_L / \omega = 500 / (2\pi \times 2.4 \times 10^9) \approx 33.2\ \text{nH}$ — any inductor of this value or larger keeps the choke's loading on the RF path below the 10% level implied by the rule of thumb.

Bias resistor: a $10\ \text{k}\Omega$ resistor in parallel with a $50\ \Omega$ system impedance combines to $(50 \times 10,000) / (50 + 10,000) \approx 49.75\ \Omega$ — a deviation of about 0.5% from the unloaded $50\ \Omega$, negligible for essentially any practical purpose, and confirming that a high-ohmic resistor is an entirely adequate choice for a near-zero-current input bias point.

Practice Problems

1. A $50\ \Omega$ system has a load impedance of $30 + j40\ \Omega$. Find the normalized impedance and the reflection coefficient Γ (in polar form).
2. For a load with $|\Gamma| = 0.5$, find the VSWR and the return loss in dB.
3. Design an L-network to match a purely resistive load of $20\ \Omega$ to a $50\ \Omega$ system at 500 MHz. Find the series inductance and shunt capacitance required.
4. An RF choke must present a reactance of at least 8 times the system impedance of $75\ \Omega$ at 900 MHz. Find the minimum required inductance.

Chapter 25 — Calculus for Electrical Engineering

Calculus has a reputation it does not deserve. Stripped of the machinery, it consists of two ideas: a derivative measures how fast something is changing, and an integral measures how much of something has accumulated. Everything else is technique for computing them. This chapter builds both ideas from the circuit behaviour that motivates them, and covers the specific results the rest of this series relies on.

25.1 The Derivative as a Rate

The derivative of a quantity with respect to time is its rate of change: how fast it is going up or down at a particular instant. Written dv/dt , it is the slope of the graph of v against t at that moment.

Circuit theory is full of derivatives because two of the three basic components are defined by them. A capacitor's current is proportional to the rate of change of its voltage, and an inductor's voltage is proportional to the rate of change of its current:

$$i = C \cdot dv/dt \quad \text{and} \quad v = L \cdot di/dt$$

Read these as physical statements rather than formulas. A capacitor passes current only while its voltage is changing — hold the voltage constant and the current stops, which is why a capacitor blocks DC. An inductor develops voltage only while its current is changing — hold the current constant and the voltage vanishes, which is why an inductor is a short circuit to DC.

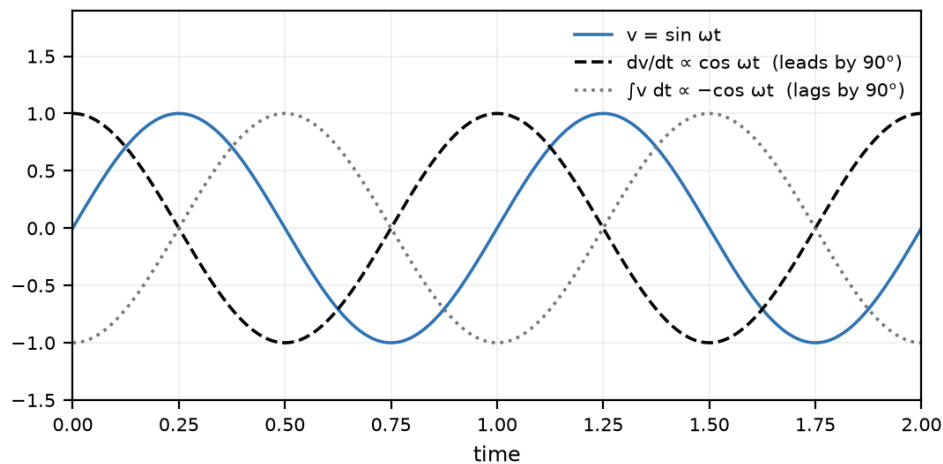


Figure 25.1 — Differentiating a sinusoid shifts its phase by 90 degrees; integrating shifts it by -90 degrees.

The derivatives you need are few:

- **A constant** has derivative zero. Nothing changing means no rate of change.
- **A straight line** $v = kt$ has derivative k , the constant slope.
- **A power** t^n has derivative $n \cdot t^{(n-1)}$.
- **The exponential** $e^{(kt)}$ has derivative $k \cdot e^{(kt)}$ — it reproduces itself, scaled. This is the defining property from Chapter 19.

- **Sine and cosine swap:** the derivative of $\sin(\omega t)$ is $\omega \cdot \cos(\omega t)$, and the derivative of $\cos(\omega t)$ is $-\omega \cdot \sin(\omega t)$. Note that each differentiation brings down a factor of ω and advances the phase by 90° , which is exactly the $j\omega$ of Chapter 23 seen without complex notation.

Common Misconception

The derivative of $\sin(\omega t)$ is $\omega \cdot \cos(\omega t)$, not $\cos(\omega t)$. The factor of ω is dropped constantly and the consequences are severe, because ω is often a large number — at mains frequency it is 377, and in RF work it is in the billions. Omitting it does not make an answer slightly wrong; it makes it wrong by orders of magnitude. The factor appears because of the chain rule: differentiating the inside function ωt gives ω , which multiplies the result.

25.2 The Integral as an Accumulation

Integration is the reverse of differentiation: given a rate, it recovers the total accumulated. Geometrically it is the area under a curve.

Rearranging the capacitor relation shows the accumulation directly. If $i = C \cdot dv/dt$, then the voltage is the accumulated current divided by capacitance:

$$v(t) = (1/C) \cdot \int i \, dt$$

This says a capacitor's voltage is proportional to the total charge that has flowed into it — the integral of current is charge, which is exactly the statement that a coulomb is an ampere-second. The capacitor is a device that remembers how much current has passed, which is why it is a memory element and why its state must be specified as an initial condition when solving a circuit.

The integrals worth knowing are the derivatives read backwards: the integral of t^n is $t^{n+1}/(n+1)$, the integral of e^{kt} is e^{kt}/k , the integral of $\cos(\omega t)$ is $\sin(\omega t)/\omega$, and the integral of $\sin(\omega t)$ is $-\cos(\omega t)/\omega$. Each integration divides by ω and retards the phase by 90° , which is the $1/j\omega$ of capacitive impedance.

25.3 Definite Integrals, Averages, and RMS

A definite integral evaluates the accumulation between two limits, and dividing by the width of the interval gives the average value:

$$\text{average of } f \text{ over } [a, b] = (1/(b-a)) \cdot \int_a^b f \, dt$$

This is where the RMS definition of Chapter 20 comes from, and it is worth seeing the calculation once. The mean square of a sinusoid over one period is

$$\text{mean of } V_m^2 \sin^2(\omega t) = V_m^2 \cdot (\text{mean of } \sin^2)$$

and the double-angle identity from Chapter 20 gives $\sin^2\theta = (1 - \cos 2\theta)/2$. The cosine term averages to zero over a whole number of cycles, leaving a mean of exactly one half. So the mean square is $V_m^2/2$, and the root mean square is $V_m/\sqrt{2}$.

That is the entire derivation of the 0.707 factor, and it makes plain why the factor is specific to sinusoids: it came from the average of $\sin^2$, and a different waveform has a different average of its square.

Worked Example 25.1

A capacitor of $4.7 \mu\text{F}$ carries a current that rises linearly from 0 to 10 mA over 5 ms, then stops. Find the voltage across it at the end of that interval, assuming it started at 0 V.

The current is $i(t) = kt$ with k chosen so that $i = 10 \text{ mA}$ at $t = 5 \text{ ms}$: $k = 0.010/0.005 = 2 \text{ A/s}$.

The voltage is the integral of current divided by capacitance:

$$v(t) = (1/C) \cdot \int_0^t k t \, dt = (1/C) \cdot k \cdot t^2/2$$

Evaluating at $t = 5 \text{ ms}$:

$$v = (1/4.7 \times 10^{-6})(2)(0.005^2/2) = (1/4.7 \times 10^{-6})(2)(1.25 \times 10^{-5}) = (2.5 \times 10^{-5})/(4.7 \times 10^{-6}) = 5.32 \text{ V}$$

A useful cross-check: the average current over the ramp is half the peak, or 5 mA, sustained for 5 ms, giving a total charge of 25 μC . Then $v = Q/C = 25 \times 10^{-6} / 4.7 \times 10^{-6} = 5.32 \text{ V}$. The two routes agree, and the second is often quicker — for a linear ramp, the area under the curve is just a triangle, and you do not need the integral at all.

Key Idea

In sinusoidal steady state, d/dt becomes $\times j\omega$ and $\int dt$ becomes $\div j\omega$. This is the whole reason Chapter 23 works. Once you accept it, no differential equation ever needs to be solved to analyse a linear circuit at a single frequency — the calculus has been done in advance, once, and packaged into the factor $j\omega$.

25.4 Differential Equations, Briefly

A differential equation relates a quantity to its own derivatives. The RC circuit produces the simplest interesting example. Kirchhoff's voltage law around a charging loop gives

$$RC \cdot dv/dt + v = V_s$$

and the solution is the exponential of Chapter 19, $v(t) = V_s(1 - e^{-(t/RC)})$. You are not expected to derive that here. What matters is recognising the pattern: a first-order differential equation produces an exponential response with a single time constant, and a second-order one produces either two exponentials or a damped oscillation, depending on the sign of the discriminant met in Chapter 16.

This is why the phasor method is such a relief. Differential equations must be solved for the complete response, including the transient. But if all you need is the steady state — the behaviour after the transients have died away, which is what most AC analysis concerns — the phasor method delivers it algebraically. Transient analysis, where the differential equation genuinely must be solved, is taken up properly in Part III.

Practice Problems

1. A capacitor of 1 μF has a voltage rising at a steady 2000 V/s. Find the current through it.
2. An inductor of 50 mH carries a current $i(t) = 3 \cdot \sin(1000t)$ A. Find the voltage across it, including its amplitude and phase relative to the current.
3. Find the derivative of $v(t) = 10 \cdot \cos(2000t + 30^\circ)$ with respect to time, and state the amplitude of the result.
4. A current of 5 mA flows into a 10 μF capacitor, initially uncharged, for 200 ms and then stops. Find the final voltage, first by integration and then by the charge method, and confirm the two agree.

Chapter 26 — Where This Book Leads Next

The mathematics is now in place. It is worth a short chapter to say explicitly where each piece gets used, both so the effort feels justified and so you know where to return when a later book assumes something you have half-forgotten.

26.1 Immediately: Part III, Circuit Theory

Part III opens with Kirchhoff's laws and resistive networks, which need only Chapters 16 and 17 — rearrangement and simultaneous equations. It then introduces capacitors and inductors, at which point Chapter 25's derivatives become necessary to state what those components do. First-order transients use the exponentials of Chapter 19 directly, including the 63% and five-time-constant rules.

The second half of Part III is sinusoidal steady-state analysis, and it uses Chapters 20 through 23 continuously and without further explanation. The phasor and impedance material of Chapter 23 is assumed knowledge from the moment AC appears. If one chapter of this book is worth revisiting before starting Part III, it is that one.

26.2 Soon: Filters, Frequency Response, and Bode Plots

Frequency response is the impedance of Chapter 23 evaluated across a range of frequencies rather than at one. The asymptotic reading of transfer functions from Chapter 18, combined with the logarithmic axes and decibels of Chapter 19, is precisely the skill set the Bode plot demands. An engineer fluent in those two chapters can sketch the response of a filter from its component values in a few seconds, without computing a single point.

26.3 Later: RF, Noise, and Communications

The RF books lean on this material harder than any others. The Smith chart is a conformal map of the complex plane, and it is unusable without genuine comfort with complex arithmetic. S-parameters are complex ratios, quoted in magnitude and phase. Link budgets are sums of decibel quantities, and the dBm conventions of Chapter 19 are used on every page. Noise figure is a ratio of ratios, expressed logarithmically, and the uncertainty reasoning of Chapter 15 governs how precisely any of it can be measured.

Modulation theory rests on the sum-of-angles identity from Chapter 20 — multiplying two sinusoids produces sum and difference frequencies, which is what a mixer does and what every superheterodyne receiver is built around.

26.4 Eventually: Transforms

The Fourier and Laplace transforms, which arrive later in the series, are direct generalisations of what you have already done. The phasor represents a signal at one frequency; the Fourier transform represents a signal as a continuous superposition of all frequencies at once. The $j\omega$ of Chapter 23 becomes the more general complex variable $s = \sigma + j\omega$ of the Laplace transform, where the real part σ describes growth or decay and the imaginary part describes oscillation.

Both transforms turn differential equations into algebraic ones, exactly as the phasor method did — they are the same trick, applied more broadly. Having understood why $j\omega$ works, you will find the transforms considerably less alien than students who meet them without this grounding.

Key Idea

Nothing later in this series depends on a fundamentally unfamiliar mathematical foundation. Later books extend these tools — complex arithmetic, logarithms, sinusoids, first-order calculus — and combine them in more demanding ways, and some topics (Fourier and Laplace transforms, probability and noise, feedback and control) can

genuinely feel harder in their own right. But none of it requires a fundamentally different mathematical starting point than what this book has already built. If this book is solid, the rest of the series is about applying that foundation, not relearning it.

Part III — DC and AC Circuit Theory

Chapter 27 — Circuit Elements and Kirchhoff's Laws

27.1 From Physics to Circuit Theory

Part I defined voltage, current, resistance, capacitance, and inductance from first physical principles. Circuit theory takes those definitions as given and asks a more restricted, more tractable question: if you know how each individual element in a network behaves — a resistor obeying $V = IR$, a capacitor obeying $C = Q/V$, an inductor obeying $\varepsilon = -L \, dI/dt$, and a source supplying a fixed voltage or current — what can you say about the voltages and currents everywhere in an arbitrary network built by wiring many such elements together?

This restriction is not a retreat from physics; it is what makes circuit analysis tractable at all. A real circuit board can have hundreds or thousands of components. Nothing about Coulomb's law or Gauss's law, applied directly, scales to a problem of that size — but the two laws introduced in this chapter, which are themselves direct consequences of charge conservation and energy conservation, scale perfectly well, because they let you reason about a circuit's topology (how things are connected) independently of the messy details of the fields inside any one component.

27.2 Nodes, Branches, and Loops

A node is any point in a circuit where two or more elements connect — physically, an entire stretch of ideal (zero-resistance) wire between components counts as a single node, since every point along it is at the same potential. A branch is a single path between two nodes, generally containing one circuit element. A loop is any closed path through the circuit that starts and ends at the same node without repeating any other node along the way.

These three words — node, branch, loop — are the vocabulary the rest of this chapter, and much of this book, is built from, and it is worth being able to identify all three correctly on a circuit diagram before proceeding, since misidentifying a node (for instance, treating two separate points joined by a plain wire as two different nodes) is one of the most common sources of error in an otherwise correct circuit analysis.

27.3 Kirchhoff's Current Law (KCL)

Kirchhoff's current law states that the sum of currents entering any node equals the sum of currents leaving it — equivalently, the algebraic sum of all currents at a node, with currents entering counted as positive and currents leaving as negative (or any other single consistent convention), is always zero:

$$\sum I_{\text{in}} = \sum I_{\text{out}} \quad (\text{equivalently: } \sum I_{\text{node}} = 0)$$

KCL is nothing more than charge conservation (Chapter 1), applied at a single point: charge cannot accumulate at an ordinary node (which has no capacity to store it), so whatever charge flows in during any interval must flow back out during that same interval. This makes KCL exact for any node, in any circuit, at any instant — it is not an approximation, and it holds regardless of what kind of elements are attached to the node or how they behave individually.

27.4 Kirchhoff's Voltage Law (KVL)

Kirchhoff's voltage law states that the algebraic sum of all voltage rises and drops around any closed loop is zero:

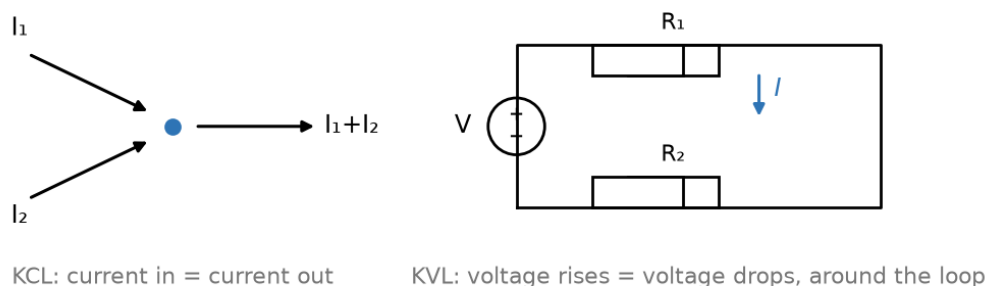


Figure 27.1 — KCL at a node and KVL around a loop, the two laws every circuit-analysis method rests on.

$$\sum V_{\text{loop}} = 0$$

KVL follows from the fact that electric potential (Chapter 5) is a well-defined function of position — travel around any closed loop, back to your starting point, and the net change in potential must be zero, since potential does not depend on the path taken between two points, only on the two points themselves. Practically, this means: pick a direction to traverse a loop, add up every voltage drop you encounter in that direction (a drop across a resistor in the direction of assumed current flow, for instance) and subtract every voltage rise (crossing a source from $-$ to $+$, for instance), and the total must equal zero. This reasoning holds for the lumped-circuit approximation used throughout this book, where any changing magnetic flux linking a loop is represented explicitly as its own circuit source — an induced EMF, as in Chapter 11 — rather than left as a distributed effect. In full electromagnetic theory, a changing magnetic flux produces a genuinely non-conservative electric field (Faraday's law, Chapter 11), and outside the lumped approximation KVL as stated here no longer holds exactly.

Key Idea

KCL is about nodes; KVL is about loops. KCL enforces charge conservation at every point where branches meet. KVL enforces that potential is a well-defined, single-valued quantity around any closed path. Together, applied systematically, they are sufficient to solve any resistive circuit — no other physical law is needed, only careful bookkeeping of these two.

27.5 Sign Conventions and the Passive Sign Convention

Before applying either law, every branch needs an assumed current direction and every element needs a labeled voltage polarity. The passive sign convention, used throughout this book and throughout the rest of this series, assigns the voltage polarity across a passive element (a resistor, capacitor, or inductor) so that the assumed current enters at the $+$ terminal — under this convention, a positive value of $P = VI$ means the element is absorbing power, and a negative value means it is delivering power, which is otherwise unusual for a passive element and typically signals either a sign error or a source hiding somewhere in what was assumed to be a passive branch.

It does not matter, in principle, which direction you assume for a branch current before solving a circuit — if you guess the wrong direction, the algebra will simply return a negative value for that current, correctly telling you the actual current flows opposite to your assumed direction. What matters is consistency: once a direction and polarity are assigned to a branch, every equation involving that branch must use them consistently throughout the entire solution.

Worked Example 27.1

Problem: A single loop contains a 12 V battery in series with a $4.0\ \Omega$ resistor and an $8.0\ \Omega$ resistor. Using KVL, find the current in the loop.

Solution: Assume current I flows clockwise, out of the battery's + terminal. Traversing the loop clockwise starting at the battery's – terminal: rise of 12 V crossing the battery, then a drop of $I(4.0)$ across the first resistor, then a drop of $I(8.0)$ across the second. KVL: $12 - 4.0I - 8.0I = 0$, so $12 = 12I$, giving $I = 1.0$ A.

Worked Example 27.2

Problem: At a node, three branches meet. Branch A carries 3.0 A into the node. Branch B carries 5.0 A into the node. Branch C leaves the node. Find the current in branch C.

Solution: By KCL, total current in equals total current out: $3.0 + 5.0 = I_C$, so $I_C = 8.0$ A, leaving the node.

Common Misconception

A very common early error is applying KVL correctly in form but inconsistently in sign — for instance, treating a resistor's voltage as a drop when traversing it in one direction during one part of a solution, and quietly treating the same resistor's voltage as a rise later in the same solution, usually because the assumed current direction was forgotten partway through. Before writing down a KVL equation, it helps to mark every assumed current direction and every resulting + / – polarity directly on the circuit diagram first, and then read the diagram rather than trying to hold the whole sign convention in your head while writing the equation.

Practice Problems

- At a node, currents of 2.0 A and 6.0 A flow in, and a single branch carries current out. Find that branch's current.
- A single loop has a 9.0 V source in series with three resistors: 2.0 Ω , 3.0 Ω , and 4.0 Ω . Find the loop current.
- In the circuit of the previous problem, find the voltage across the 4.0 Ω resistor.
- Explain, in terms of charge conservation, why KCL would be violated if current could accumulate indefinitely at an ordinary circuit node.

Chapter 28 — Equivalent Resistance and Dividers

28.1 Series and Parallel, Revisited

Part I introduced the series and parallel combination rules for resistors: resistances add directly in series, and combine by reciprocal sum in parallel. This chapter treats those rules as tools for simplifying entire networks, not just pairs of resistors, and develops the two divider circuits — voltage and current — that follow directly from them and that appear, in some form, inside an enormous fraction of practical circuits.

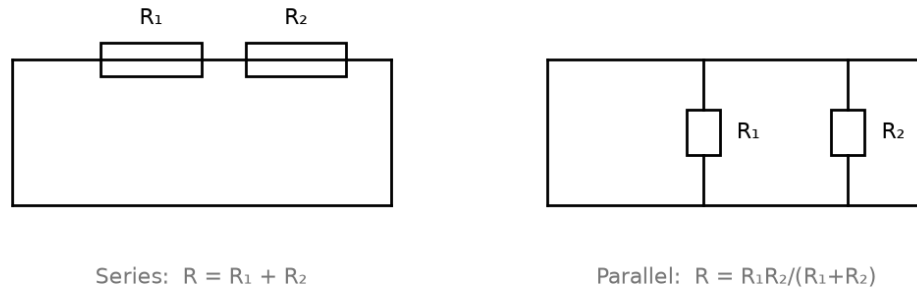


Figure 28.1 — Series and parallel resistor combinations, with their equivalent-resistance formulas.

$$R_{\text{series}} = R_1 + R_2 + \dots, \quad 1/R_{\text{parallel}} = 1/R_1 + 1/R_2 + \dots$$

28.2 Reducing a Network by Combination

Many practical circuits, though they may look complicated at first glance, can be fully solved by repeatedly identifying resistors that are purely in series or purely in parallel, combining them into a single equivalent resistor, and redrawing the (now simpler) circuit — repeating until only a single equivalent resistance remains between the source terminals. This is not a universal method (Chapters 28 and 29 introduce the systematic techniques needed once a network cannot be reduced this way, typically because of a “bridge” configuration with no purely series or parallel path), but it is the fastest method whenever it applies, and recognizing when it applies is itself a skill worth developing before reaching for heavier machinery.

28.3 The Voltage Divider

Two resistors in series across a voltage source V split that voltage in direct proportion to their resistances, since the same current flows through both:

$$V_1 = V \cdot R_1 / (R_1 + R_2), \quad V_2 = V \cdot R_2 / (R_1 + R_2)$$

The voltage divider generalizes immediately to any number of series resistors: the voltage across any one of them is the source voltage times that resistor's share of the total series resistance. It is worth noting explicitly what the voltage divider assumes: that whatever draws off the divided voltage (a downstream load) draws negligible current compared to the current already flowing through the divider itself. If that assumption fails — if the load draws a comparable or larger current — the simple divider formula no longer gives the correct output voltage, and the load's own resistance must be folded into the calculation as a resistor in parallel with whichever divider resistor it is connected across.

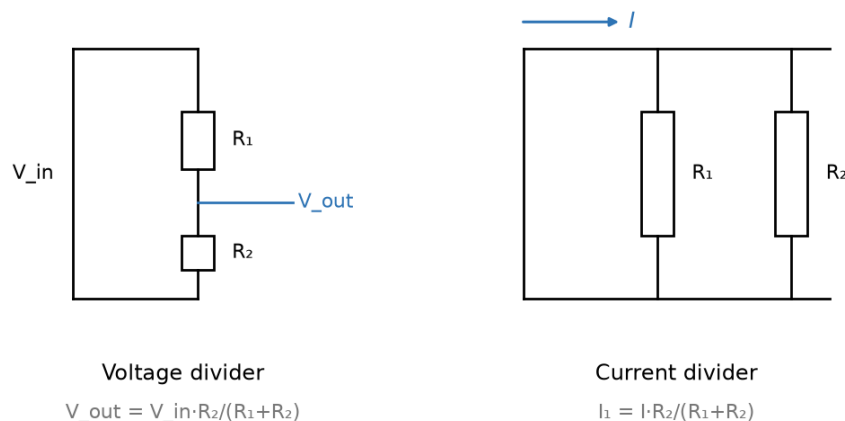


Figure 28.2 — Voltage and current dividers, with their governing formulas.

28.4 The Current Divider

Two resistors in parallel across a current source (or a branch carrying a known total current) split that current inversely in proportion to their resistances — the larger resistor carries the smaller share, since current preferentially takes the path of least resistance:

$$I_1 = I \cdot R_2 / (R_1 + R_2), \quad I_2 = I \cdot R_1 / (R_1 + R_2)$$

Notice that each resistor's current formula uses the other resistor's value in the numerator — the reverse of the voltage divider's pattern, where each resistor's voltage formula uses its own value in the numerator. This inversion is easy to state and easy to forget under exam pressure; rederiving it once from Ohm's law and KCL (each resistor sees the same voltage $V = IR_{\text{parallel}}$, so $I_1 = V/R_1$ and $I_2 = V/R_2$, and substituting R_{parallel} confirms the pattern above) is a more reliable way to keep it straight than memorizing it as an isolated fact.

Key Idea

Voltage dividers use each resistor's own value; current dividers use the other resistor's value. This asymmetry falls directly out of Ohm's law applied to series versus parallel configurations, and rederiving it once is more durable than memorizing the two formulas as unrelated facts.

Worked Example 28.1

Problem: A 15 V source drives a series combination of a 5.0 k Ω and a 10 k Ω resistor. Find the voltage across the 10 k Ω resistor.

Solution: $V_{10k} = 15 \times 10 / (5.0 + 10) = 15 \times 10 / 15 = 10$ V.

Worked Example 28.2

Problem: A current source supplies 12 mA to a parallel combination of a 2.0 k Ω and a 6.0 k Ω resistor. Find the current through the 2.0 k Ω resistor.

Solution: Using the current-divider formula (note the other resistor's value appears in the numerator): $I_{2k} = I \times R_{6k} / (R_{2k} + R_{6k}) = 12 \times 6.0 / (2.0 + 6.0) = 12 \times 0.75 = 9.0$ mA. Check: the smaller resistor should carry the larger share of the current, and 9.0 mA out of 12 mA total is indeed the larger share, confirming the arithmetic direction is right.

Common Misconception

It is a common shortcut error, once a network contains more than two resistors, to attempt to identify series and parallel groupings by eye without carefully tracing which nodes each resistor actually connects — two resistors that merely look adjacent on a drawn diagram are not necessarily in series or parallel with each other in the sense these

rules require. A resistor is in series with another only if they share exactly one node with nothing else connected there (so the same current is forced through both), and two resistors are in parallel only if both of their endpoints connect to the same pair of nodes. When a network's diagram is drawn with resistors at odd angles or overlapping, redrawing it schematically, node by node, before attempting any combination is far more reliable than trying to judge series/parallel relationships from the picture's layout alone.

Practice Problems

- A 20 V source drives a series combination of a 3.0 k Ω , 5.0 k Ω , and 2.0 k Ω resistor. Find the voltage across the 5.0 k Ω resistor.
- A current source supplies 30 mA to two parallel resistors, 4.0 k Ω and 12 k Ω . Find the current through each.
- Reduce the following network to a single equivalent resistance: a 6.0 Ω resistor in series with the parallel combination of a 4.0 Ω and a 12 Ω resistor.
- A voltage divider built from a 1.0 k Ω and a 9.0 k Ω resistor across a 10 V supply is loaded by a 9.0 k Ω resistor connected across the 9.0 k Ω divider resistor. Find the actual output voltage, and compare it to the unloaded divider prediction.

Chapter 29 — Nodal Analysis

29.1 The Idea Behind Nodal Analysis

Series/parallel reduction (Chapter 28) fails as soon as a circuit contains a configuration where no two resistors are purely in series or purely in parallel with each other — a bridge circuit is the classic example. Nodal analysis is the first of two systematic methods (mesh analysis, Chapter 30, is the second) that handles any resistive circuit whatsoever, however tangled its topology, by turning the circuit into a system of simultaneous algebraic equations solved once, together, rather than reduced step by step.

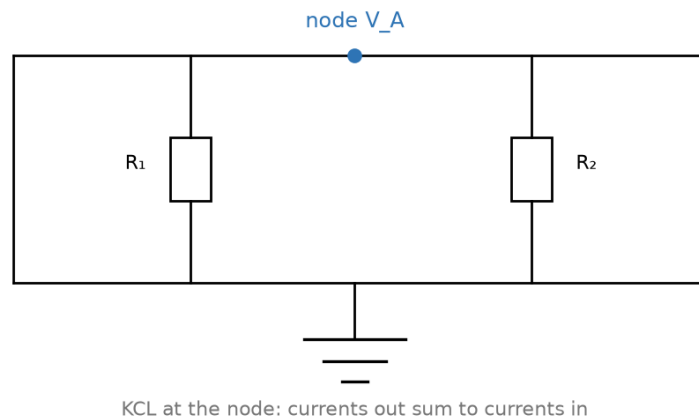


Figure 29.1 — A node voltage and the branch currents leaving it: KCL sets currents out equal to currents in.

The method treats every node's voltage (relative to a chosen reference node) as an unknown, writes one KCL equation per non-reference node, and solves the resulting system.

29.2 Choosing a Reference Node

Since only voltage differences are ever physically meaningful (Chapter 5), one node in the circuit is designated the reference node, arbitrarily assigned a voltage of exactly 0 V, usually chosen to be the node with the most connections (often literally labeled “ground” on a circuit diagram). Every other node's voltage is then understood to mean its voltage relative to this reference.

29.3 Writing the Node Equations

At every non-reference node, KCL is applied by expressing every branch current leaving that node in terms of the node voltages at its two ends and the resistance of that branch (using Ohm's law, $I = V/R$, applied across each resistor), and setting the sum of these outgoing currents equal to any current injected into the node by an independent source:

$$\sum (V_{\text{node}} - V_{\text{neighbor}}) / R_{\text{branch}} = I_{\text{injected}}$$

Repeating this for every non-reference node produces a system of linear equations — one equation per unknown node voltage — solvable by substitution, elimination, or matrix methods for larger networks.

29.4 Handling Voltage Sources: The Supernode

An independent voltage source connected directly between two non-reference nodes complicates the ordinary node equation, since the current through a voltage source cannot be written directly in terms of a single resistance the way a resistor's current can. The standard technique is the supernode: the two nodes on either side of such a source are combined into a single “supernode,” for which one KCL equation is written treating the combined current leaving both nodes together, supplemented by a second equation simply stating the known voltage difference enforced by the source itself. This pair of equations — one KCL equation for the supernode as a whole, one constraint equation from the source — replaces the two individual node equations that would otherwise be needed.

Key Idea

Nodal analysis produces exactly one equation per unknown node voltage — except at a supernode, which still produces exactly two equations for its two node voltages, just not in the ordinary single-node form. Getting the equation count right before attempting to solve is the fastest way to catch a setup error before it propagates into a wrong final answer.

Worked Example 29.1

Problem: A circuit has two non-reference nodes, A and B. A 10 A current source injects current into node A. A 2.0 Ω resistor connects node A to ground. A 4.0 Ω resistor connects node A to node B. A 5.0 Ω resistor connects node B to ground. Find V_A and V_B .

Solution: Node A: current in (10 A) equals current out through the two resistors leaving A: $10 = V_A/2.0 + (V_A - V_B)/4.0$. Node B: current leaving B through its two resistors sums to zero (no source directly at B): $(V_B - V_A)/4.0 + V_B/5.0 = 0$. Multiplying the first equation by 4.0: $40 = 2V_A + (V_A - V_B)$, so $40 = 3V_A - V_B$. Multiplying the second by 20: $5(V_B - V_A) + 4V_B = 0$, so $9V_B = 5V_A$, giving $V_B = (5/9)V_A$. Substituting: $40 = 3V_A - (5/9)V_A = (22/9)V_A$, so $V_A = 360/22 \approx 16.36$ V, and $V_B = (5/9)(16.36) \approx 9.09$ V.

29.5 Nodal Analysis for Larger Networks

For a network with many non-reference nodes, writing and solving the system of node equations by hand becomes impractical past three or four unknowns, and in practice the equations are instead assembled into a matrix equation, $GV = I$, where G is a conductance matrix built directly from the branch conductances ($1/R$ for each resistor), V is the vector of unknown node voltages, and I is a vector of known injected currents. This matrix, for networks without supernodes, has a simple, mechanical construction rule: each diagonal entry is the sum of conductances connected to that node, and each off-diagonal entry is the negative of the conductance directly connecting the two corresponding nodes. Software tools used throughout the rest of this series (and throughout real engineering practice) build and solve exactly this matrix automatically for networks with hundreds or thousands of nodes — but the underlying equations being solved are never anything more than the same KCL statement introduced in this chapter, applied node by node.

Worked Example 29.2

Problem: A single non-reference node has two resistors, 4.0 Ω and 8.0 Ω , both connecting it to ground, and a 12 A current source injecting current into it. A third resistor, 2.0 Ω , connects this node to a second non-reference node, which itself has only a 6.0 Ω resistor to ground (no other connections). Find both node voltages.

Solution: Let node A be the first node and node B the second. Node A: $12 = V_A/4.0 + V_A/8.0 + (V_A - V_B)/2.0$. Node B: $(V_B - V_A)/2.0 + V_B/6.0 = 0$. Multiplying the node A equation by 8.0: $96 = 2V_A + V_A + 4(V_A - V_B) = 7V_A - 4V_B$. Multiplying the node B equation by 6.0: $3(V_B - V_A) + V_B = 0$, so $4V_B = 3V_A$, giving $V_B = 0.75V_A$. Substituting: $96 = 7V_A - 4(0.75V_A) = 7V_A - 3V_A = 4V_A$, so $V_A = 24$ V, and $V_B = 0.75(24) = 18$ V.

Common Misconception

A frequent setup error is writing a node's KCL equation with inconsistent current directions — for instance, treating $(V_A - V_B)/R$ as current leaving node A toward B in one term, while elsewhere in the same equation treating

current from a neighboring node as if it were also automatically leaving A. The safest habit is to fix, once and for all before writing anything down, the convention that all resistor currents at a node are written as leaving that node (using the node's own voltage minus its neighbor's, divided by the connecting resistance), with any current source term signed according to whether it injects into or draws out of the node — and then to apply that same convention uniformly to every node equation in the circuit.

Practice Problems

- A single non-reference node has a 6 A current source injecting current into it, and two resistors — $3.0\ \Omega$ and $6.0\ \Omega$ — connecting it to ground. Find the node voltage.
- Write (but do not necessarily solve) the node equations for a circuit with three non-reference nodes, where node 1 has a 2 A source injected, a $4.0\ \Omega$ resistor to ground, and a $2.0\ \Omega$ resistor to node 2; node 2 has a $5.0\ \Omega$ resistor to ground and a $3.0\ \Omega$ resistor to node 3; node 3 has an $8.0\ \Omega$ resistor to ground.
- Explain, in your own words, why an independent voltage source connected between two non-reference nodes requires the supernode technique rather than an ordinary node equation.

Chapter 30 — Mesh Analysis

30.1 The Idea Behind Mesh Analysis

Mesh analysis is nodal analysis's mirror image: rather than treating node voltages as the unknowns and applying KCL, it treats loop (“mesh”) currents as the unknowns and applies KVL around each mesh. A mesh, specifically, is a loop that contains no other loop within it — the smallest possible closed paths in a planar circuit diagram, which is why mesh analysis, unlike nodal analysis, applies only to planar circuits (those that can be drawn on a flat page with no branches crossing over one another).

30.2 Assigning Mesh Currents

Each mesh is assigned its own circulating current, conventionally taken clockwise by convention (though any consistent choice works), understood as a fictitious current that flows only around that one mesh's own loop. Where two meshes share a branch, the actual physical current in that shared branch is the difference (or sum, depending on the relative directions) of the two meshes' currents — this is the key bookkeeping idea that makes mesh analysis work, and it is worth drawing out explicitly on a diagram before writing any KVL equation.

30.3 Writing the Mesh Equations

For each mesh, KVL is applied by traversing that mesh's loop in the assumed direction and summing the voltage drops across every element encountered, expressing each resistor's voltage drop in terms of the net current actually flowing through it (which may involve a neighboring mesh's current, for a shared branch), and setting independent sources' contributions on the other side of the equation:

$$\sum (I_{\text{mesh}} - I_{\text{neighbor}}) R_{\text{shared}} + \sum I_{\text{mesh}} R_{\text{own}} = \sum V_{\text{sources}}$$

As with nodal analysis, a current source shared between two meshes complicates the ordinary equation — the analogous fix here is the supermesh: the two meshes on either side of a shared current source are combined into a single supermesh, treated with one KVL equation around the combined outer loop, supplemented by a constraint equation stating the known current relationship the source enforces between the two mesh currents.

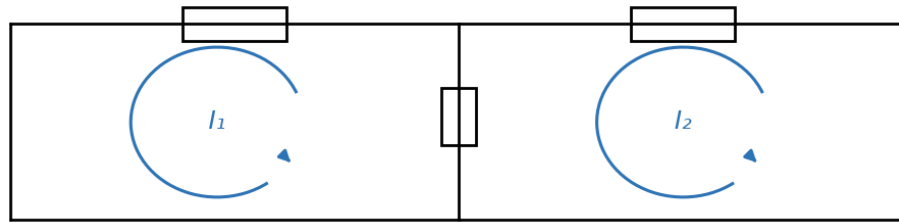
Key Idea

Nodal analysis and mesh analysis are dual methods. Nodal analysis is usually the more efficient choice when a circuit has fewer nodes than meshes (and is especially natural when current sources are involved, since they slot directly into KCL); mesh analysis is usually more efficient when a circuit has fewer meshes than nodes (and is especially natural when voltage sources dominate, since they slot directly into KVL). Both methods, correctly applied, must produce the same final answer for any quantity computed both ways — this agreement is one of the most useful checks available for catching an arithmetic or setup error.

Worked Example 30.1

Problem: A circuit has two meshes. Mesh 1 (current I_1 , clockwise) contains a 10 V source and a $2.0\ \Omega$ resistor, and shares a $4.0\ \Omega$ resistor with mesh 2. Mesh 2 (current I_2 , clockwise) contains the shared $4.0\ \Omega$ resistor and its own $6.0\ \Omega$ resistor, with no independent source. Find I_1 and I_2 .

Solution: Mesh 1 KVL (traversing clockwise, starting just after the source): $10 = 2.0I_1 + 4.0(I_1 - I_2)$, giving $10 = 6.0I_1 - 4.0I_2$. Mesh 2 KVL: the shared resistor's drop, from mesh 2's perspective, is $4.0(I_2 - I_1)$, plus the $6.0\ \Omega$ resistor's own drop $6.0I_2$, with no source: $0 = 4.0(I_2 - I_1) + 6.0I_2$, giving $0 = -4.0I_1 + 10I_2$, so $I_2 = 0.4I_1$. Substituting into the first equation: $10 = 6.0I_1 - 4.0(0.4I_1) = 6.0I_1 - 1.6I_1 = 4.4I_1$, giving $I_1 \approx 2.27\ \text{A}$, and $I_2 = 0.4(2.27) \approx 0.91\ \text{A}$. The actual current through the shared $4.0\ \Omega$ resistor is $I_1 - I_2 \approx 1.36\ \text{A}$.



the shared branch carries $I_1 - I_2$

Figure 30.1 — Two mesh currents sharing a branch: the branch carries their difference.

30.4 Source Transformations

A practical bridge between nodal and mesh analysis, worth knowing independently of either: any real voltage source in series with a resistance can be replaced by an equivalent current source in parallel with the same resistance, and vice versa, without changing anything about how the rest of the circuit responds — this is exactly the same Thevenin/Norton relationship developed fully in Chapter 31, applied here as a mid-analysis simplification tool. A source transformation can convert a voltage source awkwardly placed for mesh analysis into a current source that slots naturally into a nodal analysis instead (or the reverse), and it is often faster, for a circuit with a mix of source types, to transform every source to whichever type matches the analysis method you intend to use, before writing a single equation.

Worked Example 30.2

Problem: A single mesh contains a 15 V source in series with a $3.0\ \Omega$ resistor, in series with a $5.0\ \Omega$ resistor and a 2.0 A current source, where the current source and the $5.0\ \Omega$ resistor are in parallel with each other, forming the mesh's other branch. Use a source transformation on the voltage source and its series resistor to simplify the circuit before analyzing it.

Solution: The 15 V source in series with $3.0\ \Omega$ transforms to an equivalent current source of $15/3.0 = 5.0$ A in parallel with the same $3.0\ \Omega$ resistor. The circuit is now two current sources (5.0 A and 2.0 A) with two parallel resistors ($3.0\ \Omega$ and $5.0\ \Omega$) — depending on the sources' relative polarities, these current sources either add or subtract, and the parallel resistors combine as $(3.0)(5.0)/(3.0+5.0) = 1.875\ \Omega$, reducing the entire problem to simple parallel-circuit reasoning without ever writing a mesh equation at all.

Common Misconception

Students who have just learned mesh analysis sometimes forget that a resistor shared between two meshes appears in *both* mesh equations, each time using the difference of the two meshes' currents (with the sign of that difference flipped between the two equations, since each mesh views the same physical current from the opposite side). Omitting the neighboring mesh's contribution to a shared resistor's voltage drop — treating it as if it belonged to only one mesh — is one of the most common setup errors in an otherwise correctly structured mesh analysis, and it is worth double-checking every shared branch specifically for this before solving the resulting system.

Practice Problems

- A single-mesh circuit contains a 20 V source, a $3.0\ \Omega$ resistor, and a $7.0\ \Omega$ resistor, all in series. Find the mesh current.
- Two meshes share a $5.0\ \Omega$ resistor. Mesh 1 has its own $3.0\ \Omega$ resistor and an 8.0 V source; mesh 2 has its own $2.0\ \Omega$ resistor and no source. Set up (but need not fully solve) the two mesh equations.

- Explain why mesh analysis, unlike nodal analysis, cannot be applied directly to a non-planar circuit.

Chapter 31 — Thevenin and Norton Equivalents

31.1 Why Equivalent Circuits Matter

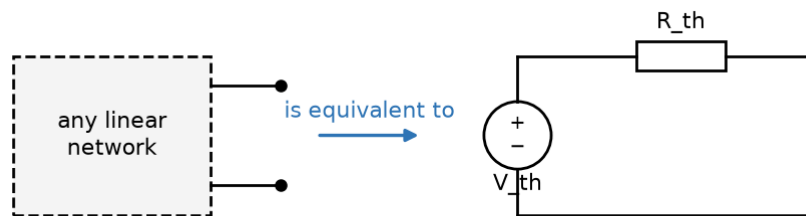
A frequent practical situation: you need to know how a complicated network of sources and resistors will behave when connected to some load — a speaker, a sensor, a motor — without needing to re-solve the entire network every time the load itself changes. Thevenin's theorem provides exactly this: any network of linear resistors and independent sources, viewed from any pair of terminals, behaves — as far as anything connected to those two terminals can tell — exactly like a single ideal voltage source in series with a single resistor.

31.2 The Thevenin Equivalent

The Thevenin equivalent of a network, as seen from a pair of terminals, consists of a voltage source V_{Th} in series with a resistance R_{Th} , found as follows:

1. V_{Th} is the open-circuit voltage at the terminals — the voltage you would measure across them with nothing connected (an open circuit), found using any of the analysis techniques from Chapters 27–30.
2. R_{Th} is the equivalent resistance seen looking back into the terminals with every independent source deactivated — voltage sources replaced by short circuits (0 V, since an ideal voltage source has zero internal resistance and forces its terminal voltage regardless of current), and current sources replaced by open circuits (since an ideal current source forces its current regardless of voltage, which an open circuit — infinite resistance — trivially satisfies at zero current).

Once V_{Th} and R_{Th} are known, any load connected to the original terminals sees exactly the same voltage and current it would see connected to this much simpler two-element equivalent — a fact that is not obvious on its own terms, but follows rigorously from the superposition property of linear circuits (developed fully in Chapter 32).



Thevenin equivalent: one source, one series resistance

Figure 31.1 — Any linear network, viewed from a pair of terminals, is equivalent to one source and one series resistance.

31.3 The Norton Equivalent

The Norton equivalent is the source-transformed twin of the Thevenin equivalent: a current source I_N in parallel with the same resistance R_{Th} (often written R_N , though it is numerically identical to R_{Th} for the same network). I_N is the short-circuit current — the current that would flow if the two terminals were directly connected together — and it relates to the Thevenin quantities by Ohm's law applied to the equivalent circuit itself:

$$I_N = V_{Th} / R_{Th}$$

Thevenin and Norton equivalents are always interchangeable via this relationship, and which one is more convenient to compute or to use downstream depends entirely on the specific problem — a network dominated by voltage sources often yields V_{Th} more directly; a network dominated by current sources often yields I_N more directly.

Key Idea

R_{Th} is always found the same way, regardless of which equivalent you ultimately want: deactivate every independent source (short voltage sources, open current sources) and find the resulting resistance looking into the terminals. This single resistance value serves both the Thevenin and Norton equivalents identically.

31.4 Maximum Power Transfer

A Thevenin equivalent makes a classical and practically important question easy to answer: given a fixed source with Thevenin resistance R_{Th} , what load resistance R_L draws the maximum possible power from it? The answer, the maximum power transfer theorem, is $R_L = R_{Th}$ — the load resistance is matched exactly to the source's internal (Thevenin) resistance. At this matched condition, exactly half the total power generated by the source is delivered to the load, and the other half is dissipated internally within R_{Th} ; this 50% efficiency ceiling is the price of maximizing delivered power rather than maximizing efficiency, and real power-delivery systems (where efficiency, not raw power transfer, is usually the goal) deliberately avoid matched-impedance operation for exactly this reason — a distinction worth keeping in mind once this series reaches power systems, where the two goals genuinely diverge.

Worked Example 31.1

Problem: A network, viewed from terminals A-B, has an open-circuit voltage of 9.0 V and a short-circuit current of 3.0 A. Find its Thevenin and Norton equivalents, and the load resistance that draws maximum power.

Solution: $V_{Th} = 9.0$ V (given directly as the open-circuit voltage). $R_{Th} = V_{Th}/I_N = 9.0/3.0 = 3.0\ \Omega$ (from the relationship between the open-circuit and short-circuit conditions). Norton equivalent: $I_N = 3.0$ A in parallel with $R_N = 3.0\ \Omega$. Maximum power transfer occurs when $R_L = R_{Th} = 3.0\ \Omega$, at which point the load voltage is exactly half of V_{Th} (by the voltage-divider relationship between R_{Th} and a matched R_L), i.e., 4.5 V, delivering $P = V^2/R_L = (4.5)^2/3.0 = 6.75$ W to the load.

31.5 Practical Measurement of Thevenin Parameters

The derivation above assumes you already know the network's internal composition well enough to compute V_{Th} and R_{Th} directly. In practice — testing an unfamiliar circuit board, or characterizing a physical device such as a battery or a sensor — both quantities can be measured directly, without any knowledge of what is inside the black box. V_{Th} is measured directly as the open-circuit voltage at the terminals of interest, using a voltmeter (whose own internal resistance is, ideally, so large that it draws negligible current and does not disturb the measurement). R_{Th} can then be found either from a separate short-circuit current measurement, via $R_{Th} = V_{Th}/I_N$ as in Section 31.3, or — often more practically, since deliberately short-circuiting a real device can be risky — by connecting a known load resistance, measuring the resulting terminal voltage, and solving the resulting voltage-divider relationship for R_{Th} . This experimental determination of a network's Thevenin equivalent is exactly how a real battery's internal resistance is characterized in practice, and it is why a battery's terminal voltage sags under heavy load: the load resistance and the battery's own (generally small, but nonzero) internal resistance form exactly the voltage-divider situation this book has already developed the tools to analyze.

Worked Example 31.2

Problem: A battery has an open-circuit terminal voltage of 9.0 V. When a $20\ \Omega$ load is connected, the terminal voltage drops to 8.0 V. Find the battery's internal (Thevenin) resistance.

Solution: The load and the battery's internal resistance form a voltage divider: $V_{\text{load}} = V_{\text{Th}} \times R_{\text{load}} / (R_{\text{load}} + R_{\text{Th}})$. Substituting known values: $8.0 = 9.0 \times 20 / (20 + R_{\text{Th}})$. Solving: $(20 + R_{\text{Th}}) = 9.0 \times 20 / 8.0 = 22.5$, so $R_{\text{Th}} = 2.5 \, \Omega$ — a small but nonzero internal resistance, entirely typical of a real battery, and the direct physical reason its terminal voltage sags noticeably under a heavy enough load.

Common Misconception

It is tempting to think maximum power transfer ($R_{\text{L}} = R_{\text{Th}}$) is always the design goal, since “maximum power” sounds unambiguously desirable. In most real power-delivery contexts it is the opposite of what you want: at the matched condition, the source itself dissipates exactly as much power internally as the load receives, a 50% efficiency ceiling that would be unacceptable for a power grid (where R_{Th} is deliberately kept as small as possible relative to the load, so that nearly all delivered power reaches the load rather than being wasted as heat inside the source). Maximum power transfer matters most in signal and communication contexts — antenna matching, for instance — where the absolute power levels are small and efficiency is a secondary concern to simply getting as strong a signal through as possible.

Practice Problems

- A network has $V_{\text{Th}} = 12 \, \text{V}$ and $R_{\text{Th}} = 4.0 \, \Omega$. Find its Norton equivalent.
- A network has a Norton equivalent of $I_{\text{N}} = 2.0 \, \text{A}$ in parallel with $R_{\text{N}} = 6.0 \, \Omega$. Find its open-circuit voltage.
- For the network in the previous problem, find the load resistance for maximum power transfer, and the power delivered to that load.
- Explain briefly why deactivating an independent voltage source means replacing it with a short circuit, referring to what an ideal voltage source's own defining behavior is.

Chapter 32 — Superposition and Linearity

32.1 Linear Circuits

A circuit is linear if it is built entirely from linear elements: resistors ($V = IR$), capacitors and inductors (whose defining relationships, from Part I, are likewise linear), and independent sources. Diodes, transistors, and other devices met later in this series are generally not linear, and none of the techniques in this chapter apply to them directly without first restricting attention to a small enough operating range that their behavior can be approximated as linear — a technique this series returns to explicitly once those devices are introduced.

32.2 The Superposition Theorem

In any linear circuit containing multiple independent sources, the voltage or current at any point can be found by considering each independent source separately — deactivating all other independent sources exactly as in the Thevenin-equivalent method of Chapter 31 (voltage sources shorted, current sources opened) — solving the resulting simpler circuit for the contribution of that one source alone, and then algebraically summing the individual contributions from every source to get the total.

Key Idea

Superposition applies to voltages and currents, never to power. Power depends on voltage and current multiplied together ($P = IV$ or $P = I^2R$), and squaring or multiplying is not a linear operation — the power delivered by two sources acting together is generally not the sum of the powers each would deliver acting alone. Superposition is a technique for finding voltages and currents; power in a multi-source circuit must always be computed afterward, from the correctly superposed total voltage or current, never by adding individually-computed powers.

32.3 Why Superposition Works

Superposition is not a special trick specific to circuits — it is the same principle already met twice in this series: as the statement that Coulomb forces and electric fields from multiple charges simply add (Chapters 2 and 3), and it holds here for exactly the same underlying reason. Every equation used to solve a linear circuit — Ohm's law, KCL, KVL, the nodal and mesh equations built from them — is linear in the sources appearing in it, meaning if you scale every source by some factor, every resulting voltage and current scales by that same factor, and if you have two separate sets of sources active simultaneously, the resulting voltages and currents are exactly the sum of what each set would produce acting alone. This is a mathematical property of linear systems in general, and circuit theory is simply one particular, especially useful, arena in which it applies.

32.4 When Superposition Is (and Is Not) Worth Using

Superposition is rarely the fastest method for finding a single specific quantity in a circuit with many sources — nodal or mesh analysis, solved once for the whole circuit, is usually less work than solving the circuit repeatedly, once per source. Superposition earns its keep in two other situations: when you specifically want to know each source's individual contribution to some quantity (useful for understanding which source dominates a circuit's behavior, or for noise analysis later in this series, where different sources of unwanted signal need to be tracked separately), and as a conceptual tool for proving other results — the Thevenin equivalent's validity, for instance, rests directly on superposition.

Worked Example 32.1

Problem: A resistor R carries current contributed by two independent sources. Acting alone (with the other source deactivated), a 10 V source produces 2.0 A through R . Acting alone, a 4 A current source produces 1.5 A through R in the same direction. Find the total current through R with both sources active.

Solution: By superposition, the total current is simply the sum of the two individual contributions (since both were found to flow in the same direction): $I_{\text{total}} = 2.0 + 1.5 = 3.5$ A. Had the two contributions been found to flow in opposite directions, they would instead have been subtracted, with the sign of each contribution (relative to a chosen reference direction) tracked explicitly throughout.

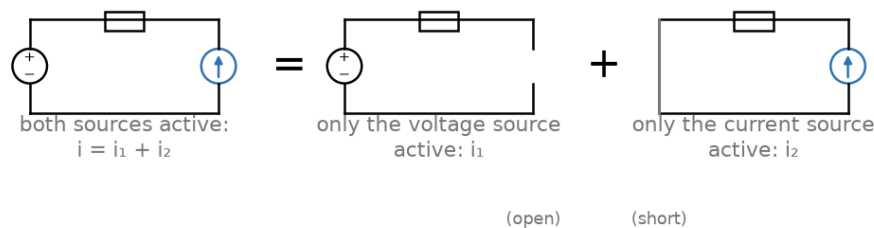


Figure 32.1 — Superposition: the response with both sources active equals the sum of each source acting alone.

32.5 Dependent Sources and Superposition

This book has, so far, only used independent sources — voltage or current sources whose value is fixed, set from outside the circuit. Many practical circuits, and essentially every transistor small-signal model this series will introduce in its semiconductor-devices volume, also contain dependent (controlled) sources, whose value is instead proportional to some other voltage or current elsewhere in the same circuit. Superposition's deactivation rule applies only to independent sources — a dependent source must never be deactivated when superposing, since doing so would discard the very relationship (to whatever voltage or current controls it) that defines it, and that relationship generally still involves contributions from every independent source active in the problem. In practice, this means a circuit with dependent sources is solved by superposing over the independent sources only, while the dependent source's own defining equation is carried through, unmodified, in every one of the individual sub-problems.

Worked Example 32.2

Problem: Acting alone, a 5.0 V independent source produces a current of 1.0 A through a resistor R , and this same current, through a dependent source's controlling relationship, produces a dependent voltage of 2.0 V elsewhere in the circuit (dependent source value = $2 \times$ the controlling current, in volts per ampere). A second independent source, a 2.0 A current source, acting alone (with the dependent source's relationship still active, only the independent 5.0 V source deactivated), produces 0.4 A through the same resistor R . Find the total current through R with both independent sources active, and the resulting dependent source value.

Solution: Superposing over the two independent sources only: total current through $R = 1.0 + 0.4 = 1.4$ A. The dependent source's value, being defined as $2 \times$ the controlling current (which is exactly this total current, since the dependent source's defining relationship was never itself superposed away), is $2 \times 1.4 = 2.8$ V — found only after the total controlling current was already assembled from both independent sources' contributions, not by superposing the dependent source's own value directly.

Common Misconception

Because superposition works cleanly for voltage and current, it is a natural (but incorrect) next guess that it should also work for power — that the total power dissipated in a resistor, with multiple sources active, is the sum of the powers each source would cause individually. It is not: power is proportional to the square of voltage or current, and $(a + b)^2 \neq a^2 + b^2$ in general (there is a cross term, $2ab$, left over). The correct procedure is always to superpose the voltages or currents first, add them algebraically to get the total, and only then compute power from that final total value — never to compute power source-by-source and add the results.

Practice Problems

- Acting alone, a voltage source produces 3.0 V across a resistor. Acting alone, a current source produces -1.0 V across the same resistor (opposite polarity). Find the total voltage with both sources active.
- Explain, using the worked example's numbers reinterpreted, why it would be incorrect to compute the power dissipated in R as the sum of the power each source alone would cause.
- A linear circuit contains three independent sources. How many simplified circuits must be solved to apply superposition fully?

Chapter 33 — Capacitors and Inductors as Circuit Elements

33.1 The v-i Relationships, Restated for Circuit Use

Part I defined capacitance and inductance from field physics. For circuit analysis, what matters is each element's voltage-current (v-i) relationship, restated here in the form used throughout the rest of this book:

$$\text{Capacitor: } i = C \, dv/dt \quad \text{Inductor: } v = L \, di/dt$$

Both relationships are fundamentally different in character from a resistor's $V = IR$: a resistor's voltage and current are related instantaneously and algebraically, while a capacitor's current and an inductor's voltage depend on the rate of change of the other quantity, making both elements inherently dynamic — their behavior depends on the circuit's history, not just its instantaneous state, which is precisely why they are called dynamic or energy-storage elements, in contrast to a resistor's purely dissipative behavior.

33.2 Continuity Conditions

Two facts follow directly from the v-i relationships above and are used constantly in transient analysis (Chapter 34): a capacitor's voltage cannot change instantaneously (an instantaneous voltage change would require $i = C \, dv/dt$ to be infinite, which no real source can supply), and an inductor's current cannot change instantaneously (for the parallel reason that an instantaneous current change would require $v = L \, di/dt$ to be infinite).

Key Idea

Capacitor voltage and inductor current are the two quantities that must remain continuous across any switching event. Everything else in a circuit — resistor voltages and currents, capacitor current, inductor voltage — can and generally does change abruptly the instant a switch operates. This single fact is the starting point for every transient-analysis problem in the next chapter.

33.3 Energy Storage, Revisited

Part I established that a capacitor stores energy $U = \frac{1}{2}Cv^2$ and, in its closing chapters, that an inductor stores energy $U = \frac{1}{2}Li^2$. Both expressions reappear constantly in circuit-level energy accounting, and both make explicit, in circuit terms, why the two continuity conditions above must hold: an instantaneous jump in capacitor voltage or inductor current would correspond to an instantaneous jump in stored energy, which would require an infinite instantaneous power delivery — physically impossible for any real, finite source.



Figure 33.1 — The defining v-i relationships for a capacitor and an inductor.

33.4 Series and Parallel Combinations, Revisited

Part I introduced the capacitor combination rules (parallel adds, series reciprocal-sums) alongside the parallel-plate capacitor derivation. Inductors, it is worth noting explicitly here, combine exactly like resistors rather than like capacitors — series inductances add directly, and parallel inductances combine by reciprocal sum:

$$L_{\text{series}} = L_1 + L_2 + \dots, \quad 1/L_{\text{parallel}} = 1/L_1 + 1/L_2 + \dots$$

This gives a useful, if slightly untidy, memory anchor across all three passive elements: resistors and inductors share one combination pattern (series adds, parallel reciprocal-sums), while capacitors follow the opposite pattern — a fact traceable, in each case, back to whether the element's defining relationship involves current or its time-integral (charge) on one side.

Worked Example 33.1

Problem: A 10 μF capacitor's voltage is changing at a rate of 500 V/s. Find the current through it.

Solution: $i = C \, dv/dt = (10 \times 10^{-6})(500) = 5.0 \times 10^{-3} \text{ A} = 5.0 \text{ mA}$.

Worked Example 33.2

Problem: A 3.0 H inductor and a 6.0 H inductor are connected (a) in series and (b) in parallel. Find the equivalent inductance in each case.

Solution: (a) Series: $L = 3.0 + 6.0 = 9.0 \text{ H}$. (b) Parallel: $1/L = 1/3.0 + 1/6.0 = 2/6.0 + 1/6.0 = 3/6.0$, so $L = 2.0 \text{ H}$ — following exactly the same combination pattern as resistors, not capacitors.

Common Misconception

Because a resistor's power is always positive (dissipated as heat, regardless of current direction, since $P = I^2R$ involves a squared term), it is easy to assume the same must be true of a capacitor's or inductor's instantaneous power. It is not: a capacitor or inductor's instantaneous power, $p(t) = v(t)i(t)$, is regularly negative — precisely during the portion of a cycle when the element is returning previously-stored energy back to the rest of the circuit rather than absorbing more of it. Averaged over a full cycle of steady sinusoidal operation, an ideal capacitor or inductor's net energy exchange is exactly zero (all the energy it absorbs during part of the cycle, it returns during another part), which is the seed of the “reactive power” concept developed fully in Chapter 36 — reactive power is, in a real sense, simply this back-and-forth energy exchange, tallied up and given its own name and unit.

Practice Problems

- A capacitor's voltage changes at a constant rate of 200 V/s, producing a current of 4.0 mA through it. Find its capacitance.
- An inductor carries a current changing at 10 A/s, producing a voltage of 2.0 V across it. Find its inductance.
- Two inductors, 4.0 H and 4.0 H, are connected in parallel. Find the equivalent inductance.
- Explain briefly why an inductor's current, unlike a resistor's, cannot change instantaneously when a switch in its circuit is closed.

Chapter 34 — First-Order Transients: RC and RL Circuits

34.1 What a Transient Is

A transient is the temporary response of a circuit as it moves from one steady condition to another, following a sudden change such as a switch closing or opening. The preceding chapters established the tools for steady-state resistive circuit analysis; this chapter is the first to ask what happens during the interval of change itself, and it is where the dynamic elements introduced in Chapter 33 — capacitors and inductors — finally do something a purely resistive circuit cannot: respond gradually, over time, rather than instantaneously.

34.2 The RC Circuit — Discharging

Consider a capacitor, initially charged to voltage V_0 , connected at time $t = 0$ across a resistor R (with no source), allowing it to discharge. Applying KVL and the capacitor's defining relationship $i = C \, dv/dt$ leads to a first-order differential equation whose solution is an exponential decay:

$$v(t) = V_0 e^{-t/RC}$$

The quantity $\tau = RC$, called the time constant, has units of time (seconds, when R is in ohms and C in farads) and sets the natural timescale of the decay: after one time constant, the voltage has fallen to about 37% ($1/e$) of its initial value; after five time constants, it has fallen below 1% and is conventionally treated as having reached its final (here, zero) value for most practical purposes.

34.3 The RC Circuit — Charging

If instead an initially uncharged capacitor is connected, at $t = 0$, in series with a resistor R across a DC source of voltage V_s , the capacitor voltage rises according to:

$$v(t) = V_s (1 - e^{-t/RC})$$

approaching V_s asymptotically, with the same time constant $\tau = RC$ governing how quickly it gets there: after one time constant, the capacitor has charged to about 63% of V_s ; after five time constants, it is within about 1% of fully charged.

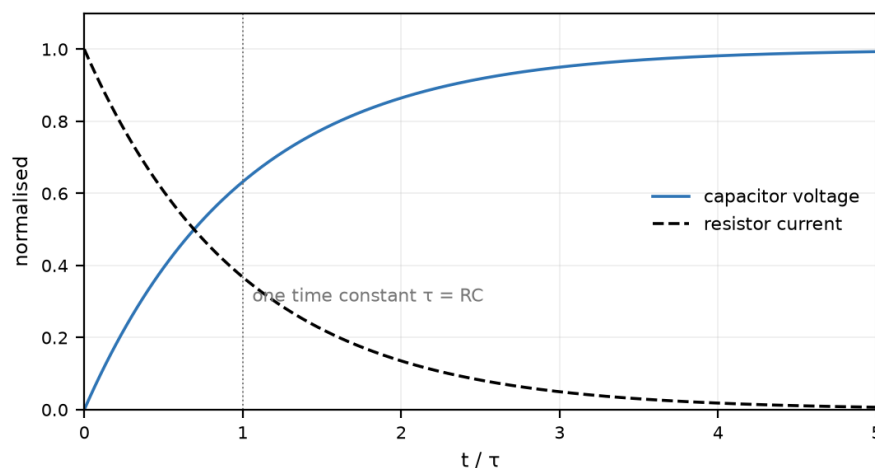


Figure 34.1 — Capacitor voltage and resistor current during RC charging, in units of the time constant.

34.4 The RL Circuit

An RL circuit behaves as the direct dual of the RC circuit, with the inductor's current — not a capacitor's voltage — being the quantity that cannot change instantaneously and that follows the exponential time course. For a series RL circuit switched onto a DC source V_s at $t = 0$, with the inductor initially carrying no current:

$$i(t) = (V_s/R) (1 - e^{-tR/L})$$

with time constant $\tau = L/R$ (units of seconds, when L is in henries and R in ohms) playing exactly the same role here that RC played for the capacitor circuit.

Key Idea

Every first-order transient follows the same universal template. Whatever the quantity of interest — capacitor voltage, inductor current, or any other voltage or current in a single-time-constant circuit — it can always be written as $x(t) = x_{\text{final}} + (x_{\text{initial}} - x_{\text{final}}) e^{-(t/\tau)}$, where x_{initial} and x_{final} are the quantity's values immediately after the switching event and infinitely long afterward, respectively, and τ is RC for a capacitive circuit or L/R for an inductive one. Memorizing this one general template is more useful than memorizing the charging and discharging formulas separately, since every specific case in this chapter is just this template with the appropriate initial value, final value, and time constant substituted in.

Worked Example 34.1

Problem: A $100\ \mu\text{F}$ capacitor, charged to $20\ \text{V}$, is discharged through a $5.0\ \text{k}\Omega$ resistor starting at $t = 0$. Find the capacitor voltage at $t = 0.50\ \text{s}$.

Solution: $\tau = RC = (5.0 \times 10^3)(100 \times 10^{-6}) = 0.50\ \text{s}$. $v(t) = V_0 e^{-(t/\tau)} = 20 e^{-(0.50/0.50)} = 20 e^{-1} \approx 7.36\ \text{V}$ — exactly the “37% of initial value” benchmark, since the elapsed time here happens to equal exactly one time constant.

Worked Example 34.2

Problem: A series RL circuit with $R = 10\ \Omega$ and $L = 2.0\ \text{H}$ is connected to a $20\ \text{V}$ source at $t = 0$, with the inductor initially carrying no current. Find the current at $t = 0.10\ \text{s}$, and the final steady-state current.

Solution: $\tau = L/R = 2.0/10 = 0.20\ \text{s}$. Final current (steady state, once $di/dt = 0$ so the inductor's voltage drop vanishes and it behaves like a plain wire): $I_{\text{final}} = V_s/R = 20/10 = 2.0\ \text{A}$. At $t = 0.10\ \text{s}$: $i(t) = I_{\text{final}}(1 - e^{-(t/\tau)}) = 2.0(1 - e^{-(0.10/0.20)}) = 2.0(1 - e^{-(0.5)}) \approx 2.0(1 - 0.607) \approx 0.79\ \text{A}$.

34.5 A Note on Second-Order Circuits

This chapter has deliberately restricted itself to first-order circuits — those containing exactly one energy-storage element (one capacitor or one inductor, possibly combined with any number of resistors that can all be reduced, by the equivalent-resistance methods of Chapter 28, to a single effective R). A circuit containing both a capacitor and an inductor together (an RLC circuit) is second-order, governed by a second-order differential equation whose solution can take three qualitatively different forms — overdamped (a sluggish, non-oscillating return to steady state, similar in character to the first-order case but slower), underdamped (an oscillating, gradually decaying return, ringing back and forth around the final value before settling), or critically damped (the fastest possible return without any oscillation, sitting exactly at the boundary between the other two behaviors). Working through the general second-order case properly requires machinery — characteristic equations, damping ratios, natural frequencies — that this book has not yet built, and it is deferred deliberately rather than treated in an abbreviated, unsatisfying way here. What this book has built, however, is not wasted: Chapter 37's resonant frequency $\omega_0 = 1/\sqrt{LC}$ is exactly the undamped natural frequency of the very same second-order RLC system being described here, found by an entirely different (sinusoidal-steady-state) route — a second confirmation, alongside the small-signal preview in Chapter 38, that the specific tools built in this book keep resurfacing, largely unchanged, in contexts this book itself does not fully develop.

Common Misconception

It is tempting to assume an inductor “blocks” current indefinitely, by loose analogy with how a capacitor blocks DC current once fully charged. In steady state (once enough time constants have elapsed that transients have died out), an inductor behaves like a plain wire — zero voltage drop, since $v = L \, di/dt$ vanishes once current is no longer changing — allowing DC current through it completely unimpeded. It is only during the transient itself, while current is actively changing, that the inductor's back-EMF has any effect; a capacitor, by contrast, does genuinely block steady DC current in its final, fully-charged steady state, the opposite long-term behavior from an inductor's.

Practice Problems

- A $50 \, \mu\text{F}$ capacitor is charged through a $2.0 \, \text{k}\Omega$ resistor from a $12 \, \text{V}$ source. Find the time constant and the capacitor voltage after one time constant has elapsed.
- An RL circuit has $R = 20 \, \Omega$ and $L = 4.0 \, \text{H}$. Find its time constant.
- For the circuit in the previous problem, connected to a $10 \, \text{V}$ source at $t = 0$ with zero initial current, find the current after two time constants have elapsed.
- Explain briefly why an inductor behaves like a plain wire (zero voltage drop) once a circuit reaches DC steady state.

Chapter 35 — Sinusoidal Steady-State Analysis

35.1 Solving AC Circuits with Phasors

With impedance defined (Chapter 23), an AC circuit at a single fixed frequency is analyzed exactly like a DC resistive circuit, with three substitutions: every resistor, capacitor, and inductor is replaced by its impedance; every sinusoidal source is replaced by its phasor; and every resulting voltage or current in the solution is itself a phasor, convertible back to a time-domain sinusoid only at the very end, using the source's known frequency ω .

35.2 Series and Parallel Impedance Combination

Impedances combine exactly like resistances — series impedances add, parallel impedances combine by reciprocal sum — with the arithmetic now performed on complex numbers rather than real ones:

$$Z_{\text{series}} = Z_1 + Z_2 + \dots, \quad 1/Z_{\text{parallel}} = 1/Z_1 + 1/Z_2 + \dots$$

The voltage and current dividers from Chapter 28 likewise carry over unchanged in form, with impedances substituted for resistances:

$$V_1 = V \cdot Z_1/(Z_1+Z_2), \quad I_1 = I \cdot Z_2/(Z_1+Z_2)$$

35.3 Nodal, Mesh, Thevenin, and Superposition in the Phasor Domain

Nodal analysis, mesh analysis, Thevenin and Norton equivalents, and superposition (Chapters 29 through 32) all carry over to phasor circuits exactly as stated, with resistances replaced by impedances throughout — the underlying laws (KCL, KVL) are statements about instantaneous charge and energy conservation that hold regardless of whether the circuit's quantities happen to be constant, transient, or sinusoidally oscillating, so nothing about the reasoning behind any of those five techniques actually depended on the circuit being purely resistive DC in the first place. This is the single biggest practical payoff of the phasor method: an engineer who has mastered Chapters 27 through 32 has, without additional conceptual work, already mastered the corresponding AC techniques — only the arithmetic (complex rather than real) is new.

Key Idea

Every systematic technique from Chapters 27–32 applies unchanged to phasor circuits. Replace every resistance with the corresponding impedance, every source with its phasor, solve using ordinary complex-number algebra, and convert the final phasor answer back to a time-domain sinusoid only as the last step.

35.4 RMS Values

A practical complication: the amplitude V_m used to define a phasor is a peak value, but most AC quantities — including household mains voltage ratings — are quoted as root-mean-square (RMS) values, defined so that an RMS voltage delivers the same average power to a resistor as a DC voltage of the same numeric value would. For a sinusoid, RMS and peak amplitude relate simply:

$$V_{\text{rms}} = V_m / \sqrt{2} \approx 0.707 V_m$$

A “120 V” household outlet in the United States, for instance, is a 120 V RMS rating, corresponding to a peak voltage of about 170 V — the actual voltage waveform swings between +170 V and –170 V every cycle, even though the quoted, practically relevant number is 120 V. Power calculations (Chapter 36) are almost always done

using RMS values directly, specifically so that AC power formulas take the same simple form as their DC counterparts.

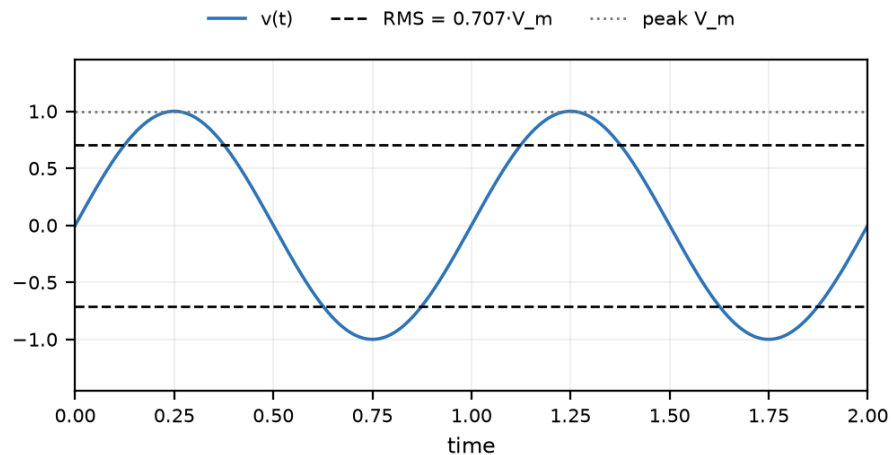


Figure 35.1 — A sinusoid's peak and RMS values.

Worked Example 35.1

Problem: A $100\angle 0^\circ$ V phasor source (peak amplitude, at some frequency ω) drives a series combination of a $30\ \Omega$ resistor and an inductor with impedance $j40\ \Omega$. Find the current phasor and the phasor voltage across the inductor.

Solution: Total impedance: $Z = 30 + j40 = 50\angle 53.1^\circ\ \Omega$ (magnitude $\sqrt{30^2 + 40^2} = 50$, angle $\arctan(40/30) \approx 53.1^\circ$). Current: $I = V/Z = 100\angle 0^\circ / 50\angle 53.1^\circ = 2.0\angle -53.1^\circ$ A. Voltage across the inductor: $V_L = I \times Z_L = (2.0\angle -53.1^\circ)(40\angle 90^\circ) = 80\angle 36.9^\circ$ V (multiplying polar-form complex numbers: multiply magnitudes, add angles).

Worked Example 35.2

Problem: A sinusoidal source has a peak voltage of 170 V. Find its RMS voltage, and identify the familiar household voltage rating this corresponds to.

Solution: $V_{\text{rms}} = V_m/\sqrt{2} = 170/1.414 \approx 120$ V — the standard nominal household voltage rating in North America, confirming that the “120 V” outlet rating is indeed an RMS value, not the actual peak voltage present on the line.

Common Misconception

Because impedance combines exactly like resistance, it is easy to slip into treating impedance magnitudes as if they combined the same simple way — for instance, assuming two impedances in series always add to give a larger magnitude in a straightforward way. Impedances are complex numbers, and complex addition does not simply add magnitudes (except in the special case where both impedances happen to have the same angle); a $30\ \Omega$ resistor in series with a $j40\ \Omega$ inductor's impedance, for instance, gives a combined magnitude of $50\ \Omega$ (via the Pythagorean-style calculation in Worked Example 35.1), not $30 + 40 = 70\ \Omega$. Always add impedances in rectangular form, term by term, and only convert to polar form (magnitude and angle) afterward, once the addition itself is complete.

Practice Problems

- A $50\angle 0^\circ$ V source drives a series RC circuit with $R = 30\ \Omega$ and $Z_C = -j40\ \Omega$. Find the total impedance in polar form.
- Using the impedance found in the previous problem, find the current phasor.
- A sinusoidal voltage has an RMS value of 240 V. Find its peak amplitude.
- Two impedances, $20\angle 0^\circ\ \Omega$ and $20\angle 90^\circ\ \Omega$, are in series. Find the combined impedance in both rectangular and polar form.

Chapter 36 — AC Power and Power Factor

36.1 Instantaneous and Average Power

For a sinusoidal voltage $v(t) = V_m \cos(\omega t)$ across an element carrying current $i(t) = I_m \cos(\omega t - \theta)$ (where θ is the phase angle by which current lags voltage, positive for an inductive load), the instantaneous power $p(t) = v(t)i(t)$ oscillates in a generally complicated way, but its time-average — the quantity that actually matters for billing, heating, and most practical purposes — reduces to a clean, simple result:

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\theta$$

The factor $\cos\theta$ is called the power factor, and it is the single most important new concept this chapter introduces: it measures what fraction of the product of RMS voltage and RMS current is actually converted to real, usable average power, as opposed to being exchanged back and forth between source and load without net transfer.

36.2 Real, Reactive, and Apparent Power

Three related power quantities are used throughout AC power engineering, and this series returns to all three, in far greater depth, in its dedicated power-systems volume:

1. Real power (P), measured in watts (W): the average power actually delivered to and consumed by the load — the only one of the three that corresponds to real, billable energy consumption or genuine heating/mechanical work.
2. Reactive power (Q), measured in volt-amperes reactive (VAR): power that is exchanged back and forth between source and load each cycle (stored temporarily in an inductor's magnetic field or a capacitor's electric field, then returned), with zero net transfer averaged over a full cycle.
3. Apparent power (S), measured in volt-amperes (VA): simply the product $V_{\text{rms}} I_{\text{rms}}$, with no $\cos\theta$ factor at all — the total power a source must be capable of supplying, real and reactive combined, even though only the real portion does useful work.

These three quantities relate through a right triangle (the “power triangle”), with apparent power as the hypotenuse:

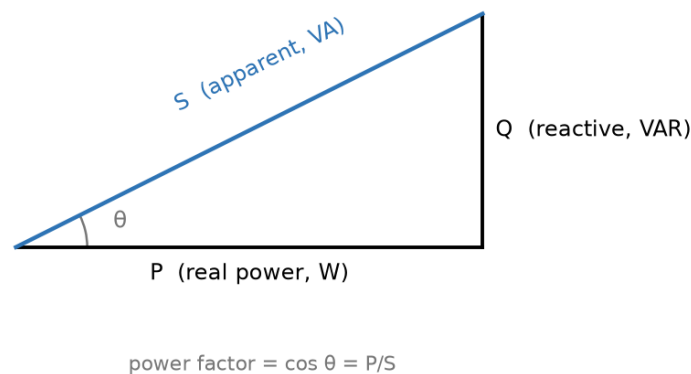


Figure 36.1 — The power triangle: real, reactive, and apparent power, related through the power-factor angle.

$$S = \sqrt{P^2 + Q^2}, \quad P = S \cos\theta, \quad Q = S \sin\theta$$

36.3 Why Power Factor Matters

A load with a low power factor ($\cos\theta$ well below 1, typical of large motors and other strongly inductive loads) draws a large current for a given amount of real power delivered, since $P = V_{\text{rms}} I_{\text{rms}} \cos\theta$ means I_{rms} must be correspondingly larger to compensate for a small $\cos\theta$ at fixed P and V_{rms} . That larger current causes larger resistive (I^2R) losses in the wiring and generating equipment supplying the load, even though those losses correspond to no additional useful work being done — which is exactly why electric utilities charge industrial customers a power-factor penalty, and why power-factor correction (typically, adding capacitors in parallel with an inductive load to cancel some of its reactive power) is standard practice in industrial electrical systems, a topic developed fully in this series' power-systems volume.

Key Idea

Power factor is the cosine of the phase angle between voltage and current, and it directly measures how much of a load's apparent power is actually useful. A purely resistive load has power factor 1 (voltage and current in phase, all apparent power is real power); a purely reactive load (an ideal inductor or capacitor alone) has power factor 0 (voltage and current 90° out of phase, all apparent power is reactive, zero real power delivered on average).

Worked Example 36.1

Problem: A load draws an RMS current of 10 A from a 120 V RMS source, with the current lagging the voltage by 36.9° . Find the real power, reactive power, and apparent power delivered to the load.

Solution: Apparent power: $S = V_{\text{rms}} I_{\text{rms}} = (120)(10) = 1200 \text{ VA}$. Real power: $P = S \cos\theta = 1200 \cos(36.9^\circ) \approx 1200(0.80) = 960 \text{ W}$. Reactive power: $Q = S \sin\theta = 1200 \sin(36.9^\circ) \approx 1200(0.60) = 720 \text{ VAR}$. Check: $\sqrt{(P^2 + Q^2)} = \sqrt{(960^2 + 720^2)} = \sqrt{(921,600 + 518,400)} = \sqrt{1,440,000} = 1200 \text{ VA}$, matching S exactly, as the power-triangle relationship requires.

36.4 Power Factor Correction

Since a low power factor arises from a load's net reactance (Chapter 35) — most commonly inductive, as in motors and transformers, which draw lagging current — the standard remedy is to add capacitors in parallel with the load, specifically sized so that the capacitor's own reactive power partially or fully cancels the load's inductive reactive power. Because reactive power, unlike real power, is not consumed but merely exchanged back and forth each cycle, a capacitor's leading reactive contribution and an inductor's lagging reactive contribution genuinely cancel in the net reactive-power total seen by the source, reducing S (and hence the current the source and wiring must supply) for the same real power P actually being delivered to the load — without needing to modify the load itself in any way, since the correcting capacitors are added entirely separately, typically right at the point where power enters an industrial facility.

Worked Example 36.2

Problem: A factory load draws 100 kW of real power at a power factor of 0.70 lagging, from a 480 V RMS supply. Find the apparent power and the reactive power drawn, and state qualitatively what adding power-factor-correcting capacitors would do to the current the supply must provide for the same 100 kW of real power delivered.

Solution: Apparent power: $S = P/\cos\theta = 100/0.70 \approx 142.9 \text{ kVA}$. Reactive power: since $\cos\theta = 0.70$ corresponds to $\sin\theta \approx 0.714$ (from $\cos^2\theta + \sin^2\theta = 1$), $Q = S \sin\theta \approx 142.9 \times 0.714 \approx 102 \text{ kVAR}$. Adding capacitors sized to supply some or all of this 102 kVAR locally means the source itself no longer needs to supply that reactive power — S decreases toward P (the power factor rises toward 1.0), and since $S = V_{\text{rms}} I_{\text{rms}}$ at fixed V_{rms} , the current I_{rms} the supply must provide decreases correspondingly, even though the same 100 kW of real power is still being delivered to the factory's actual equipment.

Common Misconception

It is easy to read “apparent power” as simply a less-accurate or approximate version of real power, since the everyday word “apparent” often means “seeming, but not actually true.” In this context the two quantities are not competing estimates of the same thing — apparent power ($S = V_{\text{rms}} I_{\text{rms}}$) is a genuinely different physical

quantity from real power ($P = S \cos\theta$), representing the full electrical burden a source and its wiring must be sized to handle, of which only the real-power portion does useful work. A generator, transformer, or set of wires must be rated for apparent power, in VA, not for real power alone in W, precisely because the reactive component still demands current-carrying capacity even though it delivers no net energy.

Practice Problems

- A load has a power factor of 0.90 and draws 1500 W of real power at 240 V RMS. Find the RMS current.
- A load draws 15 A RMS from a 100 V RMS source at a power factor of 0.60 (current lagging). Find the real, reactive, and apparent power.
- Explain briefly why a purely inductive load (power factor 0) delivers zero average real power despite drawing nonzero current.

Chapter 37 — Resonance, Frequency Response, and Transformers

37.1 The Series RLC Circuit

A series combination of a resistor, inductor, and capacitor, driven by a sinusoidal source, has total impedance:

$$Z = R + j\omega L + 1/(j\omega C) = R + j(\omega L - 1/(\omega C))$$

The imaginary part — the net reactance — depends on frequency in an interesting way: the inductor's reactance ωL grows with frequency while the capacitor's reactance $1/(\omega C)$ shrinks with frequency, so at one particular frequency, the resonant frequency, the two exactly cancel, leaving a purely real impedance equal to R alone.

37.2 Resonant Frequency

Setting $\omega L = 1/(\omega C)$ and solving for ω gives the resonant angular frequency:

$$\omega_0 = 1/\sqrt{LC}$$

At resonance, a series RLC circuit's impedance is at its minimum possible value (equal to R alone, with the reactive parts cancelling), so current is at its maximum for a given source voltage, and — since the inductor's and capacitor's voltages are equal in magnitude but opposite in phase at this exact frequency — the voltage across either one individually can substantially exceed the source voltage itself, an effect exploited deliberately in radio tuning circuits, where a resonant circuit is used to select one specific broadcast frequency out of many present simultaneously on an antenna.

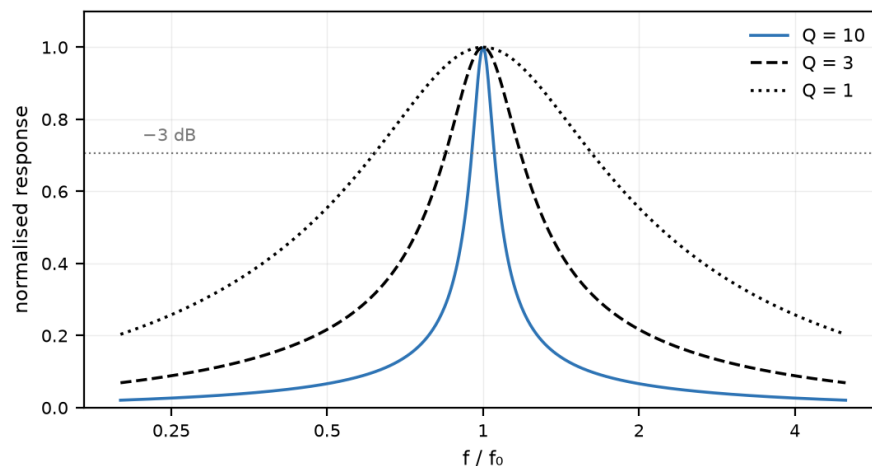


Figure 37.1 — Resonance curves at three values of Q : a higher Q gives a sharper, narrower peak.

Key Idea

Resonance is the frequency at which inductive and capacitive reactance exactly cancel. Below resonance, a series RLC circuit's net reactance is capacitive (current leads voltage); above resonance, it is inductive (current lags voltage); exactly at resonance, the circuit is purely resistive, current and voltage are in phase, and the power factor is exactly 1.

37.3 Bandwidth and Quality Factor

A resonant circuit does not respond only at exactly ω_0 — it responds over a range of frequencies around ω_0 , with the response falling off on either side. The bandwidth (the width of the frequency range over which the circuit's response stays within a standard threshold of its peak value) and the quality factor Q , which measures how sharply peaked the resonance is (a high- Q circuit has a narrow, sharply peaked response; a low- Q circuit has a broad, gentle one), are related by:

$$Q = \omega_0 L / R = 1 / (\omega_0 RC), \quad \text{Bandwidth} = \omega_0 / Q$$

A high- Q circuit — achieved with a small resistance relative to the inductor's reactance at resonance — is desirable in a radio tuner precisely because it rejects nearby, unwanted frequencies sharply while passing the desired station strongly; the same quality factor, and much of this same frequency-response machinery, reappears throughout this series' RF and microwave volumes, where filters and resonant cavities are central design elements rather than incidental circuit behaviors.

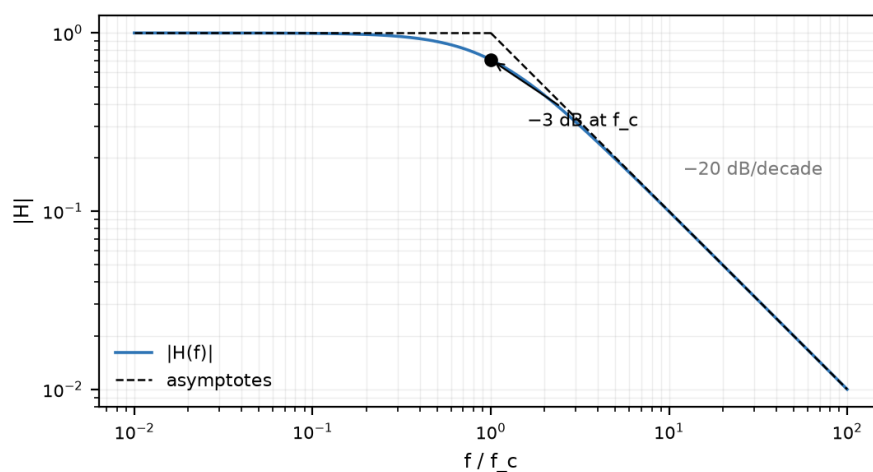


Figure 37.2 — A first-order low-pass response: flat below the corner frequency, falling at -20 dB/decade above it.

37.4 Mutual Inductance and the Transformer

Two coils placed near each other share magnetic flux, so that a changing current in one coil (Chapter 11) induces a voltage in the other — this coupling is quantified by the mutual inductance M , and it is the physical basis of the transformer, one of the most important devices in all of electrical power engineering. An ideal transformer, with N_1 turns on its primary winding and N_2 turns on its secondary, relates the two windings' voltages and currents by the turns ratio:

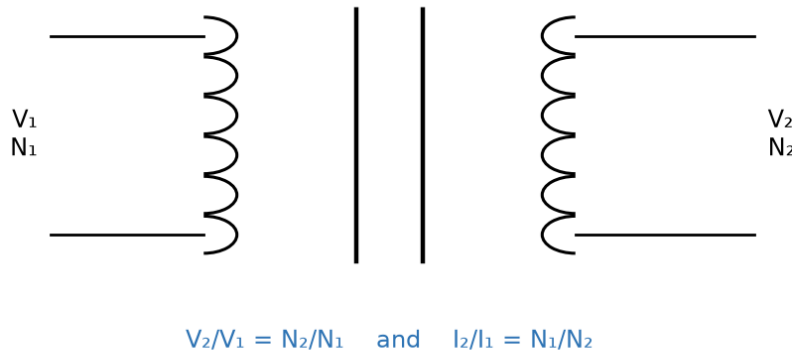


Figure 37.3 — An ideal transformer: voltage and current scale with the turns ratio, in opposite directions.

$$V_2/V_1 = N_2/N_1, \quad I_2/I_1 = N_1/N_2$$

Notice that voltage and current transform in opposite ratios — a step-up transformer (more secondary turns than primary) increases voltage but correspondingly decreases current by the same factor, and vice versa for a step-down transformer, so that an ideal transformer's input and output apparent power (Chapter 36) are exactly equal: transformers change the voltage-to-current ratio of the power delivered, but (idealized, with no internal losses) never change the total power itself. This single fact — trading voltage for current, or current for voltage, at essentially no cost — is the entire reason long-distance power transmission is done at very high voltage (minimizing current, and therefore minimizing I^2R transmission losses over the wires) before being stepped back down to safer, more usable voltages near the point of consumption, a subject this series returns to fully in its power-systems volume.

37.5 Simple Filters: Low-Pass and High-Pass RC Circuits

A single resistor and capacitor, arranged one way, form the simplest possible low-pass filter; arranged the other way, they form the simplest possible high-pass filter — and both are a direct, practical application of the frequency-dependent impedance introduced in Chapter 35. Take a series RC circuit driven by a sinusoidal source, and consider the output taken across the capacitor. By the voltage-divider relationship (Chapter 35, generalized to impedances):

$$V_{\text{out}}/V_{\text{in}} = Z_C/(R + Z_C) = 1/(1 + j\omega RC)$$

At low frequencies (ω small), $Z_C = 1/(j\omega C)$ is large compared to R , so nearly all the input voltage appears across the capacitor — the circuit passes low frequencies through to the output largely unattenuated. At high frequencies, Z_C shrinks toward zero, and the output voltage shrinks with it — high frequencies are attenuated. This is a low-pass filter, and the frequency at which the output has fallen to $1/\sqrt{2}$ (about 70.7%) of the input — the conventional definition of a filter's cutoff frequency — occurs exactly where $|Z_C| = R$, which works out to:

$$\omega_c = 1/(RC) \quad (\text{equivalently } f_c = 1/(2\pi RC))$$

Taking the output across the resistor instead of the capacitor, in the same series RC circuit, gives the complementary behavior: at low frequencies, nearly all the voltage drops across the now-large capacitor impedance, leaving little across the resistor; at high frequencies, the capacitor's impedance shrinks and the resistor's share of the voltage grows toward the full input. This is a high-pass filter, with exactly the same cutoff frequency $\omega_c = 1/(RC)$, simply measured across the other element.

Key Idea

The same RC circuit is a low-pass or a high-pass filter depending only on which element's voltage you take as the output. Both share the identical cutoff frequency $\omega_c = 1/(RC)$ — the same time constant from Chapter 34's transient analysis reappears here as a frequency-domain quantity, which is not a coincidence: a circuit's time-domain settling behavior and its frequency-domain filtering behavior are two views of the exact same underlying impedance relationships.

Worked Example 37.1

Problem: A low-pass RC filter is built with $R = 1.0 \text{ k}\Omega$ and $C = 100 \text{ nF}$. Find its cutoff frequency in Hz, and determine whether a 500 Hz signal is passed or substantially attenuated.

Solution: $\omega_c = 1/(RC) = 1/[(1.0 \times 10^3)(100 \times 10^{-9})] = 1/(1.0 \times 10^{-4}) = 1.0 \times 10^4 \text{ rad/s}$. Converting: $f_c = \omega_c/(2\pi) \approx 1592 \text{ Hz}$, about 1.6 kHz. Since 500 Hz is well below this cutoff frequency, it falls within the filter's passband and is passed through with little attenuation — only frequencies well above roughly 1.6 kHz would be substantially attenuated by this particular filter.

Worked Example 37.2

Problem: A series RLC circuit has $L = 10 \text{ mH}$, $C = 100 \text{ nF}$, and $R = 5.0 \Omega$. Find the resonant frequency (in Hz) and the quality factor.

Solution: $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{(10 \times 10^{-3})(100 \times 10^{-9})} = 1/\sqrt{1.0 \times 10^{-9}} = 1/(3.16 \times 10^{-5}) \approx 3.16 \times 10^4 \text{ rad/s}$. Converting to ordinary frequency: $f_0 = \omega_0/(2\pi) \approx 5033 \text{ Hz}$, about 5.0 kHz. Quality factor: $Q = \omega_0 L/R = (3.16 \times 10^4)(10 \times 10^{-3})/5.0 \approx 63.2$ — a fairly high-Q, sharply resonant circuit.

Worked Example 37.3

Problem: An ideal transformer has 1000 turns on its primary and 100 turns on its secondary, with 240 V RMS applied to the primary, delivering 5.0 A RMS to a load on the secondary. Find the secondary voltage and the primary current.

Solution: Turns ratio $N_2/N_1 = 100/1000 = 0.10$, a step-down transformer. Secondary voltage: $V_2 = V_1 \times (N_2/N_1) = 240 \times 0.10 = 24 \text{ V}$. For the current, the ratio inverts: $I_1 = I_2 \times (N_2/N_1) = 5.0 \times 0.10 = 0.50 \text{ A}$. Check via apparent power: primary $S = V_1 I_1 = 240 \times 0.50 = 120 \text{ VA}$; secondary $S = V_2 I_2 = 24 \times 5.0 = 120 \text{ VA}$ — equal, as required for an ideal (lossless) transformer.

Common Misconception

It is a common first-pass error to assume a step-down transformer, since it reduces voltage, must therefore also reduce the power available — as though “stepping down” meant losing capability. An ideal transformer changes voltage and current in exactly inverse proportion and delivers the same apparent power out as goes in; a step-down transformer trades a smaller voltage for a larger available current at the same total power, which is exactly the behavior wanted at the receiving end of a power line, where a lower, safer voltage is needed but the same total power must still be delivered to the load.

Practice Problems

- A series RLC circuit has $L = 5.0 \text{ mH}$ and $C = 20 \text{ nF}$. Find its resonant frequency in Hz.
- For the circuit in the previous problem, if $R = 2.0 \Omega$, find the quality factor Q .
- An ideal transformer steps 480 V RMS down to 120 V RMS. If the secondary delivers 20 A RMS to a load, find the primary current.
- Explain briefly why long-distance power transmission is done at very high voltage rather than at the voltage ultimately used by consumers.
- A high-pass RC filter uses $R = 2.2 \text{ k}\Omega$ and $C = 47 \text{ nF}$. Find its cutoff frequency in Hz.

Chapter 38 — Where This Series Goes Next

38.1 What This Book Has Built

This book took the physical definitions from Part I — voltage, current, resistance, capacitance, inductance — and built the systematic machinery needed to analyze any linear circuit built from them: Kirchhoff's two laws as the foundation; series/parallel reduction and dividers as the fastest tool when it applies; nodal and mesh analysis as universal methods for any resistive network; Thevenin and Norton equivalents for reasoning about a network's behavior independent of whatever load is eventually attached to it; superposition as both a practical technique and a conceptual justification for several of the results around it; first-order transients as the first genuinely time-dependent behavior in the book; and finally phasors and impedance, which extend every one of the DC techniques to sinusoidal steady-state AC circuits with only a change from real to complex arithmetic.

38.1a A Preview: Linearizing a Nonlinear Device

It is worth seeing, briefly and concretely, exactly how the next book's central technique connects to this one, rather than taking the connection entirely on faith. A diode's actual current-voltage relationship is strongly nonlinear — current grows roughly exponentially with voltage once the diode is forward-biased past a threshold, nothing like a resistor's straight-line $V = IR$. And yet, once a diode is biased at some fixed operating point (a fixed DC voltage and current, set by the rest of the circuit) and only a small additional signal is applied on top of that bias, the diode's response to that small extra signal is, to good approximation, linear — behaving, for that small signal only, exactly like an ordinary resistor whose value (called the diode's small-signal or dynamic resistance) depends on where along its curve the bias point happens to sit.

Concretely: if a diode's bias point sits at a current I_D , its small-signal resistance works out (from the exponential diode equation, derived properly in the next book) to $r_d \approx V_T/I_D$, where V_T is a small, roughly fixed voltage (about 26 mV at room temperature). A small AC signal riding on top of the DC bias sees the diode as nothing more than this resistance r_d — and once that substitution is made, the entire nonlinear device has been replaced, for purposes of analyzing the small signal, by an ordinary linear resistor, dropped into an otherwise perfectly ordinary circuit built from the same resistors, capacitors, and sources Part III has spent thirteen chapters learning to handle. Nodal analysis, Thevenin equivalents, phasors for an AC signal riding on the bias — every technique in this book applies directly to that small-signal picture, exactly as advertised.

Worked Example 38.1 — A Genuine Preview

Problem: A diode is biased at a DC current of 1.0 mA, giving it a small-signal resistance of $r_d = V_T/I_D = 0.026/0.001 = 26\ \Omega$. This diode's small-signal model (a $26\ \Omega$ resistor) is placed in series with a $1.0\ \text{k}\Omega$ resistor across a small 10 mV AC signal source. Find the AC voltage appearing across the diode's small-signal resistance.

Solution: This is nothing more than a voltage divider (Chapter 28), applied to the diode's linearized small-signal model exactly as if it were any other resistor: $V_{\text{diode}} = V_{\text{signal}} \times r_d/(r_d + R) = 10\ \text{mV} \times 26/(26+1000) \approx 10 \times 0.0253 \approx 0.25\ \text{mV}$. Every step here — identifying the divider, applying the ratio, computing the numeric answer — is drawn directly and without modification from Chapter 28 of this book. The only genuinely new idea, contributed by the next book in this series, was finding that $26\ \Omega$ small-signal resistance value in the first place, from the diode's underlying (and quite different-looking) nonlinear physics.

38.2 A Deliberate Limitation, and Why It Is Temporary

Every circuit in this book has been linear: built entirely from resistors, capacitors, inductors, independent sources, and (in Chapter 37) ideal transformers — elements whose defining relationships are all directly proportional, with no thresholds, no saturation, and no fundamentally different behavior in one direction of current or voltage versus

the other. Real electronic circuits are built overwhelmingly from devices that are not linear at all: diodes, which conduct current in essentially only one direction; transistors, whose entire usefulness as amplifiers and switches comes from a deliberately engineered, strongly nonlinear response. None of the techniques in this book apply directly to such devices without modification.

This is not a gap left unaddressed — it is exactly where this series goes next. The next book covers semiconductor physics and devices from the ground up: how doping creates the PN junction at the heart of every diode; how bipolar and field-effect transistors amplify and switch; and how the same nonlinear devices, once understood physically, are analyzed using a small-signal linearization technique that — not coincidentally — reduces a nonlinear device, operating near some fixed bias point, back into exactly the kind of linear circuit this book already knows how to solve. Every technique in this book — Thevenin equivalents especially — reappears directly once that linearization is performed, which is precisely why this book's systematic circuit-analysis toolkit was worth building thoroughly and rigorously before moving on: it is not merely useful for the resistor-capacitor-inductor circuits examined explicitly here, but for the local, small-signal behavior of every nonlinear device this series goes on to study.

Key Idea

Nonlinear devices are not exceptions to this book's methods — they are this book's methods, applied locally.

A transistor's small-signal model, introduced in the next volume, is itself a linear circuit (resistors, dependent sources, sometimes capacitors), and every technique from this book — nodal analysis, Thevenin equivalents, superposition — applies to that small-signal model exactly as presented here. The nonlinearity is handled once, in finding the right operating point and the right small-signal model; everything downstream of that is this book, again.

End-of-Book Review Problems

- State, in your own words, why KCL is a direct consequence of charge conservation while KVL is a direct consequence of potential being a well-defined function of position.
- Explain the practical difference between choosing nodal analysis and choosing mesh analysis for a given circuit, and identify one circuit feature that would make one method clearly preferable to the other.
- A Thevenin equivalent circuit is connected to a variable load resistor. Describe qualitatively how the load voltage and load current each change as the load resistance is increased from zero to a very large value, and identify the load resistance at which power delivered to the load is maximized.
- Explain why superposition can be used to find a voltage or current in a multi-source linear circuit, but cannot be used the same way to find power.
- A series RLC circuit is driven at a frequency well below its resonant frequency. State, with brief reasoning, whether the circuit's net reactance is inductive or capacitive at this frequency.
- Explain briefly why an ideal transformer's turns ratio affects voltage and current in inverse proportion to each other, referring to the requirement that input and output apparent power must be equal.

Appendix A — Formula Reference

Part I — The Physics of Electricity and Magnetism

SI Prefixes

These prefixes recur throughout this book and throughout the rest of this series, and it is worth having the full range fixed firmly in mind rather than only the two or three you happen to use most often — a stray factor of a thousand, misread from an unfamiliar prefix, is one of the most common sources of an answer that is wrong by several orders of magnitude despite every other step being correct.

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
(none)	—	10^0
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Electrostatics

Quantity	Formula
Coulomb's law	$F = kq_1q_2/r^2$, $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$
Electric field (point charge)	$E = kq/r^2$
Electric field, force	$E = F/q_0$
Gauss's law	$\Phi_E = Q_{\text{en}}C/\epsilon_0$
Field, infinite sheet	$E = \sigma/(2\epsilon_0)$
Field, infinite line	$E = \lambda/(2\pi\epsilon_0 r)$
Potential (point charge)	$V = kq/r$
Potential energy (pair)	$U = kqQ/r$
Field $\leftrightarrow$ potential (uniform)	$E = \Delta V/d$

Capacitance

Quantity	Formula
Capacitance	$C = Q/V$
Parallel plate	$C = \epsilon_0 A/d$ (with dielectric: $\kappa\epsilon_0 A/d$)
Energy stored	$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/(2C)$
Capacitors in parallel	$C_p = C_1 + C_2 + \dots$
Capacitors in series	$1/C_s = 1/C_1 + 1/C_2 + \dots$

Current, Resistance, and Circuits

Quantity	Formula
Current	$I = dQ/dt$
Current density, drift velocity	$J = I/A = nqv_d$
Resistance	$R = \rho L/A$

Ohm's law	$V = IR$
Power	$P = IV = I^2R = V^2/R$
Series resistors	$R_s = R_1 + R_2 + \dots$
Parallel resistors	$1/R_p = 1/R_1 + 1/R_2 + \dots$

Magnetism

Quantity	Formula
Magnetic force on charge	$F = q\mathbf{v} \times \mathbf{B}$, $F = qvB \sin\theta$
Force on current-carrying wire	$F = I\mathbf{L} \times \mathbf{B}$, $F = BIL \sin\theta$
Radius of circular motion	$r = mv/(qB)$
Field of a long straight wire	$B = \mu_0 I / (2\pi r)$, $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$
Field inside a solenoid	$B = \mu_0 n I$ (with core: $\mu_r \mu_0 n I$)
Ampere's law	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{en}} C$

Induction

Quantity	Formula
Magnetic flux	$\Phi_B = BA \cos\theta$
Faraday's law	$\mathcal{E} = -N d\Phi_B/dt$
Motional EMF	$\mathcal{E} = BLv$
Self-inductance	$\mathcal{E} = -L di/dt$
Speed of light	$c = 1/\sqrt{\epsilon_0 \mu_0} \approx 3.00 \times 10^8 \text{ m/s}$

Part II — Mathematics and Measurement

Units and Measurement

Prefix pairs that cancel: $\mu \times M = m \times k = n \times G = p \times T = 1$

Relative uncertainty of a product: add the relative uncertainties

Absolute uncertainty of a sum or difference: add the absolute uncertainties

Circuit Algebra

Series resistance: $R = R_1 + R_2 + \dots$

Parallel resistance: $1/R = 1/R_1 + 1/R_2 + \dots$, and for two, $R = R_1 R_2 / (R_1 + R_2)$

Voltage divider: $V_{\text{out}} = V_{\text{in}} \cdot R_2 / (R_1 + R_2)$

Current divider: $I_1 = I_{\text{total}} \cdot R_2 / (R_1 + R_2)$

Quadratic roots: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$

Binomial approximation, small x : $(1 + x)^n \approx 1 + nx$

Two-by-two determinant: $D = a_1 b_2 - a_2 b_1$

Exponentials, Logarithms, Decibels

Decay: $v(t) = V_0 \cdot e^{-(t/\tau)}$; Rise: $v(t) = V_{\text{f}}(1 - e^{-(t/\tau)})$

Time constants: $\tau = RC$ and $\tau = L/R$

Settling: 63% at 1τ , 86% at 2τ , 95% at 3τ , 99.3% at 5τ

Log laws: $\log(ab) = \log a + \log b$; $\log(a/b) = \log a - \log b$; $\log(a^n) = n \log a$

Power ratio: $G(\text{dB}) = 10 \cdot \log_{10}(P_2/P_1)$; Voltage ratio: $G(\text{dB}) = 20 \cdot \log_{10}(V_2/V_1)$

Key values: 3 dB = $\times 2$ power; 6 dB = $\times 2$ voltage; 10 dB = $\times 10$ power; 20 dB = $\times 10$ voltage

Absolute: 0 dBm = 1 mW; +30 dBm = 1 W; -30 dBm = 1 μ W

Trigonometry and Sinusoids

General sinusoid: $v(t) = V_{\text{m}} \cdot \sin(\omega t + \phi)$, with $\omega = 2\pi f = 2\pi/T$

Pythagorean: $\sin^2\theta + \cos^2\theta = 1$

Sum of angles: $\sin(A+B) = \sin A \cos B + \cos A \sin B$; $\cos(A+B) = \cos A \cos B - \sin A \sin B$

Double angle: $\sin 2\theta = 2 \sin \theta \cos \theta$; $\cos 2\theta = 1 - 2\sin^2\theta$

RMS of a sinusoid: $V_{\text{rms}} = V_{\text{m}}/\sqrt{2} \approx 0.707 V_{\text{m}}$ (square wave: V_{m} ; triangle: $V_{\text{m}}/\sqrt{3}$)

Complex Numbers

Rectangular: $z = a + jb$, with $j^2 = -1$

Magnitude: $|z| = \sqrt{a^2 + b^2}$; Angle: $\theta = \arctan(b/a)$, adjusted for quadrant

Conjugate: $z^* = a - jb$, and $z \cdot z^* = a^2 + b^2$

Polar product: magnitudes multiply, angles add

Euler: $e^{j\theta} = \cos \theta + j \sin \theta$; $\cos \theta = (e^{j\theta} + e^{-j\theta})/2$; $\sin \theta = (e^{j\theta} - e^{-j\theta})/2j$

Phasors and Impedance

Phasor of $V_m \cdot \cos(\omega t + \phi)$: $V = (V_m/\sqrt{2}) \angle \phi$ using RMS convention

Impedances: $Z_R = R$; $Z_L = j\omega L$; $Z_C = 1/(j\omega C) = -j/(\omega C)$

Series impedance adds; parallel impedance by product-over-sum; admittance $Y = 1/Z$ adds in parallel

Impedance angle = phase by which voltage leads current

Smith Chart and Impedance Matching

Quantity	Formula
Reflection coefficient	$\Gamma = (Z - Z_0)/(Z + Z_0) = (z - 1)/(z + 1)$
VSWR	$VSWR = (1 + \Gamma)/(1 - \Gamma)$
Return loss	$RL \text{ (dB)} = -20 \log_{10} \Gamma $
L-network series reactance ($R_L < Z_0$)	$X_s = \sqrt{R_L(Z_0 - R_L)}$
L-network shunt susceptance	$B_p = X_s/(R_L^2 + X_s^2)$
Short-circuited stub susceptance	$B = -Y_0 \cot(\beta l)$
Open-circuited stub susceptance	$B = +Y_0 \tan(\beta l)$
RF choke sizing (rule of thumb)	$X_L \geq (5 \text{ to } 10) \times Z_0$

Calculus

Component laws: $i = C \cdot dv/dt$; $v = L \cdot di/dt$

d/dt of $e^{kt} = k \cdot e^{kt}$; of $\sin(\omega t) = \omega \cos(\omega t)$; of $\cos(\omega t) = -\omega \sin(\omega t)$; of $t^n = n \cdot t^{n-1}$

$\int e^{kt} dt = e^{kt}/k$; $\int \cos(\omega t) dt = \sin(\omega t)/\omega$; $\int t^n dt = t^{n+1}/(n+1)$

In sinusoidal steady state: $d/dt \rightarrow \times j\omega$, and $\int dt \rightarrow \div j\omega$

Part III — DC and AC Circuit Theory

Kirchhoff's Laws and Basic Relations

Quantity	Formula
Kirchhoff's Current Law	$\Sigma I_{in} = \Sigma I_{out}$ at any node
Kirchhoff's Voltage Law	$\Sigma V = 0$ around any closed loop
Voltage divider	$V_1 = V \cdot R_1 / (R_1 + R_2)$
Current divider	$I_1 = I \cdot R_2 / (R_1 + R_2)$

Equivalent Circuits

Quantity	Formula
Thevenin resistance	R_{Th} : sources deactivated, look into terminals
Thevenin voltage	V_{Th} = open-circuit voltage
Norton current	I_N = short-circuit current = V_{Th} / R_{Th}
Maximum power transfer	$R_L = R_{Th}$; $P_{max} = V_{Th}^2 / (4R_{Th})$

Dynamic Elements and Transients

Quantity	Formula
Capacitor v-i relation	$i = C \, dv/dt$
Inductor v-i relation	$v = L \, di/dt$
Inductors in series/parallel	$L_s = L_1 + L_2 + \dots$; $1/L_p = 1/L_1 + 1/L_2 + \dots$
RC time constant	$\tau = RC$
RL time constant	$\tau = L/R$
General transient form	$x(t) = x_{final} + (x_{initial} - x_{final})e^{-(t/\tau)}$

Phasors and Impedance

Quantity	Formula
Impedance, resistor	$Z_R = R$
Impedance, inductor	$Z_L = j\omega L$
Impedance, capacitor	$Z_C = 1/(j\omega C) = -j/(\omega C)$
RMS from peak	$V_{rms} = V_m / \sqrt{2}$

AC Power

Quantity	Formula
Real power	$P = V_{rms} I_{rms} \cos\theta$ (W)
Reactive power	$Q = V_{rms} I_{rms} \sin\theta$ (VAR)
Apparent power	$S = V_{rms} I_{rms}$ (VA) ; $S = \sqrt{P^2 + Q^2}$
Power factor	$\cos\theta = P/S$

Resonance and Transformers

Quantity	Formula
Resonant angular frequency	$\omega_0 = 1/\sqrt{LC}$
Quality factor (series RLC)	$Q = \omega_0 L / R$
Bandwidth	$= \omega_0 / Q$
Ideal transformer	$V_2/V_1 = N_2/N_1$; $I_2/I_1 = N_1/N_2$

Appendix B — Practice Problem Solutions

Full solutions to every practice problem in the book are given below, organized by chapter. Attempt each problem in full before consulting its solution here — see the Preface for why this matters more than it might seem to.

Chapter 1 Solutions

1. $q = -8.0 \text{ nC} = -8.0 \times 10^{-9} \text{ C}$. Number of excess electrons $n = |q|/e = (8.0 \times 10^{-9})/(1.602 \times 10^{-19}) \approx 5.0 \times 10^{10}$ electrons.
2. Total charge before contact: $+6.0 + (-2.0) = +4.0 \text{ nC}$. Since the spheres are identical, this total splits evenly on contact: each sphere ends with $+2.0 \text{ nC}$.
3. Charge moved: $Q = I\Delta t = (2.0 \text{ A})(60 \text{ s}) = 120 \text{ C}$. Number of elementary charges: $n = Q/e = 120/(1.602 \times 10^{-19}) \approx 7.5 \times 10^{20}$ charges.
4. First contact: neutral sphere touches a sphere at $+9.0 \text{ nC}$. Total $+9.0 \text{ nC}$ splits evenly: each ends at $+4.5 \text{ nC}$. Second contact: that same sphere (now $+4.5 \text{ nC}$) touches a sphere at -3.0 nC . Total: $+4.5 + (-3.0) = +1.5 \text{ nC}$, splitting evenly to $+0.75 \text{ nC}$ each. Final state: first sphere $+4.5 \text{ nC}$, second sphere $+0.75 \text{ nC}$, third sphere $+0.75 \text{ nC}$.

Chapter 2 Solutions

1. $F = kq_1q_2/r^2 = (8.99 \times 10^9)(4.0 \times 10^{-6})(6.0 \times 10^{-6})/(0.30)^2 \approx 2.4 \text{ N}$, attractive (opposite signs), directed along the line joining the charges.
2. Set $F = kq^2/r^2 = 1.0 \text{ N}$ and solve for r : $r = \sqrt{(kq^2/F)} = \sqrt{[(8.99 \times 10^9)(1.0 \times 10^{-6})^2/1.0]} \approx 0.095 \text{ m}$, about 9.5 cm .
3. Force from $+2.0 \mu\text{C}$ (10 cm away) on the $-3.0 \mu\text{C}$ middle charge: attractive, pulling toward $x = 0$, magnitude $F_1 = k(2.0 \times 10^{-6})(3.0 \times 10^{-6})/(0.10)^2 \approx 5.39 \text{ N}$. Force from $+1.0 \mu\text{C}$ (10 cm away) on the middle charge: attractive, pulling toward $x = 20 \text{ cm}$, magnitude $F_2 = k(1.0 \times 10^{-6})(3.0 \times 10^{-6})/(0.10)^2 \approx 2.70 \text{ N}$. These act in opposite directions; net force is $5.39 - 2.70 \approx 2.7 \text{ N}$, directed toward $x = 0$.
4. Three forces act on each sphere: gravity (mg , downward), string tension (T , along the string), and the Coulomb repulsion from the other sphere (F_e , horizontal, pushing the spheres apart). At equilibrium, resolving tension into components: $T \cos(10^\circ)$ balances gravity, and $T \sin(10^\circ)$ balances the Coulomb force. Dividing these two equations eliminates T : $\tan(10^\circ) = F_e/(mg)$, so $F_e = mg \tan(10^\circ)$. Since $F_e = kq^2/r^2$ (with r related to the string length and the 10° angle by geometry), setting these two expressions for F_e equal gives an equation solvable for q once the string length is specified.

Chapter 3 Solutions

1. $E = kq/r^2 = (8.99 \times 10^9)(12 \times 10^{-9})/(0.10)^2 \approx 1.08 \times 10^4 \text{ N/C}$, directed radially inward (toward the negative charge).
2. Each charge is 2.0 cm from the midpoint. As in Worked Example 3.2, the fields from a $+q$ and $-q$ pair point the same direction at the midpoint between them: $E_1 = E_2 = kq/r^2 = (8.99 \times 10^9)(5.0 \times 10^{-9})/(0.020)^2 \approx 1.12 \times 10^5 \text{ N/C}$ each, adding to a total of about $2.25 \times 10^5 \text{ N/C}$, directed from the positive charge toward the negative one.
3. $F = qE = (1.60 \times 10^{-19})(500) = 8.0 \times 10^{-17} \text{ N}$. Since the electron's charge is negative, the force is opposite to E — i.e., in the $-x$ direction.
4. $\tau = pE \sin\theta = (2.0 \times 10^{-29})(3.0 \times 10^5) \sin(90^\circ) = 6.0 \times 10^{-24} \text{ N}\cdot\text{m}$.

Chapter 4 Solutions

1. By Gauss's law, $\Phi_E = Q_{\text{enc}}/\epsilon_0 = (2.0 \times 10^{-9})/(8.85 \times 10^{-12}) \approx 226 \text{ N} \cdot \text{m}^2/\text{C}$. Notice the sphere's radius (3.0 cm) is irrelevant to the answer — flux through a closed surface depends only on enclosed charge, not on the surface's size or shape.
2. $E = \sigma/(2\epsilon_0) = (4.0 \times 10^{-6})/[2(8.85 \times 10^{-12})] \approx 2.26 \times 10^5 \text{ N/C}$. Note this does not depend on the 2.0 cm distance given — the field of an ideal infinite sheet is the same at every distance.
3. If the field at a conductor's surface had any component parallel to the surface, that component would push the conductor's free surface charges to redistribute along the surface until they cancelled it out — a process that continues until no field component remains parallel to the surface, i.e., until the field is purely perpendicular to it. This is simply the definition of electrostatic equilibrium applied at the surface.
4. $E = \lambda/(2\pi\epsilon_0 r) = (5.0 \times 10^{-9})/[2\pi(8.85 \times 10^{-12})(0.040)] \approx 2.25 \times 10^3 \text{ N/C}$, directed radially away from the wire.

Chapter 5 Solutions

1. $V = kq/r = (8.99 \times 10^9)(3.0 \times 10^{-9})/(0.15) \approx 180 \text{ V}$.
2. $W = \Delta U = kqQ/r = (8.99 \times 10^9)(2.0 \times 10^{-9})(5.0 \times 10^{-9})/(0.10) \approx 8.99 \times 10^{-7} \text{ J}$.
3. $\Delta V = Ed = (2.0 \times 10^4)(0.0050) = 100 \text{ V}$.
4. Kinetic energy gained equals the magnitude of the potential energy lost: $KE = |q|\Delta V = (1.60 \times 10^{-19})(500) = 8.0 \times 10^{-17} \text{ J}$. Speed from $KE = \frac{1}{2}mv^2$: $v = \sqrt{2KE/m} = \sqrt{2(8.0 \times 10^{-17})/(9.11 \times 10^{-31})} \approx 1.33 \times 10^7 \text{ m/s}$ (about 4.4% of the speed of light — fast enough that a fully careful treatment would need relativistic corrections, though this book treats it non-relativistically, which is accurate to a few percent at this speed).

Chapter 6 Solutions

1. $Q = CV = (4.0 \times 10^{-6})(12) = 4.8 \times 10^{-5} \text{ C} = 48 \text{ } \mu\text{C}$. Energy: $U = \frac{1}{2}CV^2 = \frac{1}{2}(4.0 \times 10^{-6})(12)^2 \approx 2.88 \times 10^{-4} \text{ J}$.
2. With Q fixed and d doubled: $C = \epsilon_0 A/d$ is halved. Since $V = Q/C$ and Q is fixed while C is halved, V doubles. Energy $U = Q^2/(2C)$: with Q fixed and C halved, U doubles as well — consistent with the fact that pulling the plates apart against their mutual attraction, at fixed charge, requires doing work on the system, which appears as the increase in stored energy.
3. $C = \kappa \times 10 \text{ pF} = 3.5 \times 10 \text{ pF} = 35 \text{ pF}$.
4. Parallel capacitance: $C = 2.0 + 4.0 + 4.0 = 10.0 \text{ } \mu\text{F}$. Total charge: $Q = CV = (10.0 \times 10^{-6})(9.0) = 9.0 \times 10^{-5} \text{ C} = 90 \text{ } \mu\text{C}$.

Chapter 7 Solutions

1. $Q = I\Delta t = (3.0)(5.0) = 15 \text{ C}$.
2. $R = \rho L/A$. Tripling L multiplies R by 3; halving A multiplies R by 2 (since $R \propto 1/A$). Combined, R increases by a factor of $3 \times 2 = 6$.
3. $\Delta T = 80 - 20 = 60^\circ\text{C}$. $R(T) \approx R_0[1 + \alpha\Delta T] = 100[1 + (0.004)(60)] = 100[1.24] = 124 \text{ } \Omega$.

4. Equal length and equal resistance means $\rho_{\text{Cu}} L/A_{\text{Cu}} = \rho_{\text{Al}} L/A_{\text{Al}}$, so $A_{\text{Cu}}/A_{\text{Al}} = \rho_{\text{Cu}}/\rho_{\text{Al}} = (1.68 \times 10^{-8})/(2.65 \times 10^{-8}) \approx 0.634$. The copper wire needs roughly 63% of the aluminum wire's cross-sectional area to match its resistance — reflecting copper's lower resistivity, though aluminum's much lower density is exactly why it is still preferred for long-distance overhead power lines despite needing a larger cross-section.

Chapter 8 Solutions

1. $R = V/I = 9.0/0.30 = 30 \, \Omega$. Power: $P = IV = (0.30)(9.0) = 2.7 \, \text{W}$.
2. $R_{\text{parallel}} = R_1 R_2 / (R_1 + R_2) = (4.0)(12)/(4.0 + 12) = 48/16 = 3.0 \, \Omega$.
3. From $P = V^2/R$, solve for R : $R = V^2/P = (5.0)^2/0.25 = 100 \, \Omega$. This is a minimum resistance — any resistance smaller than $100 \, \Omega$ would dissipate more than $0.25 \, \text{W}$ at $5.0 \, \text{V}$ and exceed the rating.
4. Voltage divider: $V_{8k} = V \times R_{8k} / (R_{2k} + R_{8k}) = 10 \times 8.0 / (2.0 + 8.0) = 10 \times 0.80 = 8.0 \, \text{V}$.

Chapter 9 Solutions

1. $F = qvB = (1.60 \times 10^{-19})(3.0 \times 10^5)(0.20) \approx 9.6 \times 10^{-15} \, \text{N}$.
2. $F = BIL = (0.30)(4.0)(0.50) = 0.60 \, \text{N}$.
3. The magnetic force $F = qv \times B$ is always perpendicular to the particle's velocity v . A force perpendicular to velocity does zero work (since work depends on the force component along the direction of motion), so kinetic energy — and hence speed — cannot change; only the direction of v can be altered.
4. From $r = mv/(qB)$, solve for v : $v = rqB/m = (0.10)(1.60 \times 10^{-19})(0.80)/(1.67 \times 10^{-27}) \approx 7.66 \times 10^6 \, \text{m/s}$.

Chapter 10 Solutions

1. Parallel currents in the same direction attract.
2. $n = 600/0.30 = 2000 \, \text{turns/m}$. $B = \mu_0 n I = (4\pi \times 10^{-7})(2000)(2.0) \approx 5.0 \times 10^{-3} \, \text{T}$.
3. From $B = \mu_0 I / (2\pi r)$, solve for r : $r = \mu_0 I / (2\pi B) = (4\pi \times 10^{-7})(10) / [2\pi(1.0 \times 10^{-4})] = 0.020 \, \text{m}$, or $2.0 \, \text{cm}$.
4. Since $B \propto \mu_r$ at fixed current and turns, the new field is simply $200 \times 8.0 \, \text{mT} = 1600 \, \text{mT} = 1.6 \, \text{T}$ (again, in practice this idealized scaling would be limited by core saturation at such a high field, as noted in Worked Example 10.2, but the direct proportionality holds as stated for the purposes of this problem).

Chapter 11 Solutions

1. $\varepsilon = N(dB/dt)A = (50)(0.50)(0.020) = 0.50 \, \text{V}$.
2. $\varepsilon = BLv = (0.50)(0.30)(4.0) = 0.60 \, \text{V}$.
3. As the magnet's north pole approaches, the flux through the loop (in the direction from the magnet toward the loop) is increasing. By Lenz's law, the induced current opposes this increase, so it flows in whatever direction creates a magnetic field opposing the approaching north pole — meaning the loop's near face effectively becomes a north pole itself (like poles repel, resisting the magnet's approach). By the right-hand rule, viewed from the approaching magnet's side, the induced current flows counterclockwise.

4. $|\mathcal{E}| = L|dI/dt| = (0.20)(5.0) = 1.0 \text{ V}.$

Chapter 12 Solutions

1. The electric field at a point describes the force per unit charge a test charge would feel there — a vector, with direction. The electric potential at the same point describes the potential energy per unit charge a test charge would have there — a scalar, with no direction, only a numeric value (relative to a chosen reference point, conventionally infinity).

2. Gauss's law and Coulomb's law describe the same underlying physical fact — how a static charge distribution produces an electric field — expressed in two mathematically equivalent forms. Section 4.3 shows explicitly that applying Gauss's law to a spherically symmetric Gaussian surface around a point charge reproduces Coulomb's law exactly; more generally, either law can be derived from the other using the mathematical relationship between a field's flux through a closed surface and its local divergence, which is beyond this book's scope but is a standard result of vector calculus.

3. A resistor dissipates the energy delivered to it as heat, permanently and irreversibly converting electrical energy into thermal energy that leaves the circuit. A capacitor stores the energy delivered to it in its electric field and can return that same energy to the circuit later, undiminished (in the idealized case with no resistive losses) — charging and discharging a capacitor is, in principle, fully reversible, while dissipating energy in a resistor is not.

4. As more load is placed on a generator, more current is drawn from the coil rotating within the magnetic field. By Lenz's law, this increased induced current creates its own magnetic field that opposes the coil's rotation more strongly than before (opposing the very motion that induces it). Overcoming this stronger opposing effect requires more mechanical torque to keep the shaft turning at the same speed, which is felt directly as increased physical effort by whatever is turning it — a turbine, an engine, or a hand crank.

5. Coulomb's law, an experiment purely about the static force between charged objects, gives the value of ϵ_0 . Ampere's measurements of the force between current-carrying wires, an experiment purely about steady currents with no light or optics involved, gives the value of μ_0 . Maxwell showed mathematically that combining his four field equations predicts a self-sustaining wave that propagates through empty space at the speed $c = 1/\sqrt{(\epsilon_0\mu_0)}$ — and when the already-known values of ϵ_0 and μ_0 (measured in entirely electrical and magnetic experiments) are plugged into this formula, the result matches the independently measured speed of light, with no light-related measurement having been used anywhere in the calculation.

6. A capacitor's defining relation, $C = Q/V$, means a capacitor opposes rapid changes in voltage: forcing its voltage to change quickly requires supplying or removing charge quickly, which the surrounding circuit may or may not be able to do. An inductor's defining relation, $\mathcal{E} = -L dI/dt$, means an inductor opposes rapid changes in current: forcing its current to change quickly induces a large back-EMF resisting that change. In each case, the component “resists” rapid change in the one quantity (voltage for the capacitor, current for the inductor) that does not appear directly in its own defining constant — a genuine mirror-image relationship between the two components that becomes central once both are combined into AC circuits in the next book of this series.

Chapter 13 Solutions

1. $Q = CV = (100 \times 10^{-9})(12) = 1.2 \times 10^{-6} \text{ C} = 1.2 \mu\text{C}$, or 1200 nC.
2. L/R has units $\text{H}/\Omega = (\text{V} \cdot \text{s}/\text{A})/(\text{V}/\text{A}) = (\text{V} \cdot \text{s}/\text{A})(\text{A}/\text{V}) = \text{s}$. A time, as required.
3. $W = \frac{1}{2}LI^2$ has units $\text{H} \cdot \text{A}^2 = (\text{V} \cdot \text{s}/\text{A})(\text{A}^2) = \text{V} \cdot \text{A} \cdot \text{s} = \text{W} \cdot \text{s} = \text{J}$. Joules, as required.
4. $Z = \omega C$ would have units $(1/\text{s})(\text{A} \cdot \text{s}/\text{V}) = \text{A}/\text{V} = \text{siemens}$, which is admittance, not impedance. Since impedance must be V/A , the correct expression must have ωC in the denominator: $Z = 1/(\omega C)$, which gives V/A as required.

Chapter 14 Solutions

1. 15 nF; 68 k Ω ; 2.5 mA; 1.5 GHz.
2. $V = (4.7 \text{ k}\Omega)(3.2 \text{ mA}) = 15.04 \text{ V}$, since kilo $\times$ milli = 1. $P = VI = (15.04)(3.2 \text{ mA}) = 48.1 \text{ mW}$.
3. Convert all to nanofarads: 470 pF = 0.47 nF, 1.2 nF stays, 0.01 $\mu\text{F} = 10 \text{ nF}$. Total = 11.67 nF.
4. $f = 1/T = 1/(20 \times 10^{-9}) = 50 \text{ MHz}$. $\omega = 2\pi f = 3.142 \times 10^8 \text{ rad/s}$.

Chapter 15 Solutions

1. 0.00250 has three; 1200 is ambiguous (two, three, or four); 1.200×10^3 has four; 100.0 has four; 0.0001 has one.
2. $V = (4.7 \text{ k}\Omega)(2.0 \text{ mA}) = 9.4 \text{ V}$. Relative uncertainties add for a product: $5\% + 2\% = 7\%$, so $9.4 \text{ V} \pm 0.66 \text{ V}$.
3. Difference = 0.5 mm. Absolute uncertainties add: $\pm 0.4 \text{ mm}$. That is 80% relative uncertainty — the difference of two nearly equal measurements, each good to 0.2%, is known only to 80%.
4. At 10.000 V: 0.05% of 10 V = 5 mV, plus 2 counts = 2 mV, total 7 mV, which is 0.07% of reading. At 0.100 V: 0.05% of 0.1 V = 0.05 mV, plus the same 2 mV, total 2.05 mV, which is 2.05% of reading — thirty times worse, because the fixed count term now dominates.

Chapter 16 Solutions

1. $R = V/I = 3.3/(150 \times 10^{-6}) = 22000 \Omega = 22 \text{ k}\Omega$.
2. Reciprocal method: $1/R = 1/1000 + 1/2000 + 1/3000 = 11/6000$, so $R = 545.5 \Omega$. Product-over-sum twice: $1\text{k}\parallel 2\text{k} = 2 \times 10^6/3000 = 666.7 \Omega$; then $666.7\parallel 3\text{k} = 2 \times 10^6/3666.7 = 545.5 \Omega$. They agree.
3. $2.5 = 9 \times 10\text{k}/(R_1 + 10\text{k})$, so $R_1 + 10\text{k} = 90\text{k}/2.5 = 36 \text{ k}\Omega$, giving $R_1 = 26 \text{ k}\Omega$.
4. $\sqrt{1.06} \approx 1 + 0.03 = 1.030$ against an exact 1.02956, an error of +0.04%. $1/0.98 = (1 - 0.02)^{-1} \approx 1 + 0.02 = 1.020$ against an exact 1.02041, an error of -0.04%.

Chapter 17 Solutions

1. From the second equation $x = y + 1$. Substituting: $2(y + 1) + 3y = 12$, so $5y = 10$, $y = 2$ and $x = 3$.
2. Multiply the first by 2 and the second by 5: $8a + 10b = 44$ and $15a - 10b = 25$. Adding gives $23a = 69$, so $a = 3$, and then $b = 2$.

3. $D = (6)(1) - (3)(2) = 0$. The determinant vanishes, so there is no unique solution. Inspecting the equations, the second is exactly half the first — they are the same equation written twice, so there are infinitely many solutions.
4. $D = (100)(90) - (-40)(-40) = 9000 - 1600 = 7400$. $D_1 = (5)(90) - (0)(-40) = 450$, so $I_1 = 60.8$ mA. $D_2 = (100)(0) - (-40)(5) = 200$, so $I_2 = 28.0$ mA. Both are positive, meaning both mesh currents flow in the directions originally assumed.

Chapter 18 Solutions

1. A straight line through the origin. Slope $= 1/R = 1/2000 = 0.5$ mS.
2. At 100 Hz, $H = 1/\sqrt{1 + 0.01} = 0.995$. At 1 kHz, $H = 1/\sqrt{2} = 0.707$. At 10 kHz, $H = 1/\sqrt{101} = 0.0995$.
3. $V = 12 - 50I$. Setting $V = 8$ gives $50I = 4$, so $I = 80$ mA.
4. A factor of 100 in magnitude per decade of frequency is a slope of -2 on log-log axes, which is 40 dB per decade. That indicates a second-order response.

Chapter 19 Solutions

1. $\tau = RC = (22 \times 10^3)(100 \times 10^{-9}) = 2.2$ ms. At $t = 5$ ms, $v = 9 \cdot e^{(-5/2.2)} = 9 \cdot e^{(-2.273)} = 9(0.1031) = 0.928$ V.
2. $2 \rightarrow 3.01$ dB; $10 \rightarrow 10$ dB; $100 \rightarrow 20$ dB; $0.5 \rightarrow -3.01$ dB; $1000 \rightarrow 30$ dB.
3. $20 \cdot \log_{10}(250) = 47.96$ dB, near enough 48 dB. After a 6 dB attenuator, 41.96 dB, which as a voltage ratio is $250/2 = 125$.
4. -95 dBm $= 10^{(-9.5)}$ mW $= 3.16 \times 10^{-10}$ mW $= 3.16 \times 10^{-13}$ W. Adding 12 dB gives -83 dBm $= 5.01 \times 10^{-9}$ mW $= 5.01 \times 10^{-12}$ W.

Chapter 20 Solutions

1. $45^\circ = \pi/4 = 0.785$ rad; $90^\circ = \pi/2 = 1.571$ rad; $120^\circ = 2\pi/3 = 2.094$ rad; $270^\circ = 3\pi/2 = 4.712$ rad. And $\pi/6 = 30^\circ$; $2\pi/3 = 120^\circ$; $3\pi/2 = 270^\circ$.
2. Amplitude 2.5 A; $f = 1000$ Hz; $T = 1$ ms; phase -45° . At $t = 0$, $i = 2.5 \cdot \sin(-45^\circ) = -1.77$ A.
3. Amplitude $= 10/2 = 5$ V. RMS $= 5/\sqrt{2} = 3.54$ V.
4. Because they are 90° apart, they add in quadrature: $\sqrt{3^2 + 4^2} = 5$ V. Adding 3 and 4 directly would assume they peak at the same instant, which they do not — the quarter-cycle offset means one is at zero when the other is at its maximum.

Chapter 21 Solutions

1. Sum $= 6 - j2$. Difference $= 2 + j8$. Product $= (4 + j3)(2 - j5) = 8 - j20 + j6 - j^2 15 = 8 + 15 - j14 = 23 - j14$.
2. $|5 + j12| = 13$, angle $\arctan(12/5) = 67.4^\circ$, so $13 \angle 67.4^\circ$. And $10 \angle -60^\circ = 10\cos(-60^\circ) + j10\sin(-60^\circ) = 5 - j8.66$.
3. Multiply above and below by $(4 + j)$: numerator $(2 + j3)(4 + j) = 8 + j2 + j12 - 3 = 5 + j14$; denominator $16 + 1 = 17$. So the answer is $0.294 + j0.824$. Checking: $(0.294 + j0.824)(4 - j) = 1.176 + 0.824 + j(3.296 - 0.294) = 2.00 + j3.00$. Correct.

4. $j^2 = -1$ (two quarter-turns, a half-turn). $j^3 = -j$ (three quarter-turns). $j^4 = 1$ (full turn, back to start). $1/j = -j$ (a quarter-turn backwards). In general j^n repeats with period 4, so j^n depends only on n modulo 4.

Chapter 22 Solutions

1. $e^{j\pi} = \cos \pi + j \sin \pi = -1$. $e^{j\pi/2} = j$. $e^{j2\pi} = 1$.
2. $|3 + j4| = 5$, angle $\arctan(4/3) = 0.927 \text{ rad} = 53.13^\circ$. So $5e^{j0.927}$.
3. Squaring and adding the exponential forms: $[(e^{j\theta} + e^{-j\theta})/2]^2 + [(e^{j\theta} - e^{-j\theta})/(2j)]^2$. The first expands to $(e^{j2\theta} + 2 + e^{-j2\theta})/4$; the second, noting $(2j)^2 = -4$, to $-(e^{j2\theta} - 2 + e^{-j2\theta})/4$. Adding, the $e^{j2\theta}$ terms cancel and the constants give $(2 + 2)/4 = 1$.
4. $v(t) = 10 \cdot \cos(500t + \pi/4)$. Frequency $f = 500/2\pi = 79.6 \text{ Hz}$. At $t = 2 \text{ ms}$ the argument is $(500)(0.002) + \pi/4 = 1 + 0.785 = 1.785 \text{ rad}$, and $\cos(1.785) = -0.213$, so $v = -2.13 \text{ V}$.

Chapter 23 Solutions

1. For the voltage: $\text{RMS} = 170/\sqrt{2} = 120.2 \text{ V}$, already a cosine, so $V = 120.2 \angle 30^\circ \text{ V}$. For the current: $\sin(377t) = \cos(377t - 90^\circ)$, and $\text{RMS} = 4.24/\sqrt{2} = 3.00 \text{ A}$, so $I = 3.00 \angle -90^\circ \text{ A}$.
2. At 2 kHz , $\omega = 12566 \text{ rad/s}$. $Z_L = j\omega L = j(12566)(0.022) = j276 \Omega$, that is $276 \angle 90^\circ$. $Z_C = -j/(\omega C) = -j/(12566 \times 470 \times 10^{-9}) = -j169 \Omega$, that is $169 \angle -90^\circ$.
3. $\omega = 2\pi(400) = 2513 \text{ rad/s}$, so $Z_L = j251 \Omega$ and $Z = 470 + j251 \Omega$. $|Z| = \sqrt{470^2 + 251^2} = 533 \Omega$ at $\arctan(251/470) = +29.1^\circ$. $I = 24/533 = 45.0 \text{ mA}$, lagging the voltage by 29.1° .
4. Equal magnitudes require $R = 1/(\omega C)$, so $\omega = 1/RC = 1/(1000 \times 10^{-7}) = 10^4 \text{ rad/s}$, giving $f = 1592 \text{ Hz}$. There $Z = 1000 - j1000$, whose angle is -45° , and $|Z| = 1414 \Omega$. The output across R is $R/|Z| = 1000/1414 = 0.707$ of the input — the -3 dB point, as expected.

Chapter 24 Solutions

1. Normalized impedance: $z = (30 + j40)/50 = 0.6 + j0.8$. $\Gamma = (z-1)/(z+1) = (-0.4 + j0.8)/(1.6 + j0.8)$. Multiplying numerator and denominator by the conjugate of the denominator gives $\Gamma = j1.6/3.2 = j0.5$, i.e., $\Gamma = 0.5 \angle 90^\circ$.
2. $\text{VSWR} = (1+0.5)/(1-0.5) = 1.5/0.5 = 3.0$. Return loss $= -20 \log_{10}(0.5) \approx 6.02 \text{ dB}$.
3. Since $R_L = 20 \Omega < Z_0 = 50 \Omega$, a series reactance is added first: $X_s = \sqrt{[R_L(Z_0 - R_L)]} = \sqrt{(20 \times 30)} = \sqrt{600} \approx 24.49 \Omega$, giving $L = X_s/\omega = 24.49/(2\pi \times 5 \times 10^8) \approx 7.80 \text{ nH}$. The remaining susceptance is $B_p = X_s/(R_L^2 + X_s^2) = 24.49/1000 \approx 0.02449 \text{ S}$, cancelled by a shunt capacitor $C = B_p/\omega \approx 7.80 \text{ pF}$.
4. $X_L \geq 8 \times 75 = 600 \Omega$. $L_{\min} = X_L/\omega = 600/(2\pi \times 9 \times 10^8) \approx 106.1 \text{ nH}$.

Chapter 25 Solutions

1. $i = C \cdot dv/dt = (1 \times 10^{-6})(2000) = 2 \times 10^{-3} = 2 \text{ mA}$.
2. $v = L \cdot di/dt = (0.05)(3)(1000)\cos(1000t) = 150 \cdot \cos(1000t) \text{ V}$. The amplitude is 150 V and, since cosine leads sine by 90° , the voltage leads the current by 90° , as it must for an inductor.
3. $dv/dt = -(10)(2000) \cdot \sin(2000t + 30^\circ) = -20000 \cdot \sin(2000t + 30^\circ)$. The amplitude is 20000 V/s , the factor of $\omega = 2000$ having come down from the chain rule.

4. By integration: $v = (1/C) \int i \, dt = (1/10^{-5})(5 \times 10^{-3})(0.2) = (1/10^{-5})(10^{-3}) = 100 \text{ V}$. By charge: $Q = It = (5 \text{ mA})(200 \text{ ms}) = 1 \text{ mC}$, and $v = Q/C = 10^{-3}/10^{-5} = 100 \text{ V}$. The two agree.

Chapter 27 Solutions

1. By KCL: $2.0 + 6.0 = I_{\text{out}}$, so $I_{\text{out}} = 8.0 \text{ A}$.
2. Total resistance: $2.0 + 3.0 + 4.0 = 9.0 \Omega$. Current: $I = V/R = 9.0/9.0 = 1.0 \text{ A}$.
3. $V = IR = (1.0)(4.0) = 4.0 \text{ V}$.
4. If current could accumulate at a node, that node would be gaining or losing net charge over time — but an ordinary node is just a junction of ideal wires with no capacity to store charge (unlike a capacitor plate). Charge conservation (Chapter 1) forbids charge from simply appearing or vanishing, so whatever charge arrives at a node in any interval must depart in that same interval, which is exactly KCL.

Chapter 28 Solutions

1. Total series resistance: $3.0 + 5.0 + 2.0 = 10 \text{ k}\Omega$. $V_{s_k} = 20 \times 5.0/10 = 10 \text{ V}$.
2. Total current 30 mA splits: $I_{4k} = 30 \times 12/(4.0 + 12) = 30 \times 0.75 = 22.5 \text{ mA}$; $I_{12k} = 30 \times 4.0/16 = 7.5 \text{ mA}$. (Check: $22.5 + 7.5 = 30 \text{ mA}$.)
3. Parallel combination: $(4.0)(12)/(4.0 + 12) = 48/16 = 3.0 \Omega$. Adding the series 6.0Ω : $R_{\text{total}} = 6.0 + 3.0 = 9.0 \Omega$.
4. Unloaded divider would give $V_{\text{out}} = 10 \times 9.0/10 = 9.0 \text{ V}$. With the $9.0 \text{ k}\Omega$ load in parallel with the $9.0 \text{ k}\Omega$ divider resistor, the parallel combination is $(9.0)(9.0)/(9.0 + 9.0) = 4.5 \text{ k}\Omega$. Actual output: $V_{\text{out}} = 10 \times 4.5/(1.0 + 4.5) = 10 \times 0.818 \approx 8.18 \text{ V}$ — noticeably lower than the unloaded 9.0 V prediction, illustrating why the simple divider formula requires a lightly-loaded assumption.

Chapter 29 Solutions

1. KCL: $6 = V/3.0 + V/6.0 = V(1/3.0 + 1/6.0) = V(0.5)$, so $V = 12 \text{ V}$.
2. Node 1: $2 = (V_1 - 0)/4.0 + (V_1 - V_2)/2.0$. Node 2: $(V_2 - V_1)/2.0 + (V_2 - 0)/5.0 + (V_2 - V_3)/3.0 = 0$. Node 3: $(V_3 - V_2)/3.0 + V_3/8.0 = 0$.
3. The current through an ideal voltage source cannot be expressed directly as (voltage difference)/(resistance), since the source itself, not a resistor, sits in that branch — an ordinary node equation needs exactly this kind of resistor-based current expression for every branch at the node. The supernode technique sidesteps this by writing one combined KCL equation for both nodes together (so the unknown source current cancels out of the total, since it leaves one node and enters the other), supplemented by the known voltage-difference constraint the source itself provides.

Chapter 30 Solutions

1. Single mesh: $I = V/R_{\text{total}} = 20/(3.0 + 7.0) = 2.0 \text{ A}$.
2. Mesh 1: $8.0 = 3.0I_1 + 5.0(I_1 - I_2)$. Mesh 2: $0 = 5.0(I_2 - I_1) + 2.0I_2$.
3. Mesh analysis relies on being able to draw the circuit on a flat page with meshes as clearly identifiable, non-overlapping “windowpanes.” A non-planar circuit has at least one branch that must cross another without an actual electrical connection, so there is no unambiguous, non-overlapping way to define its meshes in the first place — the method's basic bookkeeping breaks down before any equations can even be written.

Chapter 31 Solutions

1. $I_N = V_{Th}/R_{Th} = 12/4.0 = 3.0$ A, in parallel with $R_N = 4.0\ \Omega$.
2. $V_{Th} = I_N \times R_N = (2.0)(6.0) = 12$ V.
3. $R_L = R_{Th} = 6.0\ \Omega$. Power: $P = V_{Th}^2/(4R_{Th}) = (12)^2/(4 \times 6.0) = 144/24 = 6.0$ W.
4. An ideal voltage source, by definition, maintains its specified voltage across its terminals regardless of the current drawn from it — it behaves, in effect, as if it had zero internal resistance, since no amount of current changes its terminal voltage. Replacing it with a short circuit ($0\ \Omega$) when deactivating sources correctly reproduces this zero-internal-resistance behavior in the equivalent-resistance calculation.

Chapter 32 Solutions

1. Total voltage: $3.0 + (-1.0) = 2.0$ V.
2. Power depends on the square of the total (superposed) voltage or current, not on the sum of individually-computed powers, because squaring (or multiplying two quantities together) is not a linear operation: $(a+b)^2 = a^2 + 2ab + b^2$, which includes a cross term $2ab$ that is lost if you compute a^2 and b^2 separately and simply add them. The correct approach is always to superpose voltages/currents first, then square (or multiply) the resulting total exactly once.
3. Three simplified circuits, one for each source acting alone with the other two deactivated.

Chapter 33 Solutions

1. $C = i/(dv/dt) = (4.0 \times 10^{-3})/(200) = 2.0 \times 10^{-5}$ F = 20 μ F.
2. $L = v/(di/dt) = 2.0/10 = 0.20$ H.
3. Parallel: $1/L = 1/4.0 + 1/4.0 = 2/4.0$, so $L = 2.0$ H.
4. An inductor's defining relation $v = L di/dt$ means an instantaneous jump in current would require di/dt (and hence v) to be infinite — no real source can supply infinite voltage, so the current physically cannot change instantaneously; it must instead change continuously over time, exactly the behavior captured by the exponential transient formulas of Chapter 34.

Chapter 34 Solutions

1. $\tau = RC = (2.0 \times 10^3)(50 \times 10^{-6}) = 0.10$ s. After one time constant, $v = V_s(1 - e^{-1}) = 12(1 - 0.368) \approx 7.58$ V.
2. $\tau = L/R = 4.0/20 = 0.20$ s.
3. Final current: $I_{final} = 10/20 = 0.50$ A. After two time constants ($t/\tau = 2$): $i = 0.50(1 - e^{-2}) = 0.50(1 - 0.135) \approx 0.43$ A.
4. Once a circuit reaches DC steady state, no quantity in it is changing anymore, so $di/dt = 0$ for every inductor — and since an inductor's voltage is $v = L di/dt$, this forces $v = 0$ regardless of how much current is flowing, which is exactly the behavior of a plain (zero-resistance) wire.

Chapter 35 Solutions

1. $Z = 30 - j40$. Magnitude: $\sqrt{(30^2+40^2)} = 50$. Angle: $\arctan(-40/30) \approx -53.1^\circ$. $Z = 50\angle-53.1^\circ \Omega$.
2. $I = V/Z = 50\angle 0^\circ / 50\angle-53.1^\circ = 1.0\angle 53.1^\circ \text{ A}$.
3. $V_m = V_{\text{rms}} \times \sqrt{2} = 240 \times 1.414 \approx 339 \text{ V}$.
4. Rectangular form: $20\angle 0^\circ = 20 + j0$; $20\angle 90^\circ = 0 + j20$. Sum: $20 + j20$. Polar form: magnitude $= \sqrt{(20^2+20^2)} = 20\sqrt{2} \approx 29.3$, angle $= 45^\circ$. Combined impedance $\approx 29.3\angle 45^\circ \Omega$.

Chapter 36 Solutions

1. $I_{\text{rms}} = P/(V_{\text{rms}} \cos\theta) = 1500/[(240)(0.90)] \approx 6.94 \text{ A}$.
2. $S = V_{\text{rms}} I_{\text{rms}} = (100)(15) = 1500 \text{ VA}$. $P = S \cos\theta = 1500(0.60) = 900 \text{ W}$. $Q = S \sin\theta = 1500(0.80) = 1200 \text{ VAR}$ (using $\sin\theta = 0.80$ from a 3-4-5-style triangle for $\cos\theta = 0.60$).
3. A purely inductive load has voltage and current exactly 90° out of phase ($\theta = 90^\circ$), so $\cos\theta = \cos(90^\circ) = 0$. Since average real power is $P = V_{\text{rms}} I_{\text{rms}} \cos\theta$, a power factor of exactly zero means $P = 0$ regardless of how large V_{rms} and I_{rms} individually are — current still flows, and energy is still exchanged back and forth each cycle (as reactive power), but none of it is delivered as net, usable average power.

Chapter 37 Solutions

1. $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{[(5.0 \times 10^{-3})(20 \times 10^{-9})]} = 1/\sqrt{(1.0 \times 10^{-10})} = 1/(1.0 \times 10^{-5}) = 1.0 \times 10^5 \text{ rad/s}$. $f_0 = \omega_0/(2\pi) \approx 15,915 \text{ Hz}$, about 15.9 kHz.
2. $Q = \omega_0 L/R = (1.0 \times 10^5)(5.0 \times 10^{-3})/2.0 = 500/2.0 = 250$.
3. Turns ratio (secondary/primary) $= 120/480 = 0.25$. Apparent power must match: $I_1 = I_2 \times (N_2/N_1) = 20 \times 0.25 = 5.0 \text{ A}$.
4. Transmitting at very high voltage minimizes the current needed to deliver a given amount of power (since $P = VI$, a larger V allows a smaller I for the same P), and resistive losses in the transmission wires scale as $I^2 R$ — so minimizing current, even at the cost of a very high and otherwise impractical voltage on the transmission line itself, dramatically reduces wasted power over long distances. Transformers then step that high voltage back down to safe, usable levels at each end, exploiting exactly the voltage-for-current tradeoff this chapter describes.
5. $\omega_c = 1/(RC) = 1/[(2.2 \times 10^3)(47 \times 10^{-9})] \approx 9671 \text{ rad/s}$. $f_c = \omega_c/(2\pi) \approx 1539 \text{ Hz}$.

Chapter 38 (End-of-Book Review) Solutions

1. KCL enforces that charge cannot accumulate at an ordinary node, a direct restatement of charge conservation applied at a single point. KVL enforces that traveling around any closed loop back to your starting point produces zero net change in potential, which follows because electric potential (Chapter 5) is a well-defined function of position — its value at a point does not depend on the path taken to reach that point, so a round trip back to the same point must return to the same potential.
2. Nodal analysis is generally preferable when a circuit has fewer independent nodes than meshes, and is especially natural when current sources are present, since a current source's value slots directly into a node's KCL equation as a

known injected current. Mesh analysis is generally preferable when a circuit has fewer meshes than nodes, and is especially natural when voltage sources dominate, since a voltage source's value slots directly into a mesh's KVL equation.

3. As load resistance increases from zero, load voltage rises from zero toward V_{Th} and load current falls from its short-circuit maximum toward zero. Power delivered to the load is maximized at $R_L = R_{Th}$, per the maximum power transfer theorem — not at either extreme, but at this specific intermediate matching condition.

4. Superposition relies on linearity: each source's contribution to a voltage or current can be found independently and added, because the equations relating sources to voltages/currents (Ohm's law, KCL, KVL) are all linear in the sources. Power depends on the square of voltage or current (or their product), and squaring/multiplying two independently-superposed quantities is not the same as summing their individually-computed squares/products, due to the cross term that this nonlinear operation introduces — so superposition, valid for voltage and current, does not carry over to power.

5. Below resonance, ωL (inductive reactance) is smaller than $1/(\omega C)$ (capacitive reactance), since inductive reactance grows with frequency while capacitive reactance shrinks with it — so at a frequency below where they are exactly equal (resonance), the capacitive term still dominates, making the net reactance capacitive.

6. An ideal transformer delivers the same apparent power ($S = VI$) on its secondary as it receives on its primary — no power is created or destroyed by an ideal transformer, only redistributed between voltage and current. If V is scaled up by some factor on one side, I must be scaled down by that same factor on the other side to keep the product VI (and hence S) unchanged, which is exactly why voltage and current transform in mutually inverse proportion to the turns ratio.

Appendix C — Glossary

Accuracy — closeness of a measurement to the true value. Distinct from precision and from resolution.

Admittance — the reciprocal of impedance, $Y = 1/Z$, in siemens. Adds for parallel branches.

Ampere (A) — SI unit of electric current; one coulomb per second.

Ampere's law — Relates the circulation of the magnetic field around a closed loop to the current enclosed by that loop.

Amplitude — the peak departure of a sinusoid from zero, written V_m . Not the same as the RMS value.

Apparent power (S) — The product of RMS voltage and RMS current, measured in volt-amperes (VA), combining real and reactive power.

Argand diagram — the complex plane, with real part plotted horizontally and imaginary part vertically.

Branch — A single path between two nodes in a circuit, generally containing one circuit element.

Capacitance (C) — The ratio of charge stored to voltage applied across a pair of conductors; measured in farads.

Conductor — A material in which charge (typically electrons) moves freely in response to an electric field.

Conjugate — the complex number with the sign of its imaginary part reversed, written z^* . Used to make denominators real.

Coulomb (C) — SI unit of electric charge.

Coulomb's law — The force between two point charges is proportional to the product of the charges and inversely proportional to the square of their separation.

Decibel — a logarithmic ratio. $10 \cdot \log_{10}$ for power ratios, $20 \cdot \log_{10}$ for voltage ratios. Dimensionless unless a reference is stated, as in dBm.

Derivative — the instantaneous rate of change of a quantity, written dv/dt .

Determinant — a scalar computed from a square matrix. A zero determinant signals that a system has no unique solution.

Dielectric — An insulating material placed between a capacitor's plates to increase its capacitance.

Dimensional analysis — checking that both sides of an equation reduce to the same combination of base units. Blind to dimensionless constants.

Drift velocity — The average velocity of charge carriers in the direction of current flow, superimposed on much larger random thermal motion.

Electric field (E) — Force per unit charge at a point in space; the physical condition set up by charge that a second charge responds to.

Electric flux — A measure of the electric field passing through a given surface.

Electric potential (V) — Potential energy per unit charge; commonly called voltage.

Electromotive force (EMF) — The energy per unit charge supplied by a source (battery, generator, or induction) that drives current around a circuit.

Elementary charge (e) — The smallest unit of free charge, carried by the proton (+e) and the electron (−e).

Engineering notation — scientific notation restricted to exponents that are multiples of three, so each maps onto an SI prefix.

Equipotential surface — A surface on which every point is at the same electric potential; always perpendicular to the local electric field.

Euler's formula — $e^{j\theta} = \cos \theta + j \sin \theta$. The link between exponentials and sinusoids, and the foundation of phasor analysis.

Faraday's law — A changing magnetic flux through a loop induces an electromotive force around that loop.

Ferromagnetic material — A material (such as iron) whose internal atomic-scale current loops align strongly with an applied magnetic field, greatly amplifying it.

Gauss's law — The total electric flux through a closed surface is proportional to the net charge enclosed.

Impedance (Z) — The complex-number generalization of resistance, relating voltage and current phasors: $V = IZ$.

Inductance (L) — A coil's characteristic constant relating the induced EMF opposing a change in its own current to the rate of that current's change.

Insulator — A material in which electrons remain bound to individual atoms and do not conduct appreciable current under ordinary fields.

Integral — the accumulated total of a quantity, equivalently the area under its curve.

j — the imaginary unit, defined by $j^2 = -1$. Written i by mathematicians; electrical engineers use j because i denotes current.

Kirchhoff's Current Law (KCL) — The sum of currents entering any node equals the sum of currents leaving it.

Kirchhoff's Voltage Law (KVL) — The sum of voltage rises and drops around any closed loop is zero.

Lenz's law — The direction of an induced current always opposes the change in magnetic flux that produced it.

Magnetic field (B) — The field that exerts a force on moving charge; measured in tesla.

Magnetic flux — A measure of the magnetic field passing through a given surface; measured in webers.

Mesh — A loop in a planar circuit that contains no other loop within it.

Mesh analysis — A systematic circuit-analysis method using loop (mesh) currents as unknowns, governed by KVL.

Mutual inductance (M) — A measure of the coupling between two coils, by which a changing current in one induces a voltage in the other.

Nodal analysis — A systematic circuit-analysis method using node voltages as unknowns, governed by KCL.

Node — A point in a circuit where two or more elements connect, all at the same potential.

Norton equivalent — A simplified representation of a linear network as a current source in parallel with a resistance.

Ohm's law — For an ohmic material, voltage across it is proportional to the current through it: $V = IR$.

Passive sign convention — The convention assigning a passive element's voltage polarity so assumed current enters at its + terminal.

Permeability of free space (μ_0) — The constant relating magnetic field to the current that produces it in vacuum.

Permittivity of free space (ϵ_0) — The constant relating electric field to the charge that produces it in vacuum.

Phase — the angular offset of a sinusoid from a reference, in degrees or radians.

Phasor — a complex number encoding the amplitude and phase of a sinusoid, with the time dependence factored out.

Power factor — The cosine of the phase angle between voltage and current; the fraction of apparent power that is real power.

Precision — the repeatability of a measurement. An instrument can be precise and inaccurate simultaneously.

Quality factor (Q) — A measure of how sharply peaked a resonant circuit's response is.

Reactance — the imaginary part of impedance. Positive for inductors, negative for capacitors. Stores energy rather than dissipating it.

Reactive power (Q) — Power exchanged between source and load each cycle with zero net average transfer, measured in VAR.

Real power (P) — The average power actually delivered to and consumed by a load, measured in watts.

Resistance (R) — The opposition a conductor presents to current flow; measured in ohms.

Resistivity (ρ) — An intrinsic material property describing how strongly a material resists current flow, independent of its shape.

Resolution — the smallest change an instrument can display. Independent of accuracy.

Resonant frequency — The frequency at which a circuit's inductive and capacitive reactance exactly cancel.

RMS (root-mean-square) — The value of a sinusoid that delivers the same average power to a resistor as a DC value of the same number.

Significant figures — the digits of a number that carry real information about its precision.

Superconductivity — A state, reached below a critical temperature in certain materials, of exactly zero electrical resistivity.

Supermesh — A combined mesh technique used when a current source is shared between two meshes.

Supernode — A combined node technique used when a voltage source connects two non-reference nodes.

Superposition — The principle that the net force or field from multiple sources is the vector sum of the contributions from each source individually.

Tesla (T) — SI unit of magnetic field.

Thevenin equivalent — A simplified representation of a linear network as a voltage source in series with a resistance.

Time constant (τ) — The characteristic timescale of a first-order transient, equal to RC or L/R .

Tolerance — the guaranteed range within which a component value lies, usually a percentage of the marked value.

Transformer — A device using mutual inductance between two windings to change voltage and current by a fixed turns ratio.

Transient — The temporary response of a circuit as it moves from one steady condition to another.

Volt (V) — SI unit of electric potential; one joule per coulomb.

Voltage divider — A pair of series resistors that splits an applied voltage between them in proportion to their resistances.

Worst-case analysis — evaluating a design with every component at its least favourable tolerance extreme simultaneously.

Appendix D — Chapter Summaries

The following summaries are meant for review after finishing the book, not as a substitute for reading each chapter in full — they compress each chapter to its central thread, deliberately leaving out the worked examples, misconceptions, and much of the connective reasoning that make the ideas stick on a first read. Use them the way you would use a set of flashcards: to check, a week or a month after finishing a chapter, whether its core idea has actually stayed with you, and to see at a glance how the thirty-eight chapters connect into a single, continuous argument rather than a list of unrelated topics.

Chapter 1 — Electric Charge

Charge is a fundamental property of matter, quantized in units of the elementary charge e , and always conserved. Objects become charged not by creating charge but by moving electrons between materials — by friction, by contact, or by induction. Conductors allow their electrons to move freely; insulators do not. This distinction, purely a matter of how tightly electrons are bound, is the seed from which the entire rest of electrical engineering grows: every subsequent chapter is, in one way or another, a study of what charge does once it is free to move, or of what happens when a field acts on charge that cannot move.

Chapter 2 — Coulomb's Law and the Electric Force

Coulomb's law quantifies the force between two point charges: proportional to the product of the charges, inversely proportional to the square of their separation. The inverse-square form reflects the geometry of a spreading sphere and reappears throughout physics. Superposition — that forces from multiple sources simply add as vectors — is what makes any charge configuration, however complex, tractable in principle, even though the bookkeeping quickly becomes the practical bottleneck rather than the physics itself.

Chapter 3 — The Electric Field

Rather than charges acting on each other directly across empty space, each charge sets up a field that fills the surrounding space, and a second charge responds only to the local field at its own position. This reframing — force as a two-step process, source to field to second charge — resolves the “action at a distance” puzzle and sets up field lines as a way of visualizing both direction (the tangent to the line) and strength (how closely lines are spaced) at a glance.

Chapter 4 — Electric Flux and Gauss's Law

Gauss's law relates the total electric flux through any closed surface to the net charge enclosed, and is exactly equivalent to Coulomb's law — but for symmetric charge distributions (spheres, infinite lines, infinite sheets), it turns what would be a difficult integral into simple algebra by letting you choose a Gaussian surface that matches the distribution's own symmetry. Applied to a conductor, it explains why excess charge sits only on the outer surface and why the field inside a conductor in equilibrium is always exactly zero.

Chapter 5 — Electric Potential Energy and Voltage

Electric potential — voltage — is potential energy per unit charge, a scalar quantity that is almost always more convenient to work with than the field directly, since it adds algebraically with no direction to track. Field and potential are two views of the same physics: the field points in the direction potential falls off fastest, and for a uniform field this reduces to the simple, practically important relation $E = \Delta V/d$. Only potential differences, never an absolute potential, are ever physically meaningful.

Chapter 6 — Conductors, Capacitance, and Dielectrics

A capacitor is any pair of conductors that stores equal and opposite charge at some voltage between them, with capacitance $C = Q/V$ depending only on geometry. The parallel-plate formula $C = \epsilon_0 A/d$ (or $\kappa \epsilon_0 A/d$ with a dielectric) is derived directly from the field of a charged sheet and the field-to-voltage relation of the previous chapter. Capacitors store real energy in their electric field, and — in a rule that inverts the resistor pattern from the next chapter — combine in parallel by simple addition and in series by reciprocal sum.

Chapter 7 — Electric Current and Resistance

Current is the rate at which charge moves past a point, conventionally taken in the direction positive charge would flow, even though in metals it is electrons moving the opposite way. Individual charge carriers drift remarkably slowly; what makes a circuit respond quickly is the field's own rapid propagation, not the carriers' speed. Resistance, $R = \rho L/A$, depends on both geometry and an intrinsic material property, resistivity, which itself depends on temperature — increasing for metals, decreasing for semiconductors, a contrast that hints at the very different physics underlying each.

Chapter 8 — Ohm's Law and a First Look at Circuits

Ohm's law, $V = IR$, is an empirical rule (not a fundamental law) that holds well for metals and manufactured resistors, though it fails deliberately for diodes and transistors later in this series. Power dissipated in a resistor, $P = IV = I^2 R = V^2/R$, is converted irreversibly to heat. Resistors combine by direct addition in series and reciprocal sum in parallel — the opposite pattern from capacitors — and a series pair forms a voltage divider, one of the most common sub-circuits in all of electronics.

Chapter 9 — Magnetism and the Magnetic Field

Unlike the electric field, the magnetic field exerts a force only on moving charge, given by $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$: always perpendicular to both velocity and field, and therefore never capable of doing work or changing a particle's speed — only its direction. A charge moving perpendicular to a uniform field travels in a circle; a current-carrying wire, being a great many moving charges together, feels a corresponding force $\mathbf{F} = I\mathbf{L} \times \mathbf{B}$, the mechanism behind every electric motor.

Chapter 10 — Sources of the Magnetic Field

Moving charge does not just respond to magnetic fields — it produces them, unifying electricity and magnetism into a single phenomenon. A long straight wire produces a field falling off as $1/r$; two such wires attract if their currents run the same direction and repel if opposite, the reverse of the static-charge rule. Ampere's law provides the same kind of symmetry-exploiting shortcut for magnetic fields that Gauss's law provides for electric fields, most usefully in deriving the uniform field inside a solenoid, whose strength can be amplified enormously by a ferromagnetic core.

Chapter 11 — Electromagnetic Induction

A changing magnetic flux through a loop induces an electromotive force, by Faraday's law, $\mathcal{E} = -N d\Phi_B/dt$ — the single fact underlying every electric generator and transformer. Lenz's law, encoded in that minus sign, guarantees the induced current always opposes the change that produced it, a direct consequence of energy conservation. A coil's own self-inductance, quantifying its resistance to changes in its own current, is the first hint of the inductor as a third fundamental circuit element, alongside the resistor and capacitor.

Chapter 12 — Toward Maxwell's Equations

Maxwell's four equations — two versions of Gauss's law, Faraday's law, and a corrected Ampere's law — unify everything in this book into one self-consistent system. Maxwell's own addition, the displacement current, completes a symmetry that predicts self-sustaining electromagnetic waves, propagating at a speed computable from ϵ_0 and μ_0 alone, that turns out to match the measured speed of light exactly — revealing light itself as an electromagnetic wave, and setting up every later book in this series as, in one way or another, a special case of these same four equations.

Chapter 13 — Quantities, Units, and the SI System

A quantity is a number times a unit, and the multiplication is real, so units obey algebra. Seven SI base units; the electrical derived units are built from four of them. Dimensional analysis catches structural errors for free but is blind to dimensionless constants, including the 2π that separates hertz from radians per second.

Chapter 14 — Prefixes, Scientific Notation, and Engineering Notation

Prefixes multiply by powers of ten and case is significant. Engineering notation restricts exponents to multiples of three so they map onto prefixes. Reciprocal prefix pairs cancel: micro times mega is one, milli times kilo is one. Exponents must match before mantissas can be added.

Chapter 15 — Precision, Significant Figures, and Uncertainty

A measurement includes its uncertainty. Significant figures describe how to report a result, not how to compute one, so do not round intermediate steps. Relative uncertainties add for products; absolute uncertainties add for sums. Subtracting nearly equal numbers destroys precision. Accuracy, precision, and resolution are three different things.

Chapter 16 — Algebra for Circuit Analysis

The reciprocal of a sum is not the sum of the reciprocals. Product-over-sum handles exactly two parallel elements. Voltage divides in proportion to resistance, current in inverse proportion. The discriminant of a quadratic tells you whether a second-order circuit is overdamped, critically damped, or oscillatory.

Chapter 17 — Systems of Linear Equations

Circuit equations from Kirchhoff's laws are linear, so systematic solution methods apply. Elimination scales better than substitution. A zero determinant indicates a physically impossible circuit, not an arithmetic mistake. The matrix form $\mathbf{G} \cdot \mathbf{V} = \mathbf{I}$ is what simulators actually solve.

Chapter 18 — Functions, Graphs, and Limiting Behaviour

Read a transfer function at its limits before plotting it: low-frequency asymptote, high-frequency asymptote, and behaviour at the corner give the shape. Log axes turn power laws into straight lines whose slope reveals the order of the system. A logarithmic axis has no zero.

Chapter 19 — Exponentials, Logarithms, and the Decibel

The exponential is the function whose rate of change equals itself, which is why it governs every first-order circuit. Logarithms convert multiplication to addition, which is the entire reason decibels exist. Use $10 \cdot \log$ for power and $20 \cdot \log$ for voltage. dB values add along a chain; dBm values do not.

Chapter 20 — Trigonometry and the Sinusoid

A sinusoid is fixed by three numbers: amplitude, angular frequency, and phase. Radians are required for calculus; degrees are convenient for phase. The Pythagorean identity produces impedance magnitudes and the sum-of-angles identity produces mixing. RMS is defined by equal heating, and the 0.707 factor applies only to sinusoids.

Chapter 21 — Complex Numbers

A complex number carries two quantities at once, which for engineers means magnitude and phase. Rectangular form suits addition; polar form suits multiplication. The conjugate makes denominators real. Multiplying by j is a 90° rotation, and $j^2 = -1$ is that fact applied twice.

Chapter 22 — Euler's Formula and the Complex Exponential

Euler's formula identifies $e^{j\theta}$ with a unit arrow at angle θ , bridging rotation and oscillation. A real sinusoid is the real part of a rotating complex exponential. Differentiating that exponential multiplies it by $j\omega$, which converts calculus into arithmetic.

Chapter 23 — Phasors and Impedance

In a linear circuit at one frequency, only amplitude and phase vary, so the frequency can be factored out into a phasor. Impedance is the resulting complex voltage-to-current ratio: R , $j\omega L$, and $1/(j\omega C)$. Every DC circuit technique carries over with R replaced by Z . Component voltages add as phasors, so they may exceed the supply voltage.

Chapter 24 — The Smith Chart, Impedance Matching, and Bias Networks

Matching a complex load to a real system impedance requires a conjugate match, not merely equal resistances. The Smith chart plots the reflection coefficient Γ over a grid of constant-resistance circles and constant-reactance arcs, letting an L-network or a microstrip stub be designed graphically: series elements move a point along a circle of constant r , shunt elements along a circle of constant g . RF chokes and high-value resistors bias a stage without disturbing its matching networks, because one presents a high reactance and the other draws negligible current, at the frequency that actually matters.

Chapter 25 — Calculus for Electrical Engineering

A derivative is a rate and an integral is an accumulation. Capacitors and inductors are defined by derivatives, which is why a capacitor blocks DC and an inductor passes it. Differentiating a sinusoid brings down a factor of ω and advances the phase by 90° , which is exactly the $j\omega$ of phasor analysis.

Chapter 26 — Where This Book Leads Next

Part III uses Chapters 16, 17, 19, and 25 for transients and Chapters 20 to 24 continuously for AC. Filters use Chapters 18 and 19. RF work leans on complex arithmetic and decibels throughout. The Fourier and Laplace transforms generalise the phasor idea, and nothing later in the series depends on a fundamentally unfamiliar mathematical foundation, even though later material builds on these tools in more demanding ways.

Chapter 27 — Circuit Elements and Kirchhoff's Laws

KCL (charge conservation at a node) and KVL (potential as a well-defined function of position, applied around a loop) are the two laws sufficient, together with each element's own v - i relationship, to solve any circuit. The passive sign convention fixes voltage polarity relative to assumed current direction so that power calculations come out with a consistent, interpretable sign.

Chapter 28 — Equivalent Resistance and Dividers

Series/parallel reduction, whenever it applies, is the fastest path to a solution. The voltage divider splits voltage in proportion to each resistor's own value; the current divider splits current in proportion to the other resistor's value — an inversion worth rederiving rather than memorizing.

Chapter 29 — Nodal Analysis

Treating node voltages as unknowns and applying KCL at each non-reference node produces a system solvable for any resistive network, planar or not. Voltage sources between two non-reference nodes require the supernode technique.

Chapter 30 — Mesh Analysis

Treating mesh currents as unknowns and applying KVL around each mesh is nodal analysis's dual, applicable only to planar circuits. Current sources shared between meshes require the supermesh technique. Choosing between nodal and mesh analysis is a practical efficiency decision, not a question of correctness — both, done right, agree.

Chapter 31 — Thevenin and Norton Equivalents

Any linear network, viewed from two terminals, reduces to a single source and a single resistance — a voltage source in series (Thevenin) or a current source in parallel (Norton), related by $I_N = V_{Th}/R_{Th}$. Maximum power transfer to a load occurs when the load resistance matches R_{Th} , at a cost of only 50% efficiency.

Chapter 32 — Superposition and Linearity

In a linear circuit with multiple sources, each source's individual contribution to a voltage or current can be found separately and summed — a direct consequence of the linearity of Ohm's law, KCL, and KVL. Superposition does not extend to power, since power depends nonlinearly (via squaring or multiplication) on voltage and current.

Chapter 33 — Capacitors and Inductors as Circuit Elements

A capacitor's current and an inductor's voltage depend on the rate of change of the other quantity, making both elements dynamic rather than instantaneous like a resistor. Capacitor voltage and inductor current cannot change instantaneously — the starting point for transient analysis. Inductors combine like resistors (series adds, parallel reciprocal-sums); capacitors combine the opposite way.

Chapter 34 — First-Order Transients

RC and RL circuits, following a switching event, relax exponentially toward a new steady state, governed by the time constant $\tau = RC$ or $\tau = L/R$. Every such transient follows the same general form, $x(t) = x_{final} + (x_{initial} - x_{final})e^{(-t/\tau)}$, regardless of which specific quantity or circuit is involved.

Chapter 35 — Sinusoidal Steady-State Analysis

With impedance defined, every DC technique from Chapters 27–32 — KVL/KCL, series/parallel combination, dividers, nodal and mesh analysis, Thevenin/Norton, superposition — applies unchanged to AC circuits, using complex impedances in place of real resistances. RMS values, not peak amplitudes, are the practical standard for quoting AC voltages and currents.

Chapter 36 — AC Power and Power Factor

Average AC power is $P = V_{rms} I_{rms} \cos\theta$, where $\cos\theta$, the power factor, measures what fraction of the apparent power ($S = V_{rms} I_{rms}$) is real, useful power (P) rather than reactive power (Q) that is exchanged but not net-delivered. Low power factor means larger current for the same real power, and correspondingly larger transmission losses — the practical reason power factor correction matters.

Chapter 37 — Resonance, Frequency Response, and Transformers

A series RLC circuit's inductive and capacitive reactances cancel exactly at the resonant frequency $\omega_0 = 1/\sqrt{LC}$, leaving a purely resistive, current-maximizing impedance; the quality factor Q measures how sharply peaked this

resonance is. An ideal transformer trades voltage for current (or vice versa) in exact proportion to its turns ratio, keeping apparent power unchanged — the basis of high-voltage long-distance power transmission.

Chapter 38 — Where This Series Goes Next

Every technique in this book applies only to linear circuits. The next book introduces the nonlinear semiconductor devices — diodes, transistors — that make up the bulk of real electronics, and shows how their small-signal, linearized behavior around a fixed operating point reduces, once again, to exactly the linear circuit analysis this book has built.

Appendix E — A Practical Circuit-Analysis Checklist

This appendix collects, in one place, the habits this book has recommended chapter by chapter — not new material, but a single reference to run through when a solution comes out wrong, or before trusting one that came out right.

Before Writing Any Equations

1. Identify every node, and confirm you have not accidentally split a single node (joined by plain wire) into two, or merged two genuinely distinct nodes into one.
2. Assign a current direction and voltage polarity to every branch, using the passive sign convention, and mark them directly on the diagram rather than trying to hold them in your head.
3. For nodal analysis, choose a reference node — usually the one with the most connections — and set it to 0 V.
4. For mesh analysis, confirm the circuit is planar before proceeding; if it is not, nodal analysis is the only systematic option available.
5. Check for supernodes (a voltage source directly between two non-reference nodes) or supermeshes (a current source shared between two meshes) before writing ordinary node or mesh equations.

While Solving

1. Count your equations against your unknowns before solving — one KCL equation per non-reference node (or one combined equation per supernode), or one KVL equation per mesh (or one per supermesh, plus its constraint equation).
2. For AC (phasor) problems, confirm every impedance is written correctly — R for a resistor, $j\omega L$ for an inductor, $1/(j\omega C)$ or $-j/(\omega C)$ for a capacitor — and that ω has been computed correctly from the source frequency before any impedance is evaluated numerically.
3. Add and subtract complex numbers in rectangular form; multiply and divide them in polar form. Converting to the wrong form for a given operation is one of the most common sources of arithmetic error in phasor problems.

After Solving

1. If a branch current came out negative, that is not necessarily an error — it means the actual current flows opposite to your originally assumed direction, and the magnitude is still correct.
2. Check KCL and KVL directly against your final answer: plug the solved voltages and currents back into the original node and loop equations and confirm they balance, rather than trusting the algebra that produced them.
3. Where two independent methods are available for the same quantity (for instance, nodal analysis and a direct Thevenin-equivalent reduction), solving the same problem both ways is one of the most reliable checks available — genuine agreement between two structurally different methods is strong evidence against a hidden error in either one.
4. For power calculations in multi-source circuits, confirm you superposed voltages/currents first and computed power only once, from the final total — never by summing individually-computed powers from each source.
5. For transient problems, confirm the initial and final conditions separately before writing the general exponential solution — an error in either endpoint propagates through the entire time-domain answer even if the time constant itself is correct.

A Final Habit Worth Keeping

Every method in this book — nodal analysis, mesh analysis, Thevenin equivalents, superposition, phasor analysis — is built on the same two laws from Chapter 27. When a solution feels uncertain, the most reliable recovery is rarely to search for a more advanced technique; it is to return to KCL and KVL directly, applied carefully and patiently to the specific circuit in front of you, and to let the rest of the book's machinery earn back your trust one correctly-checked step at a time.

Appendix F — Further Reading

This book derives and verifies every result it uses from first principles, so none of the following is required to read it. They are listed for readers who want to go beyond a single introductory chapter in a given area — organised by part, with a dedicated list for Chapter 24's specialised topic, impedance matching and the Smith chart, which draws on a body of RF and microwave engineering well beyond this book's scope.

Part I — The Physics of Electricity and Magnetism

David J. Griffiths, *Introduction to Electrodynamics* — the standard next step in rigor beyond this book's algebra-based treatment, developing Maxwell's equations in their full vector-calculus form.

Daniel Fleisch, *A Student's Guide to Maxwell's Equations* — a friendlier companion to Griffiths, walking through each of the four equations term by term — arguably the better next step for a reader who liked this book's approachable style specifically.

David Halliday, Robert Resnick, and Jearl Walker, *Fundamentals of Physics* — a broader general-physics reference, useful if you want more worked problems at this same introductory level across all of physics, not only electricity and magnetism.

Part II — Mathematics and Measurement

Erwin Kreyszig, *Advanced Engineering Mathematics* — the standard engineering-mathematics reference; it covers every topic in this part in more depth and continues into differential equations, transforms, and vector calculus.

K. A. Stroud (with Dexter J. Booth), *Engineering Mathematics* — a gentler, more example-driven companion if Kreyszig's pace feels too fast.

Part III — DC and AC Circuit Theory

Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits* — the standard next-level circuits textbook, with far more worked problems and a treatment of Laplace-domain circuit analysis that this book only opens the door to in Chapter 25.

Paul Horowitz and Winfield Hill, *The Art of Electronics* — not a substitute for a formal circuits course, but the standard reference for the practical, component-level electronics that the next book in this series builds on.

For Chapter 24 — The Smith Chart, Impedance Matching, and Bias Networks

David M. Pozar, Microwave Engineering — the standard graduate-level microwave engineering textbook; its treatment of the Smith chart, impedance matching, and microstrip design is the most complete and rigorous available, and the single source this chapter leaned on most heavily.

Phillip H. Smith, Electronic Applications of the Smith Chart — written by the chart's originator, often referred to simply as the Smith Chart Handbook; the most thorough single-topic treatment of the chart itself and its graphical constructions.

Chris Bowick, RF Circuit Design — a shorter, design-oriented companion, strong on practical lumped-element matching networks and filter design.

Guillermo Gonzalez, Microwave Transistor Amplifiers: Analysis and Design — extends this chapter's bias-network treatment into full small-signal amplifier design.

Ulrich L. Rohde and David P. Newkirk, RF/Microwave Circuit Design for Wireless Applications — a further, more advanced treatment of matching networks and RF circuit design in a systems context.

